

(12) **United States Patent**
Harmon et al.

(10) **Patent No.: US 12,403,769 B2**
(45) **Date of Patent: Sep. 2, 2025**

(54) **MODULAR VEHICLE HMI**

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Dearborn, MI (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

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(21) Appl. No.: **18/471,817**

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(65) **Prior Publication Data**
US 2025/0100379 A1 Mar. 27, 2025

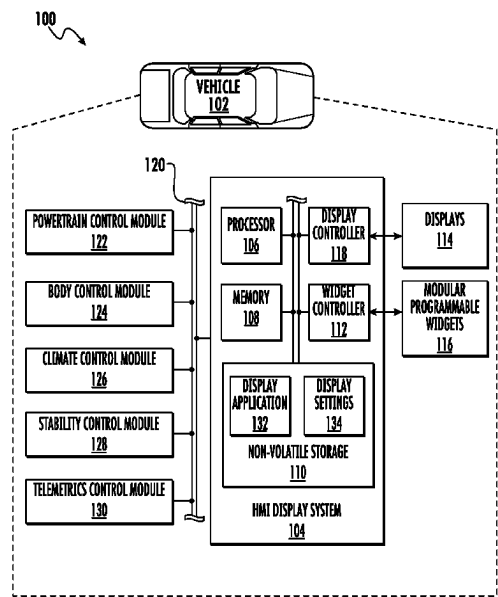
(51) **Int. Cl.**
B60K 35/10 (2024.01)

(52) **U.S. Cl.**
CPC **B60K 35/10** (2024.01)

(58) **Field of Classification Search**
CPC G06F 3/0227; G06F 3/0202; G06F 3/0362; G06F 3/0482; G06F 3/04847; G06F 3/0488; B60K 35/10; B60K 2360/126; B60K 2360/131; B60K 35/00
See application file for complete search history.

(57) **ABSTRACT**
A modular programmable widget for a vehicle includes a front housing and a back housing defining an enclosure; one or more physical controls provided on a front face of the enclosure; a contact pad exposed by the enclosure for electrical connection to a dock, the contact pad providing for transmission of data and power between the modular programmable widget and the dock; and one or more features for mechanical connection and securing of the modular programmable widget to the dock.

24 Claims, 54 Drawing Sheets



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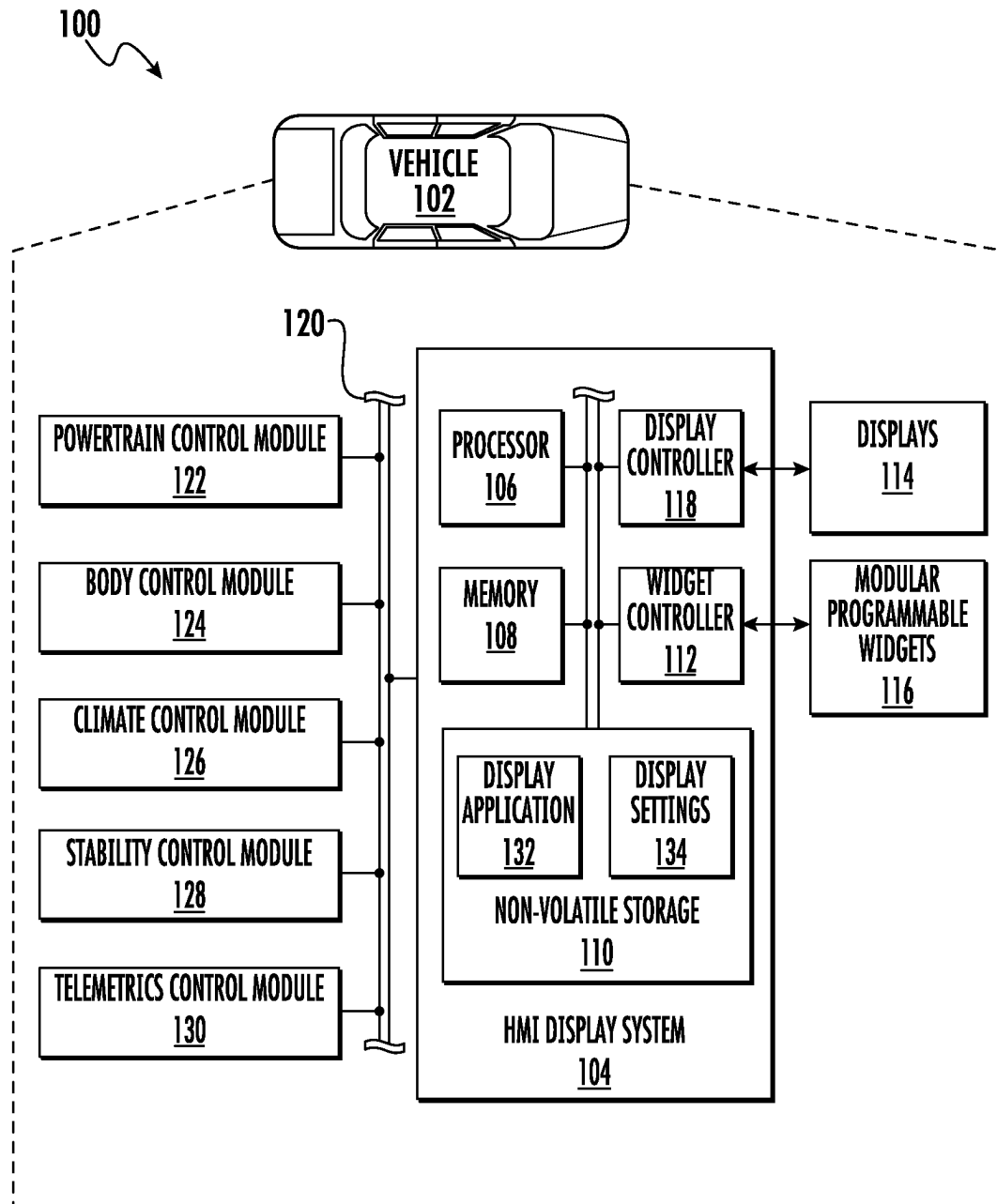


FIG. 1

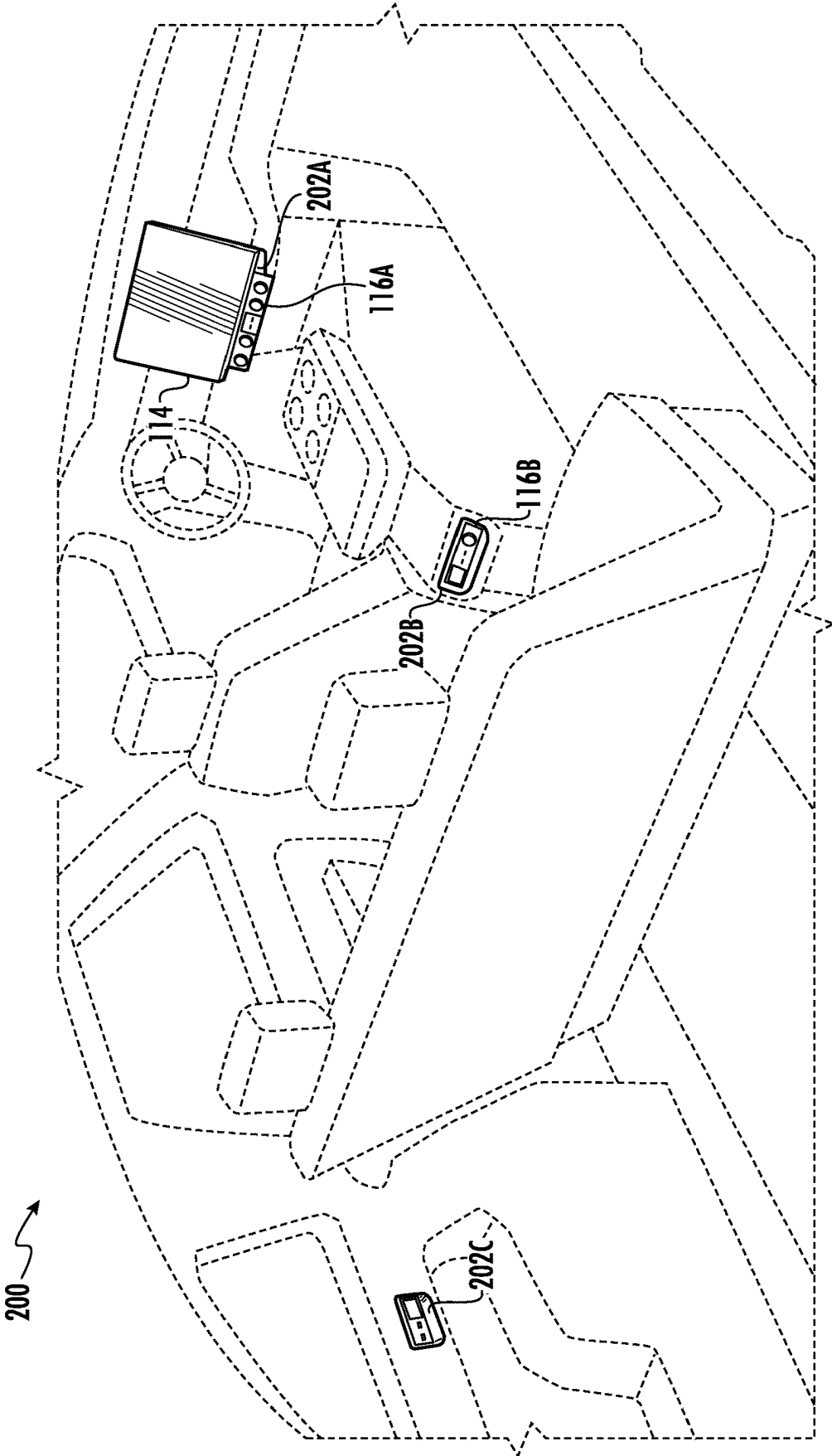
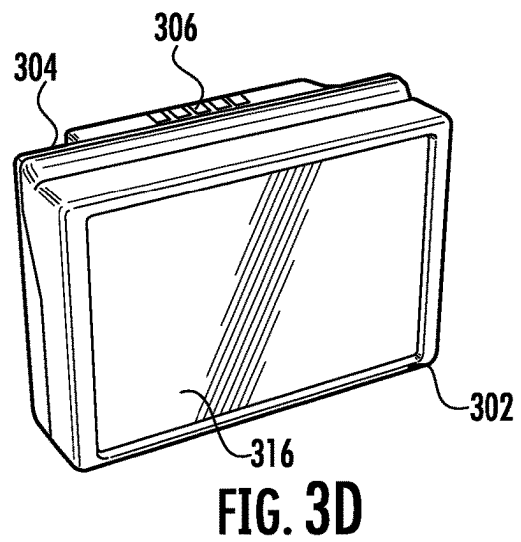
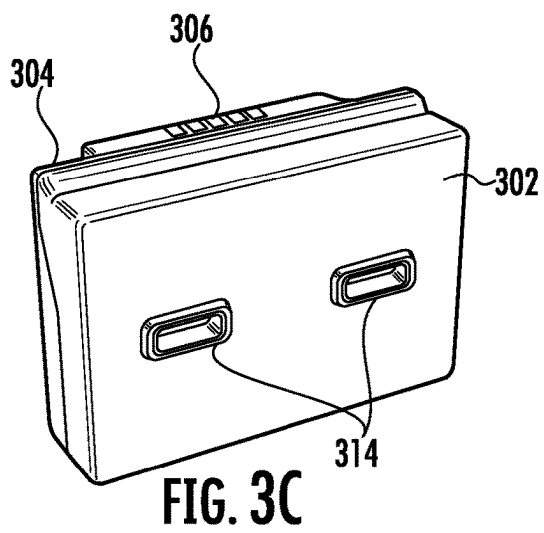
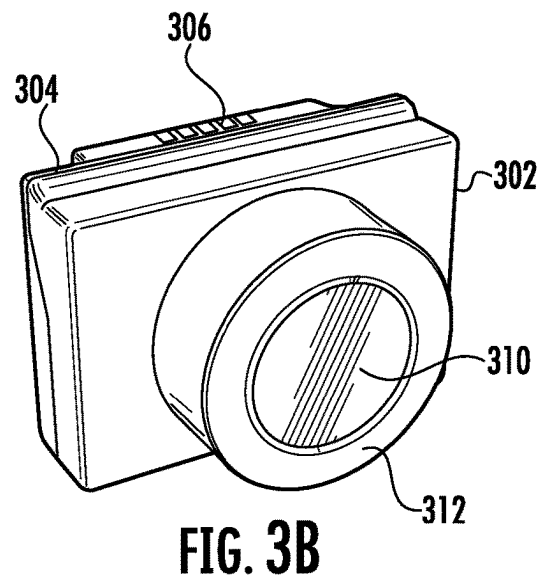
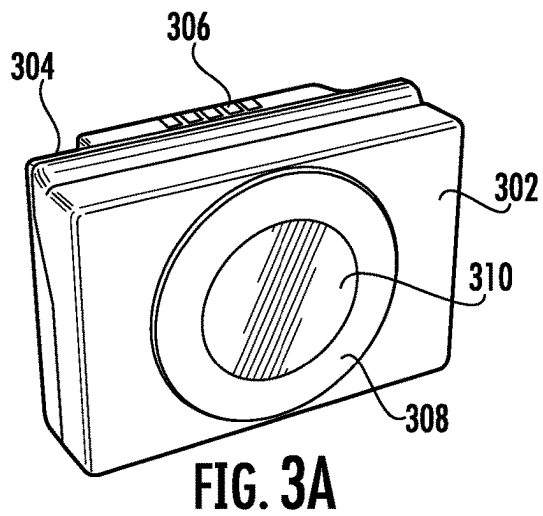
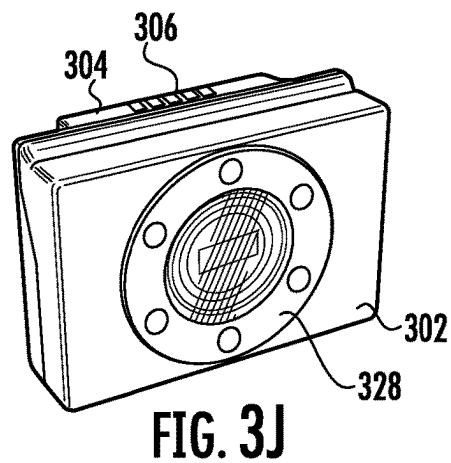
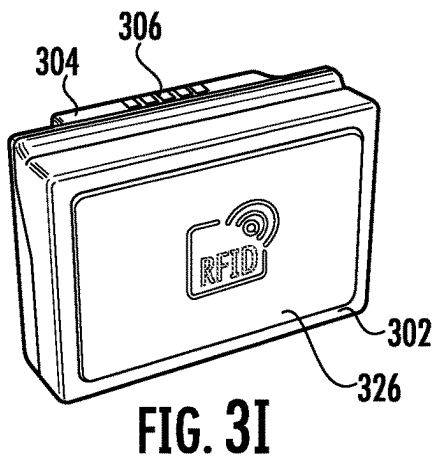
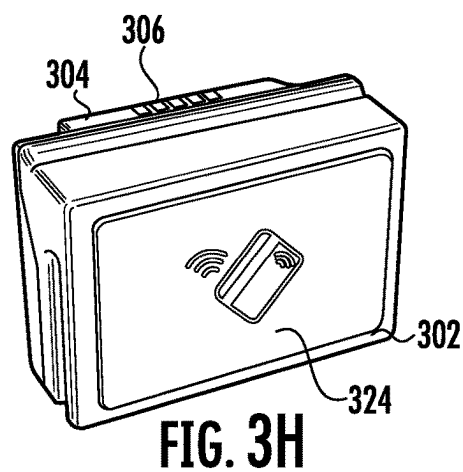
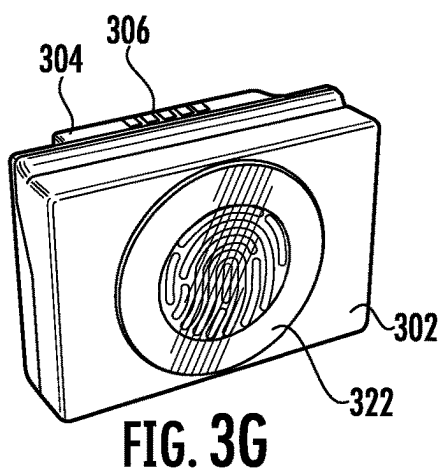
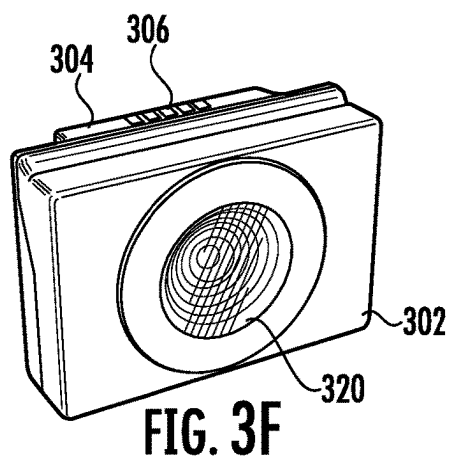
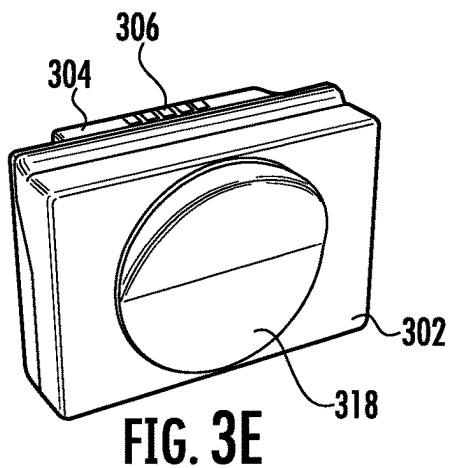


FIG. 2





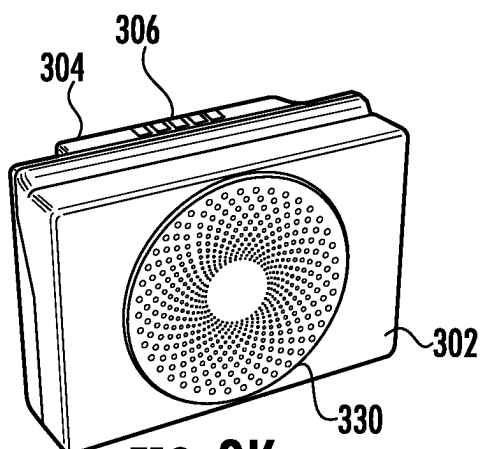


FIG. 3K

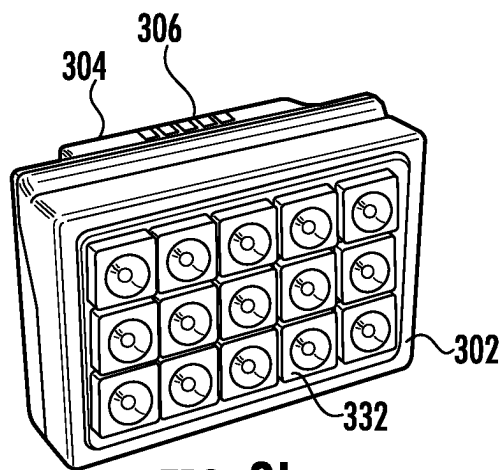


FIG. 3L

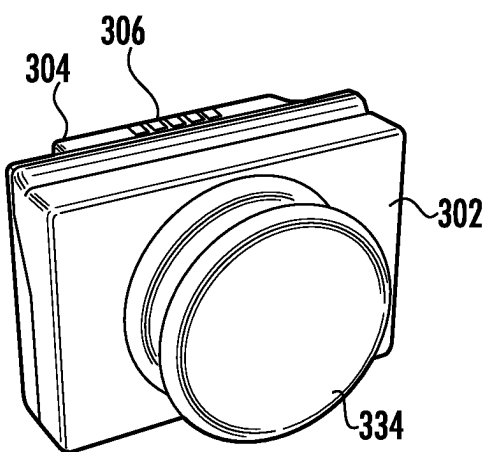


FIG. 3M

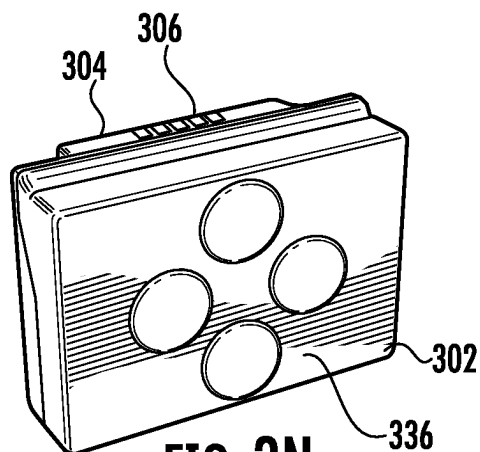


FIG. 3N

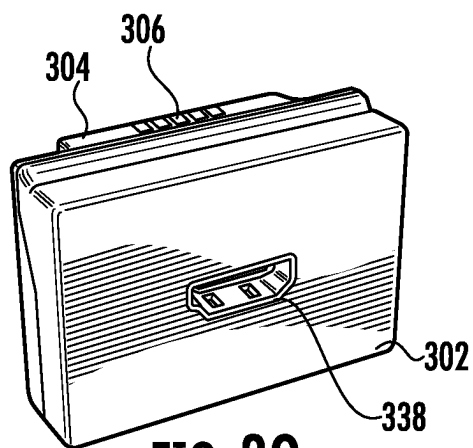


FIG. 3O

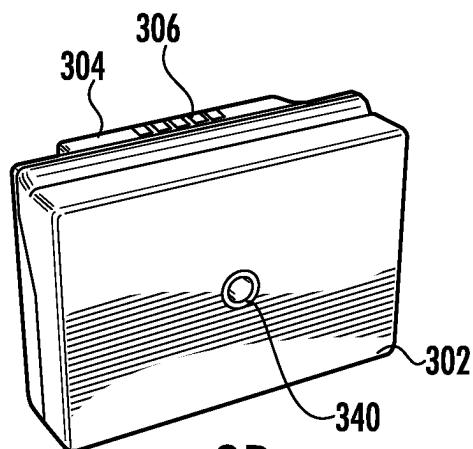


FIG. 3P

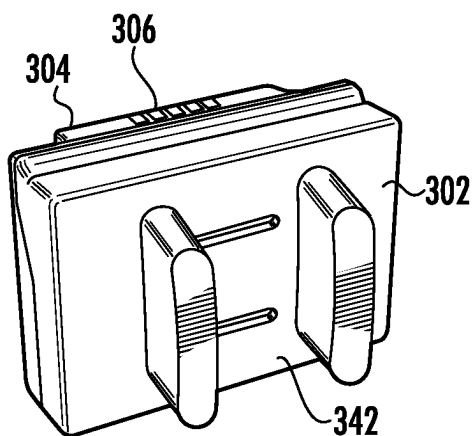


FIG. 3Q

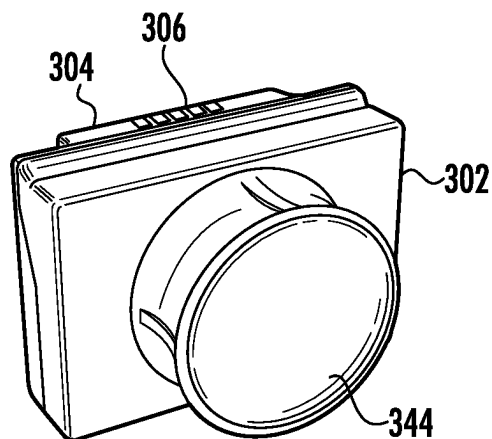


FIG. 3R

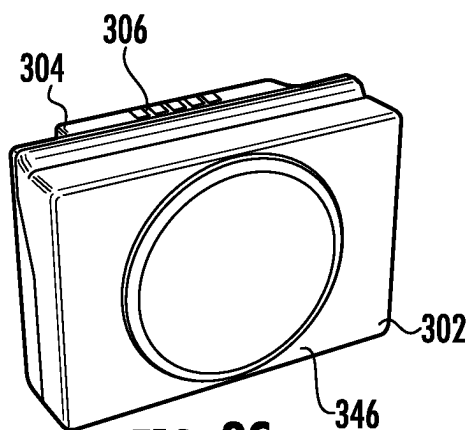


FIG. 3S

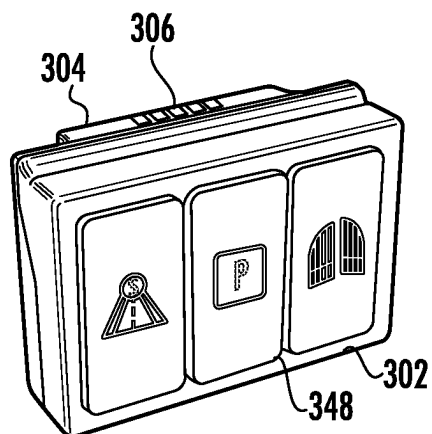


FIG. 3T

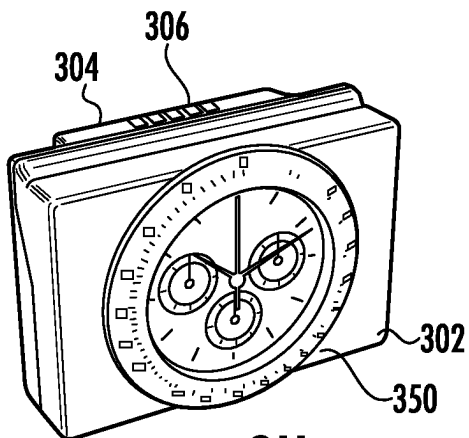


FIG. 3U

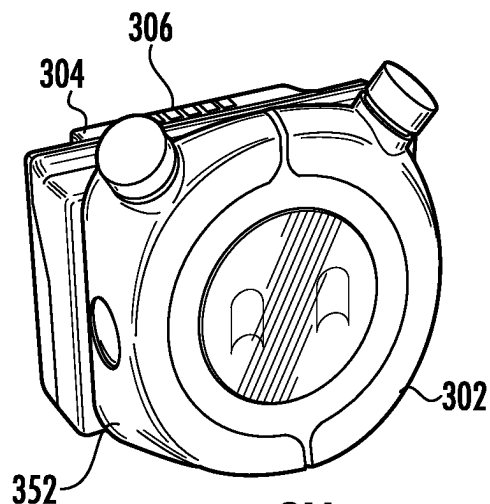
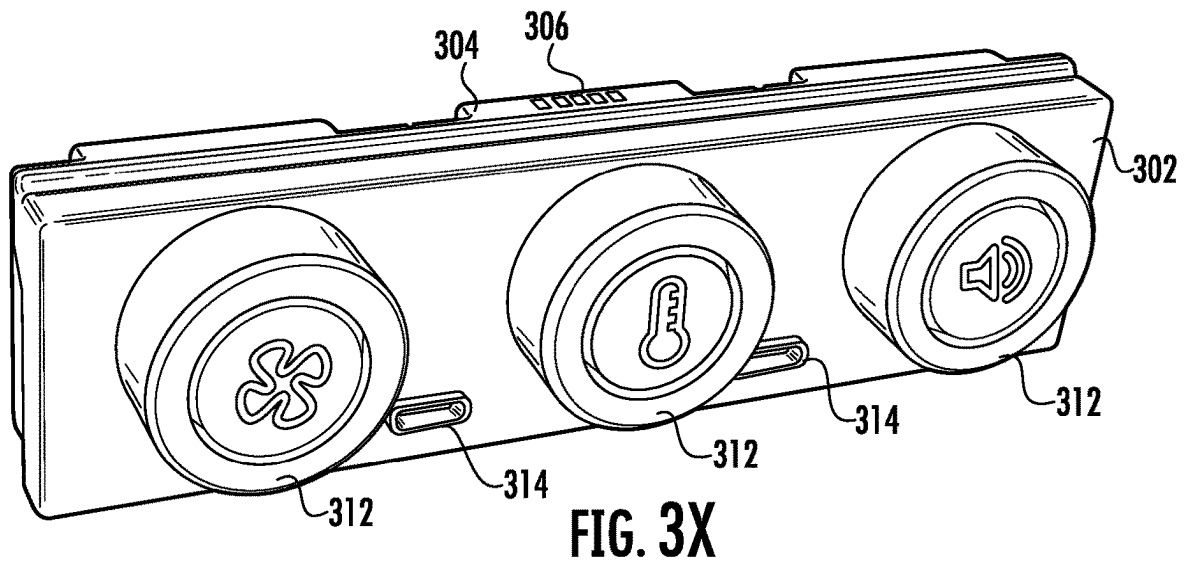
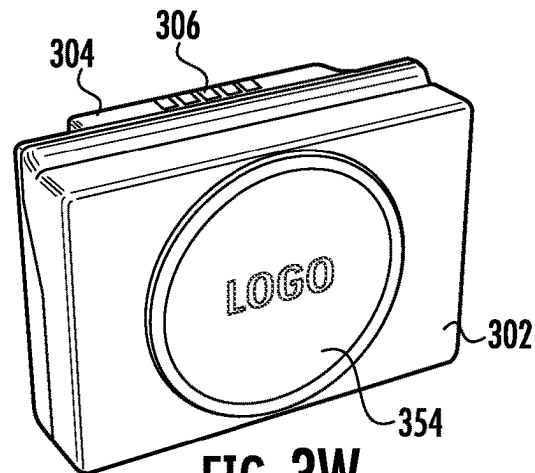


FIG. 3V



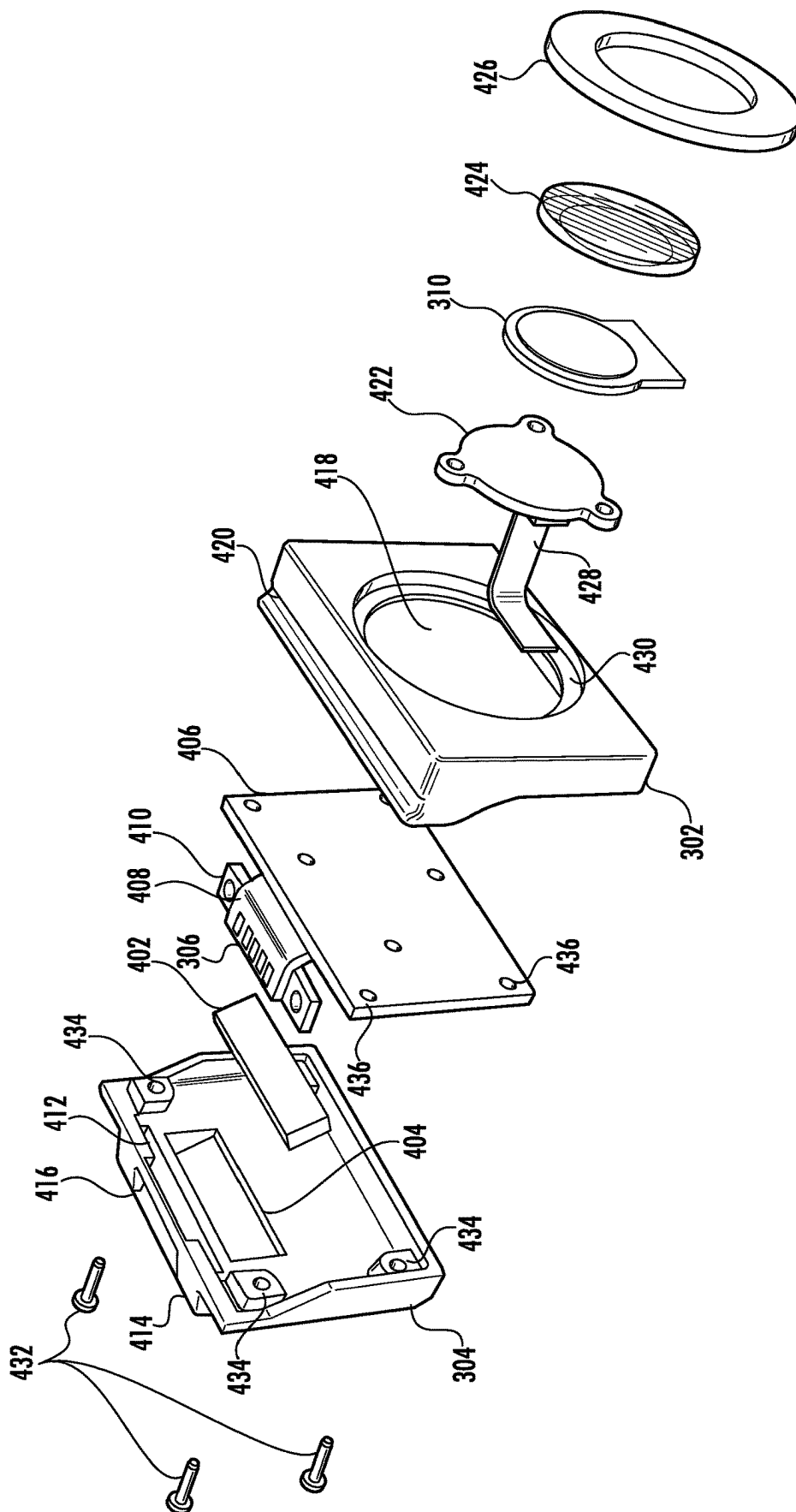


FIG. 4A

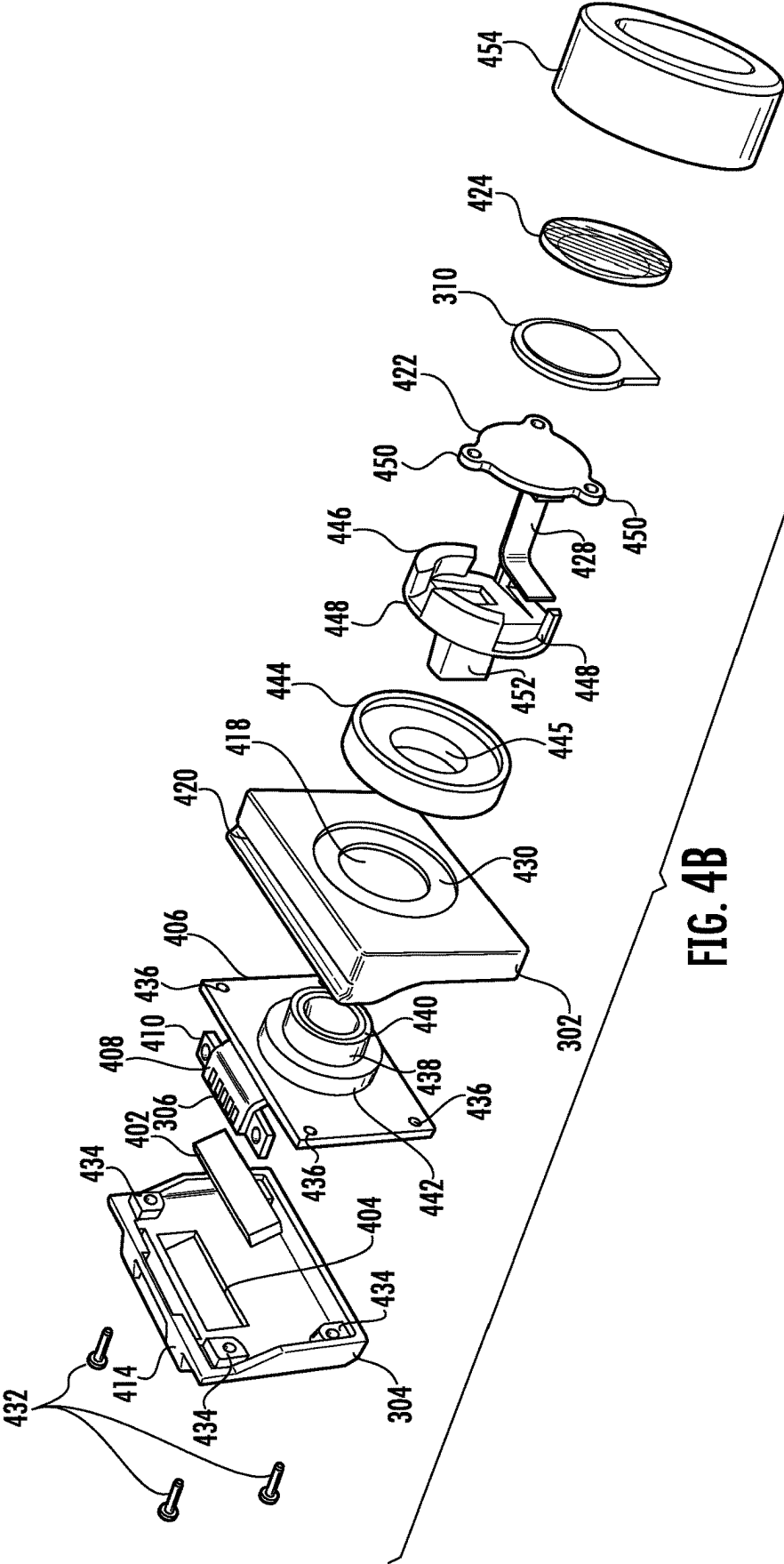
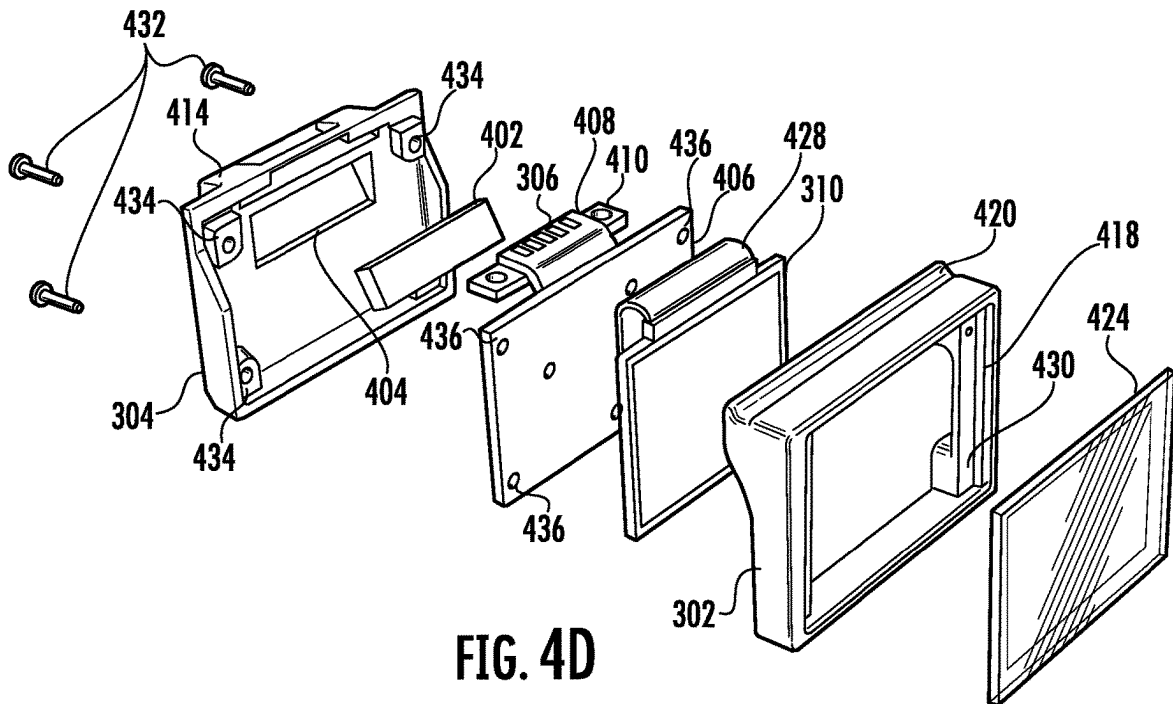
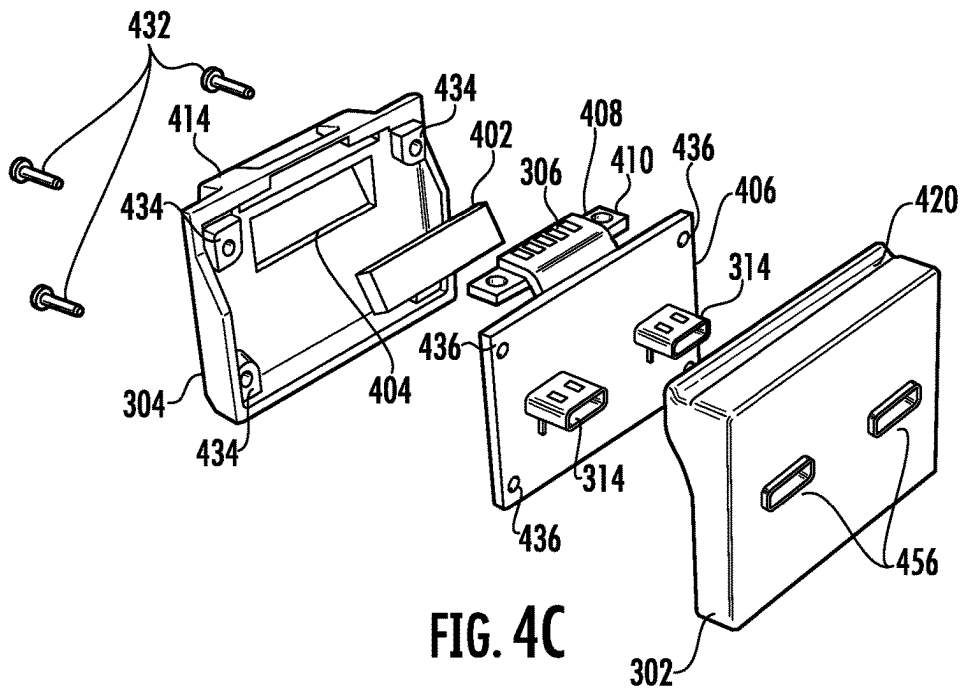


FIG. 4B



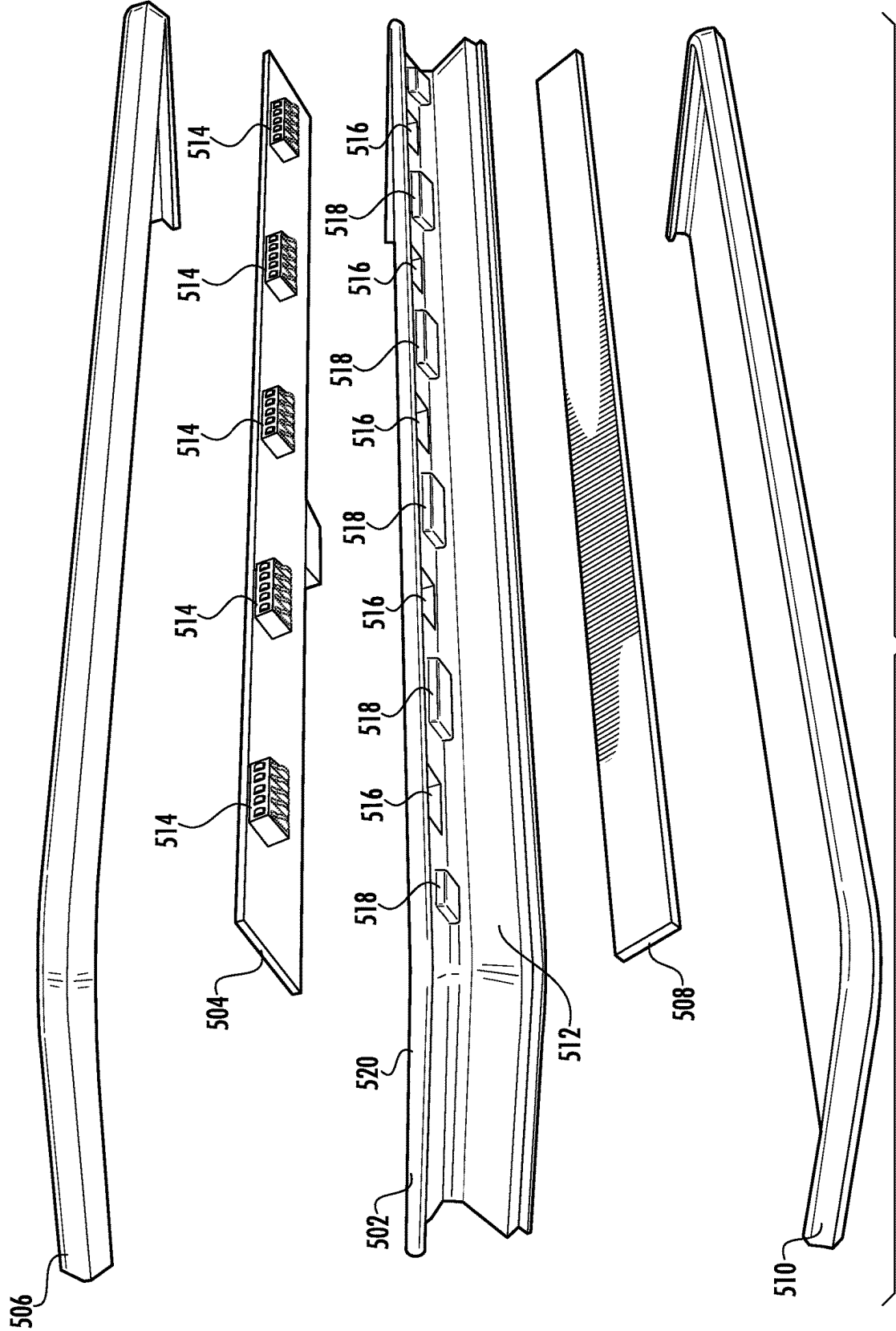
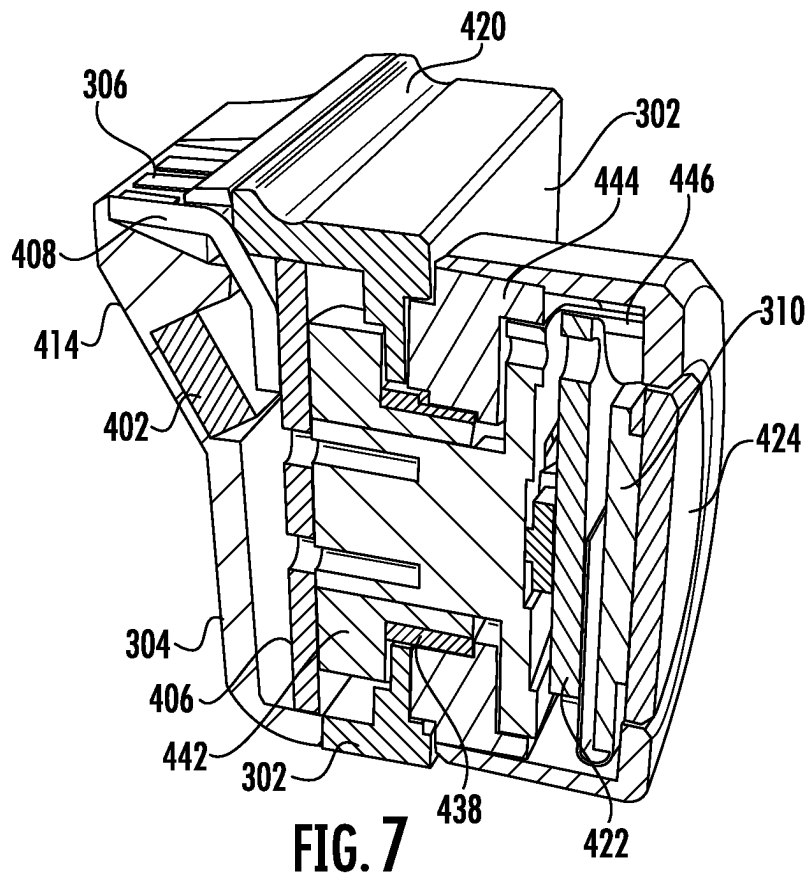
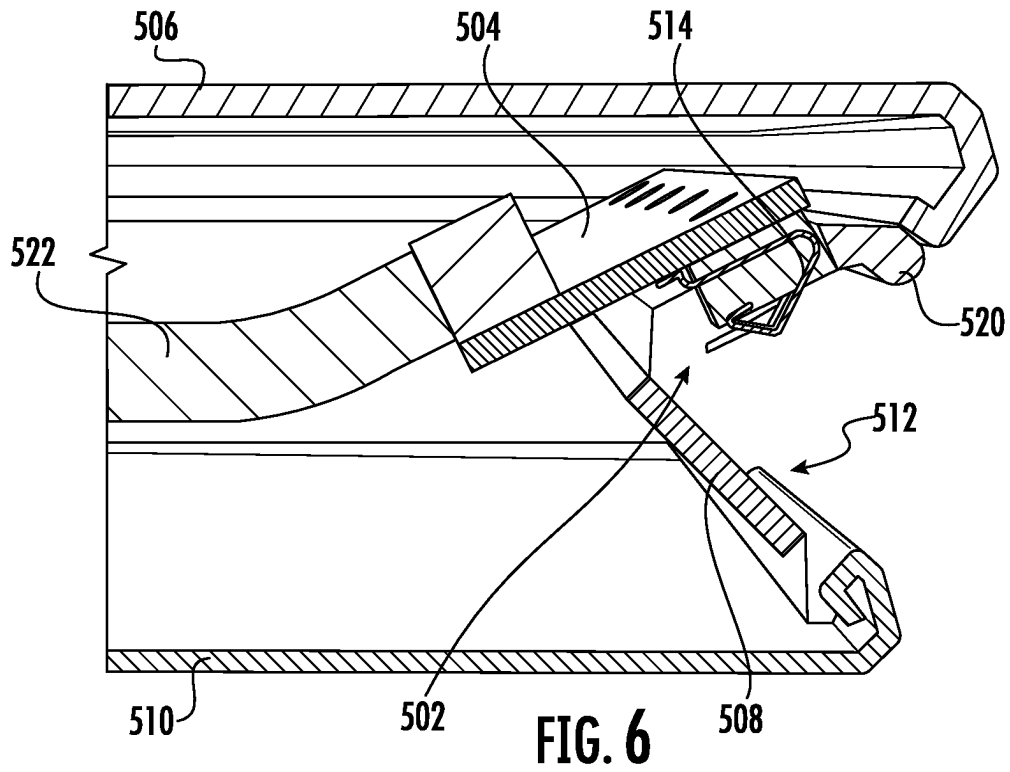
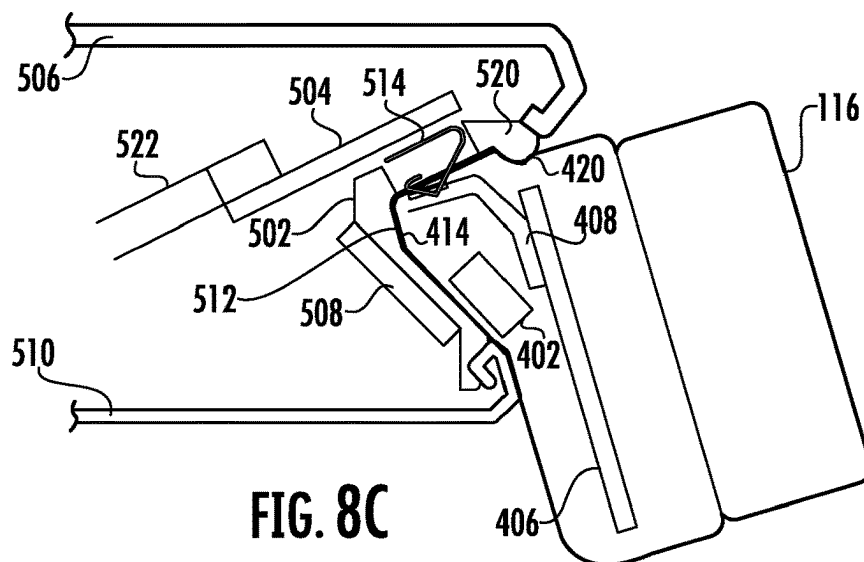
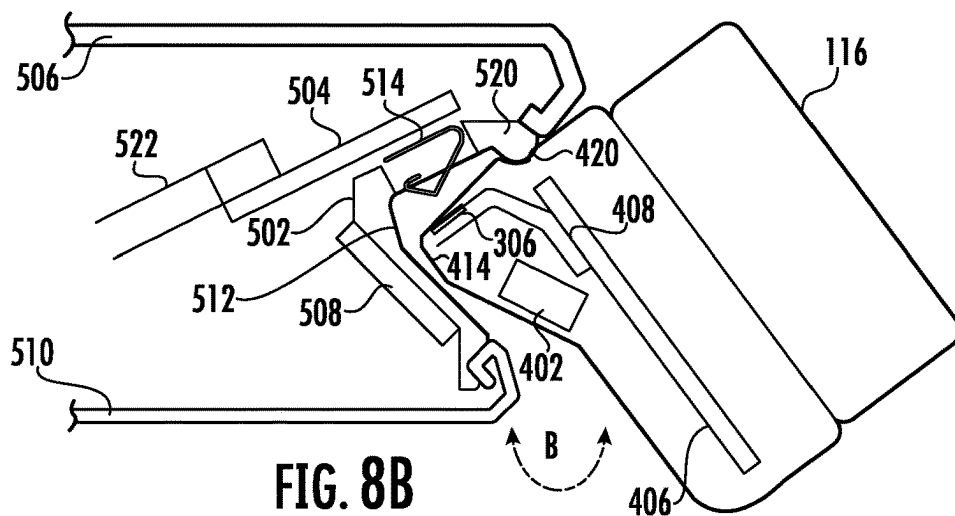
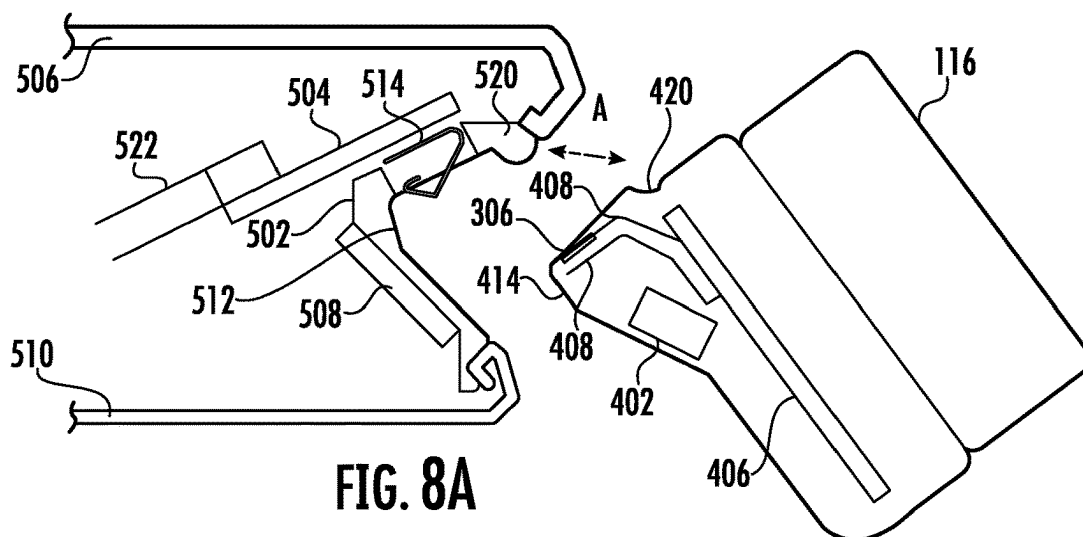


FIG. 5





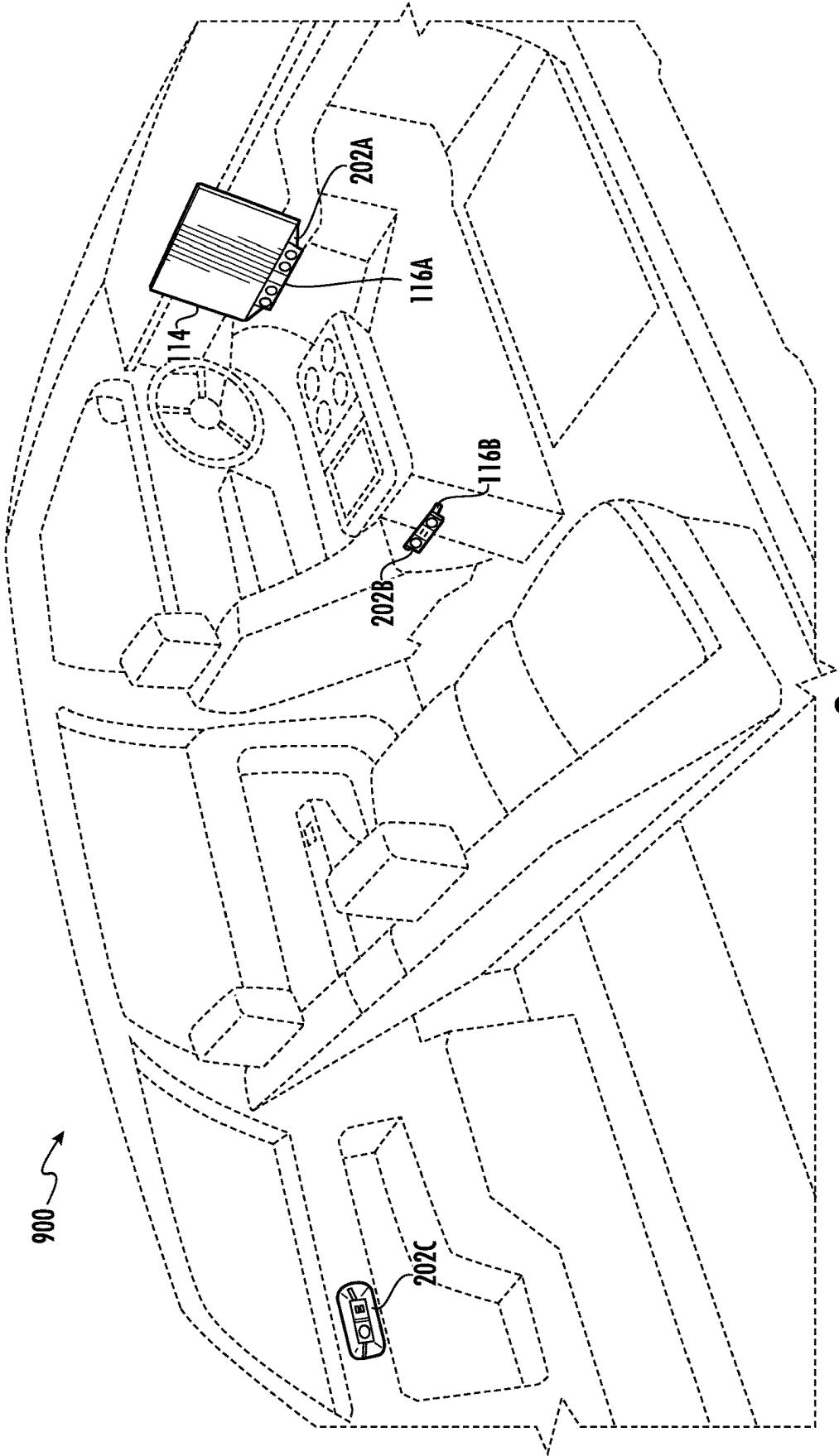


FIG. 9

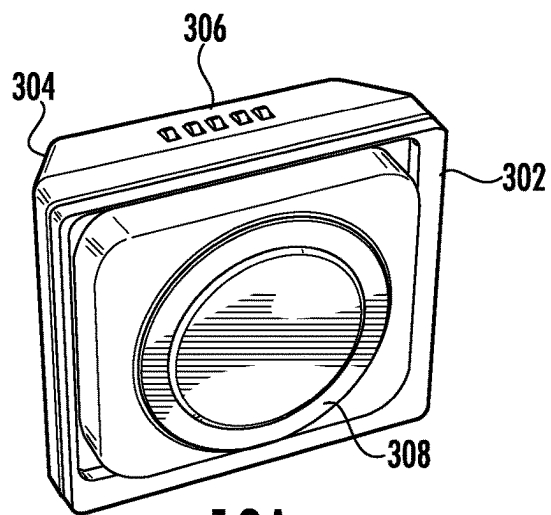


FIG. 10A

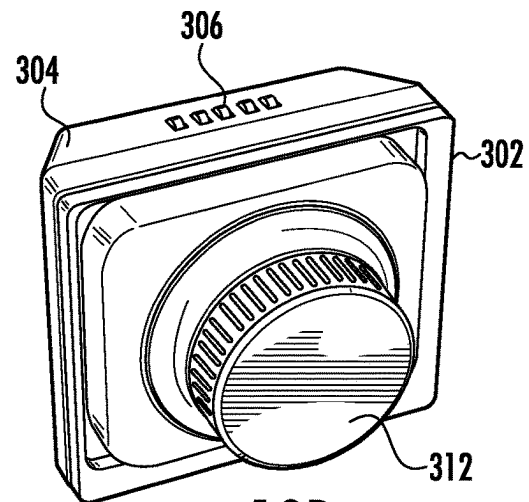


FIG. 10B

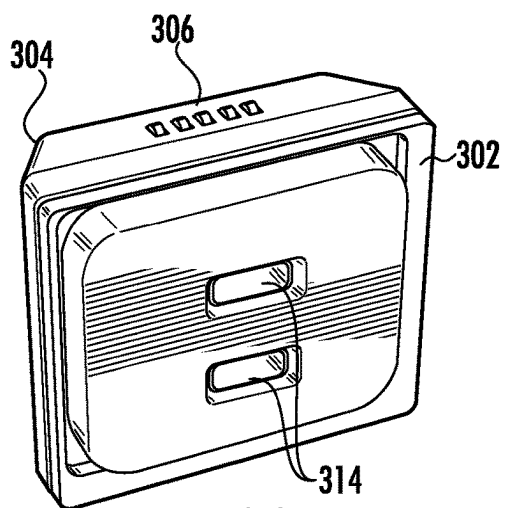


FIG. 10C

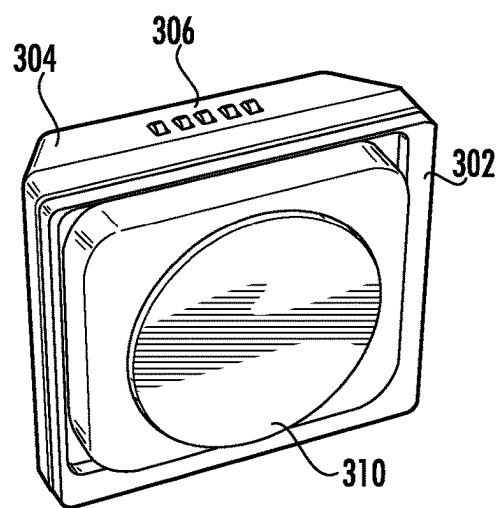


FIG. 10D

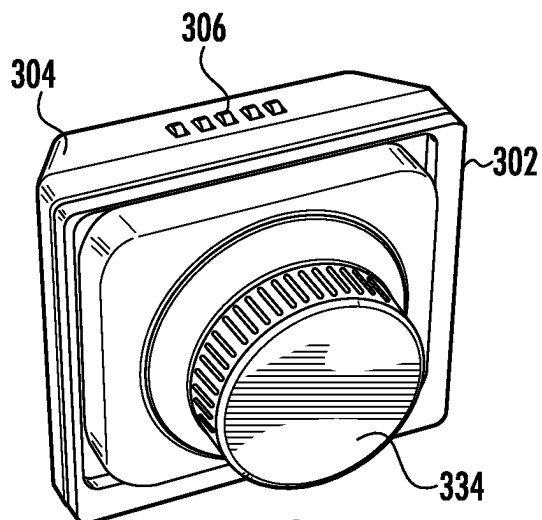


FIG. 10E

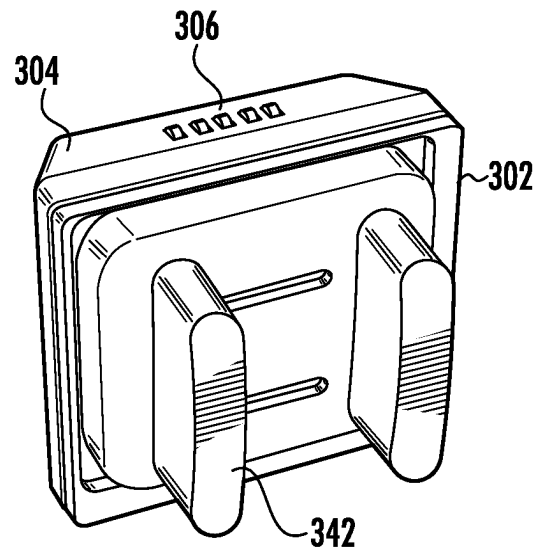


FIG. 10F

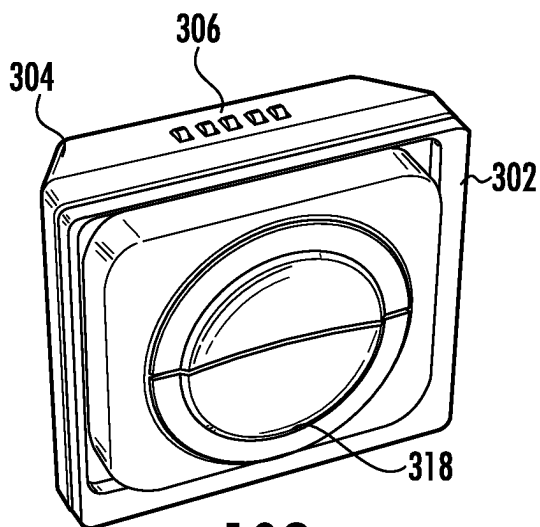


FIG. 10G

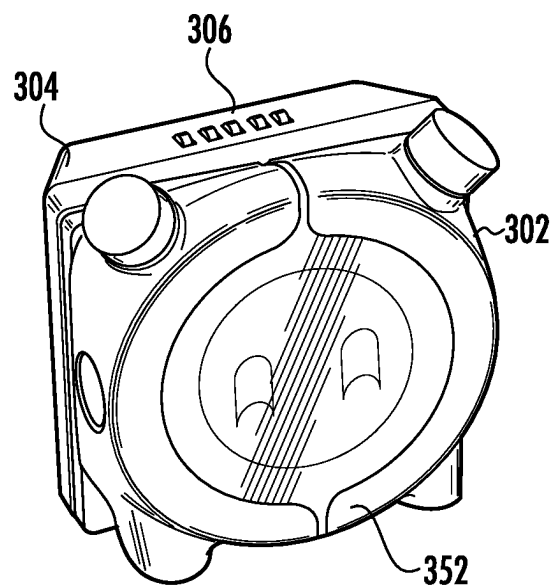


FIG. 10H

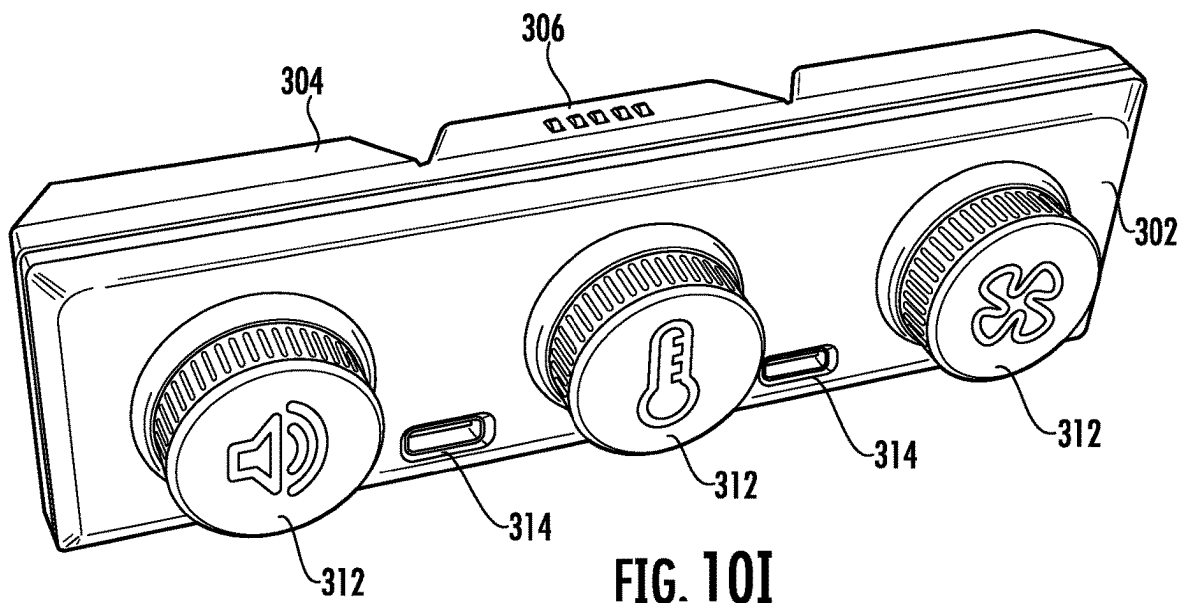


FIG. 10I

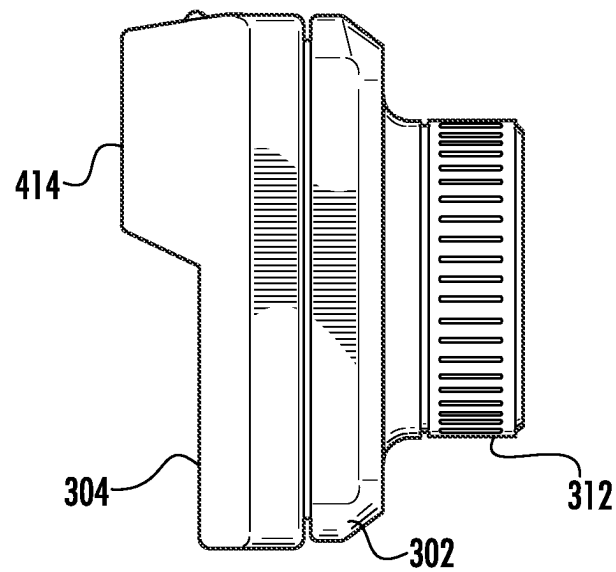


FIG. 11

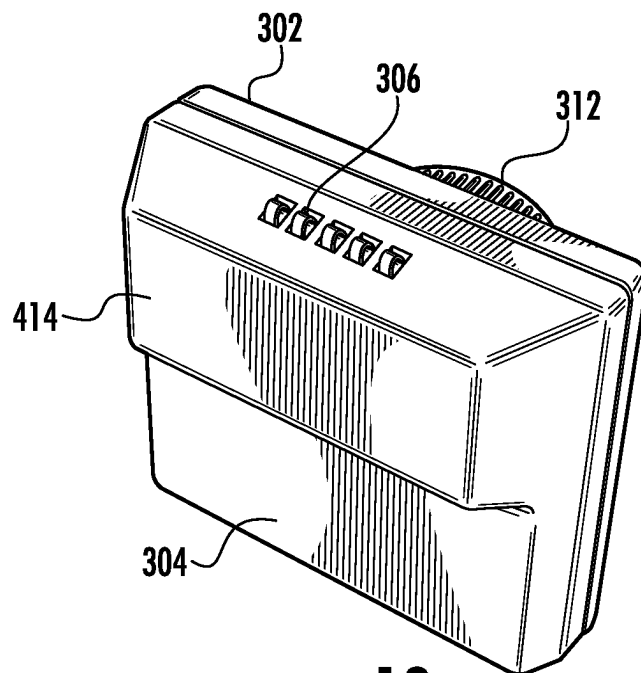


FIG. 12

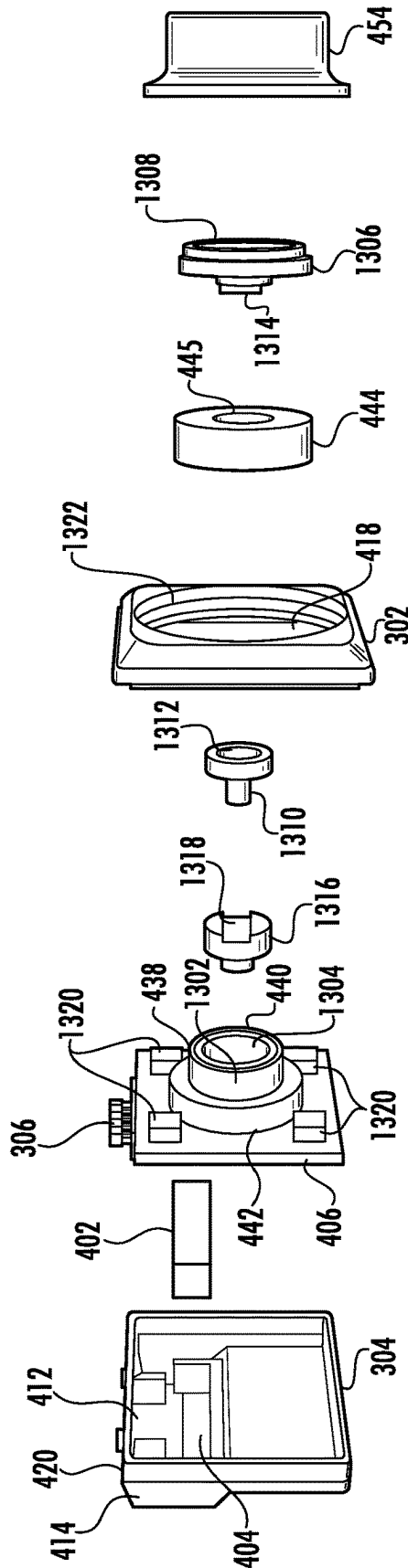


FIG. 13

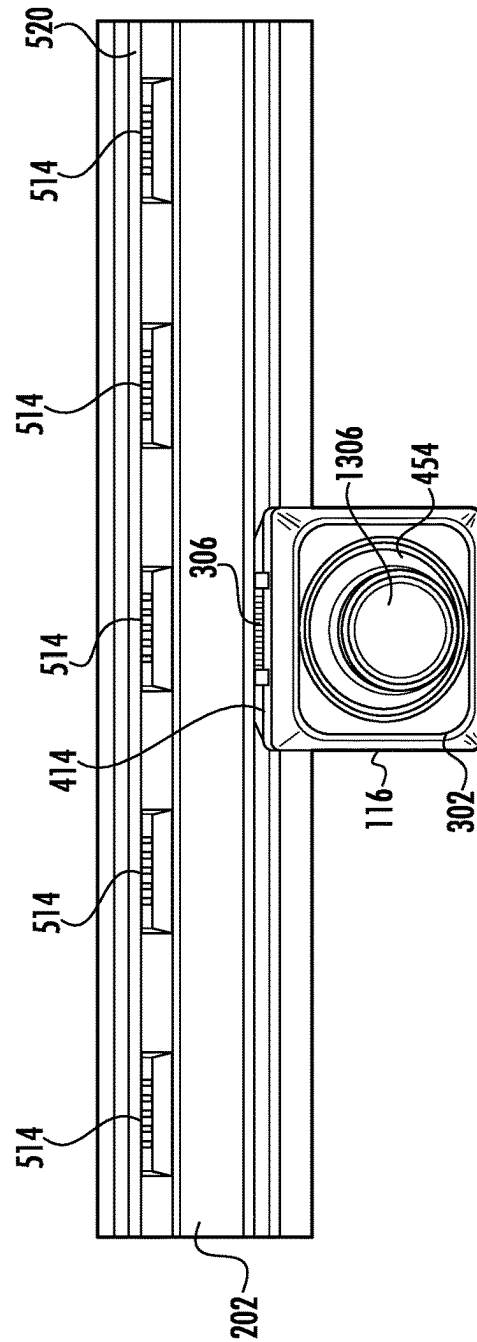


FIG. 14

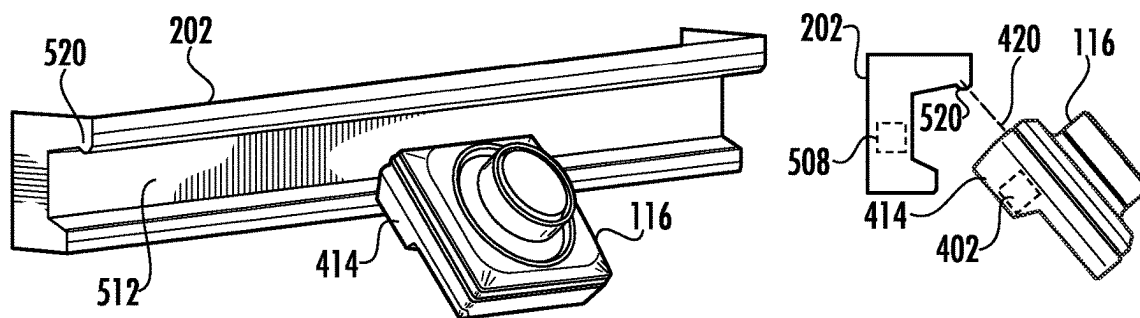


FIG. 15A

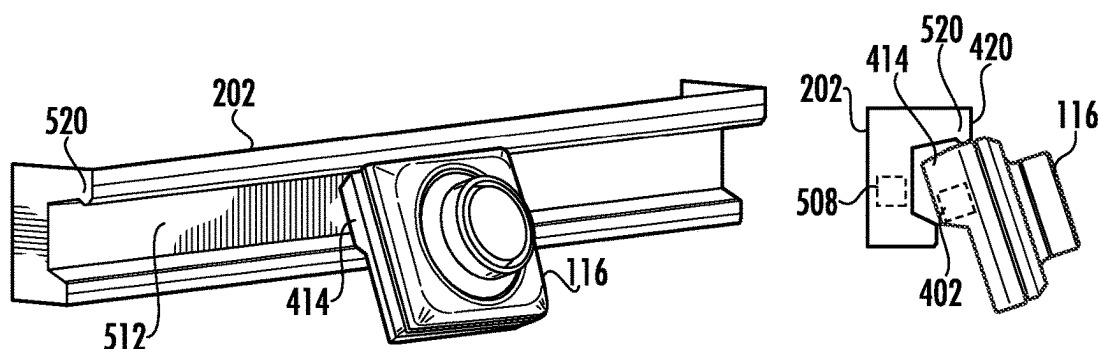


FIG. 15B

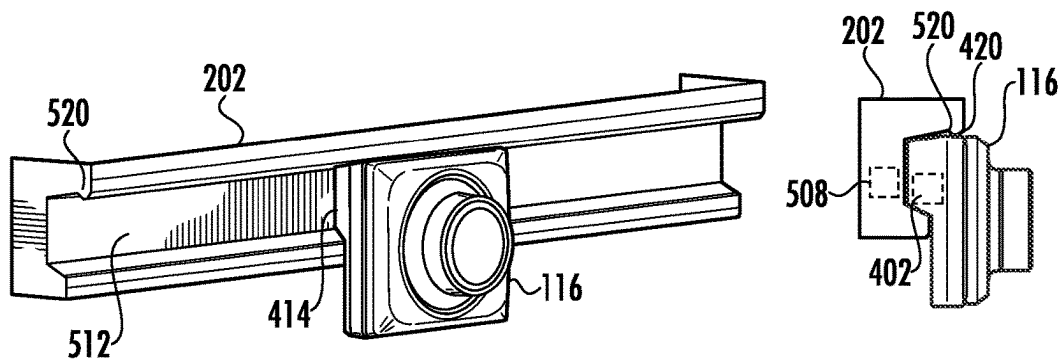
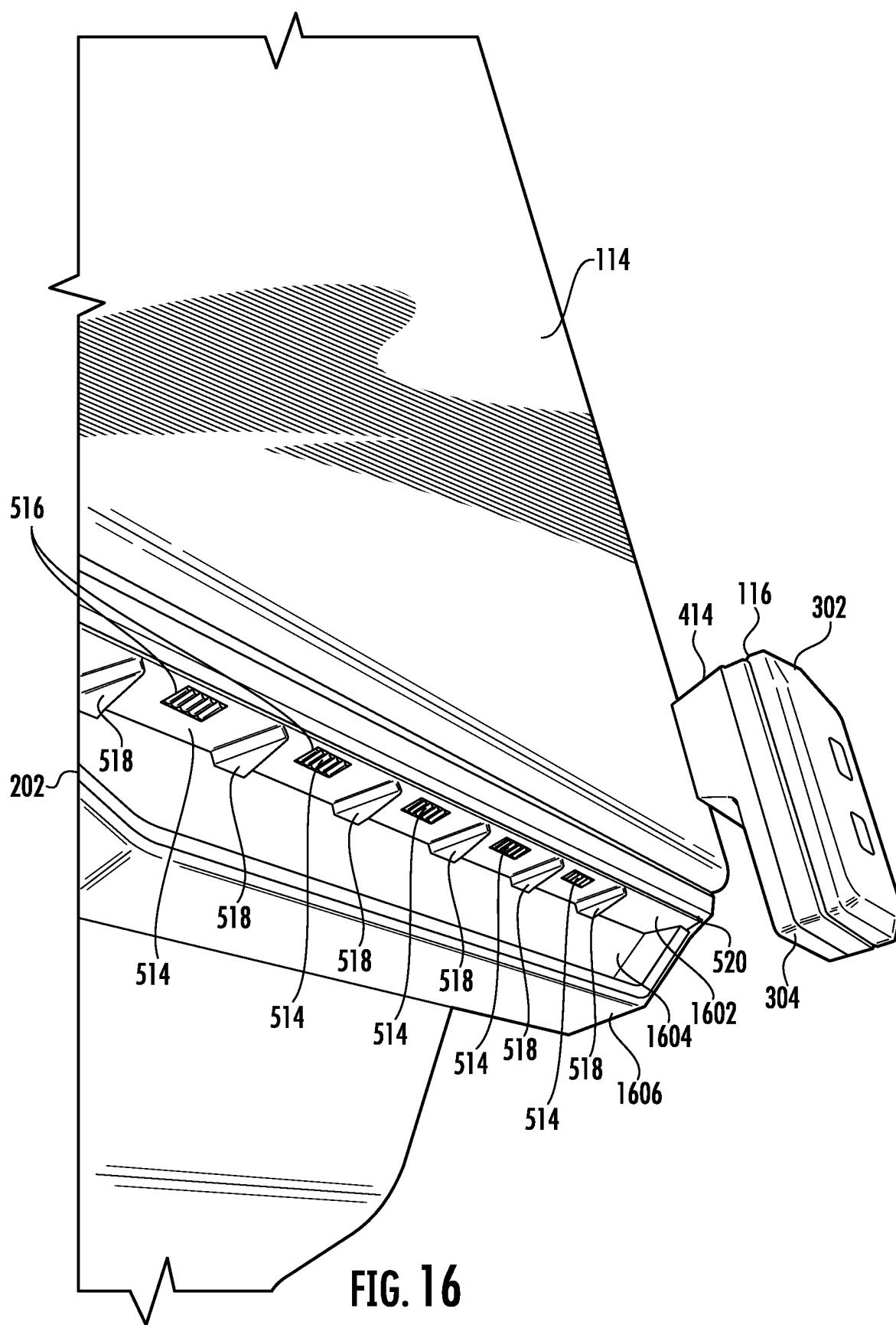


FIG. 15C



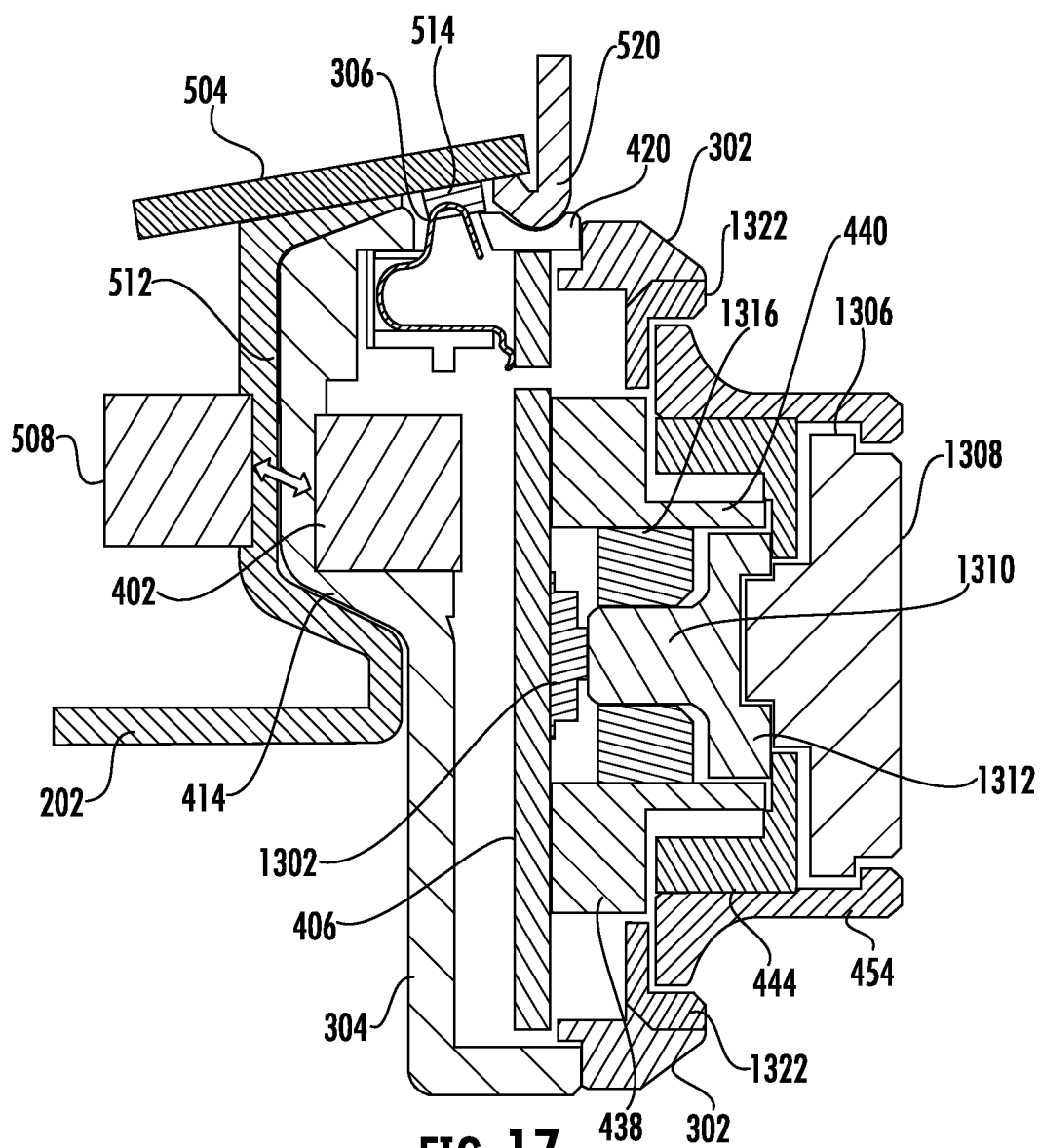
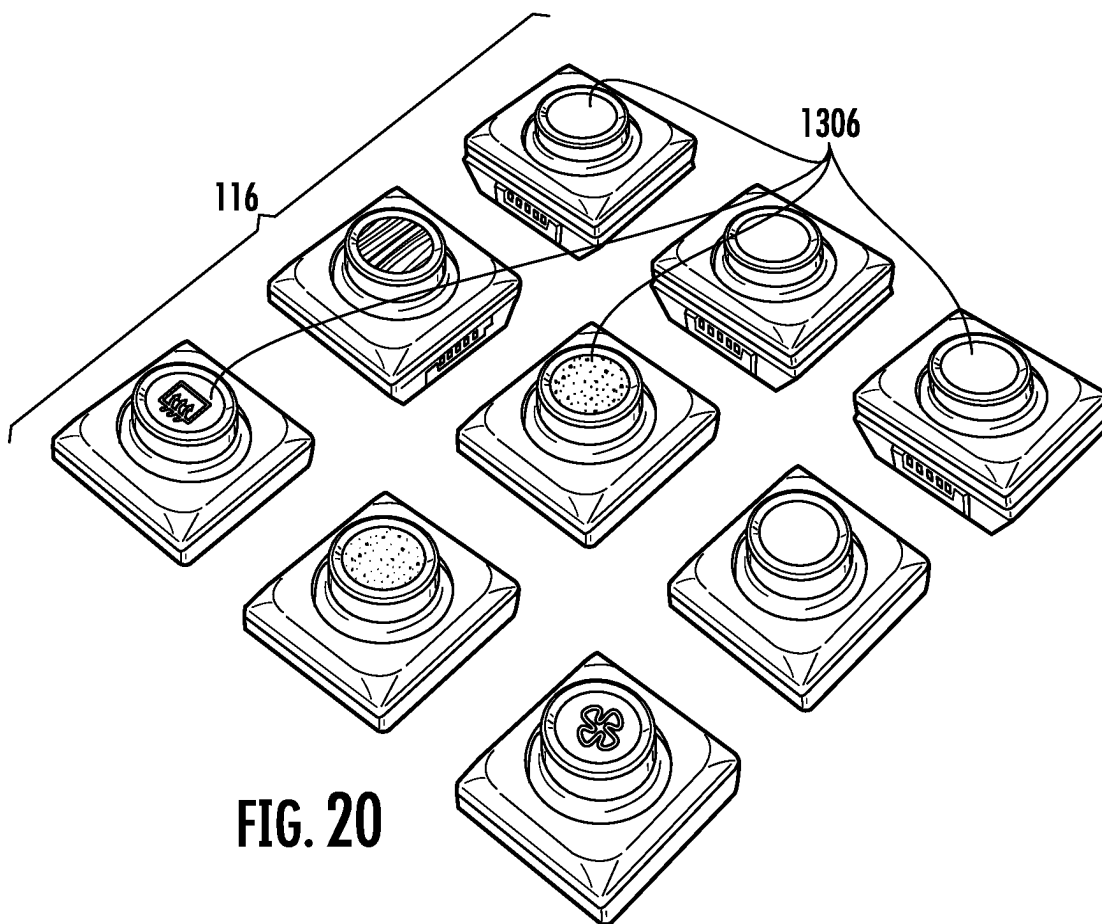
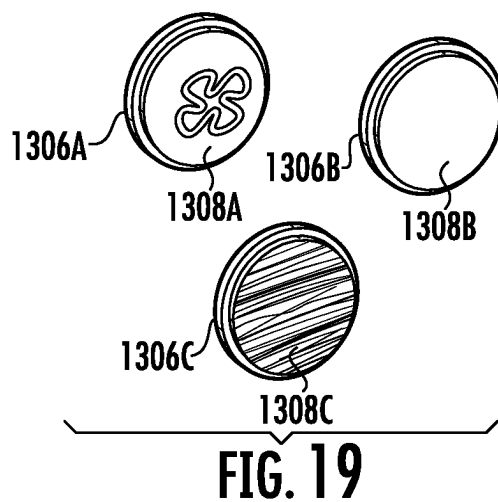
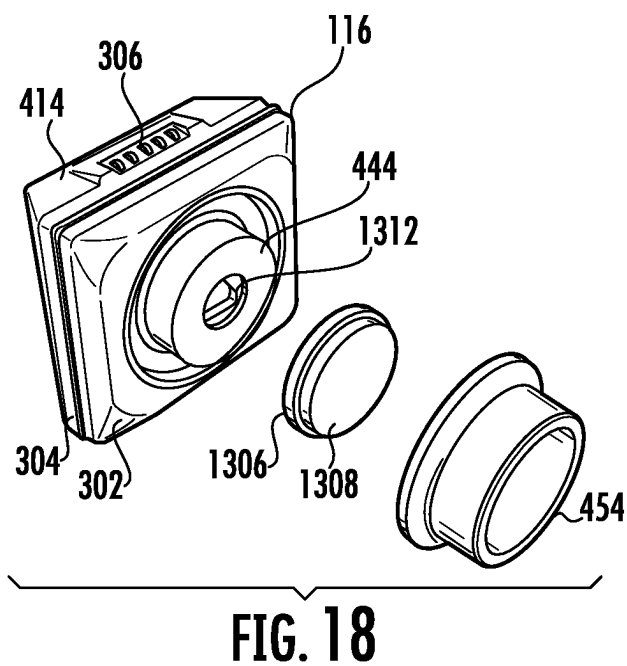
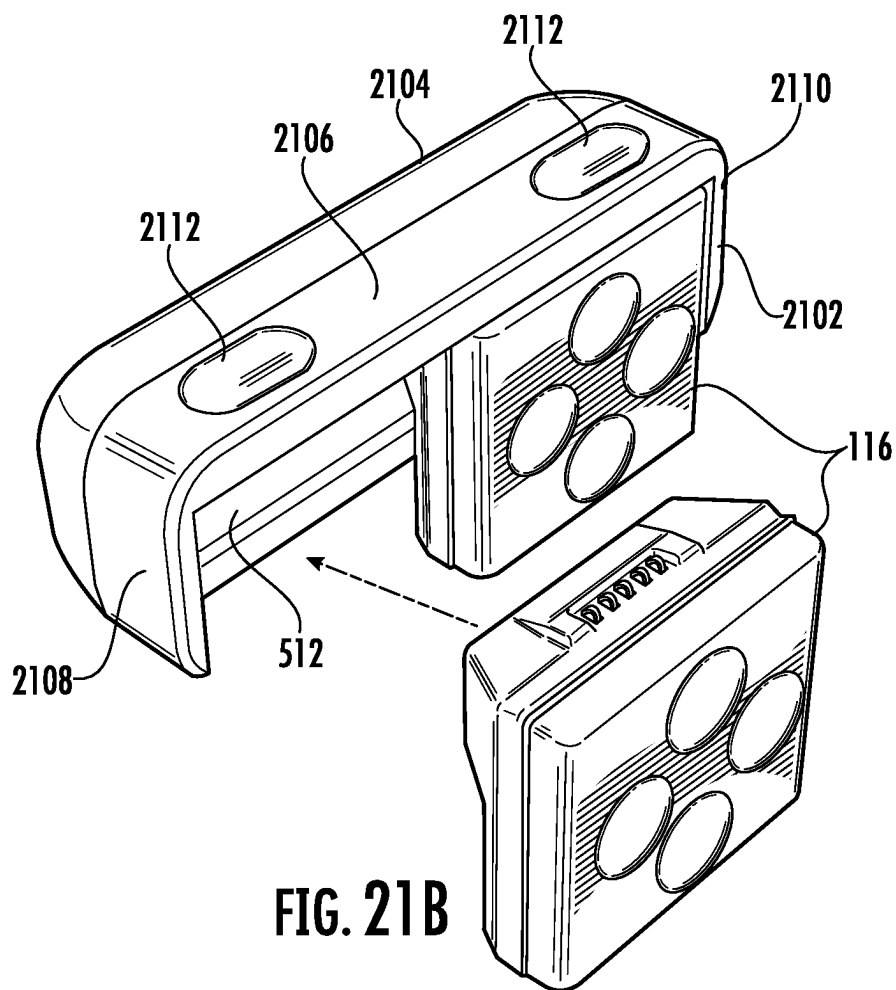
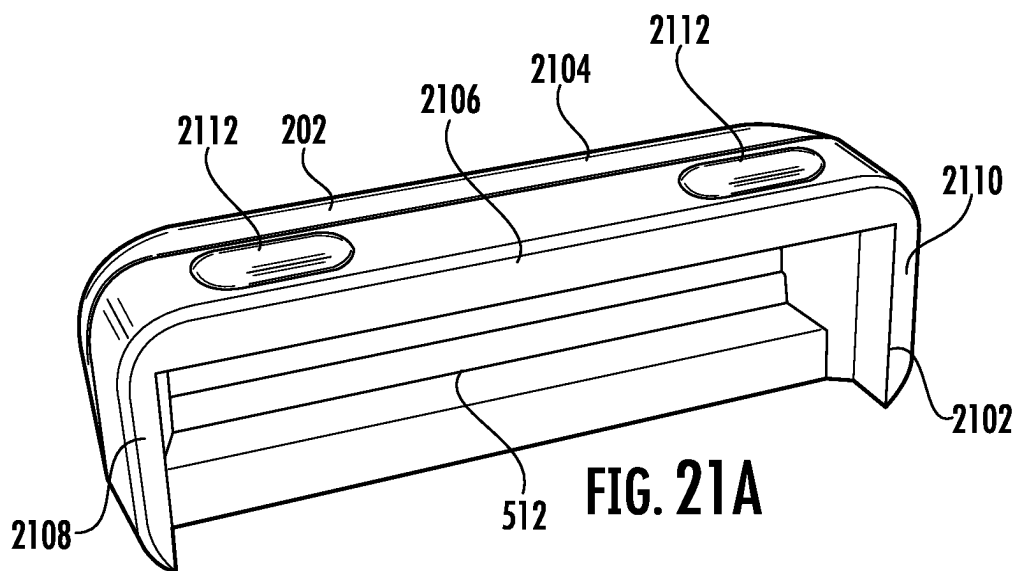
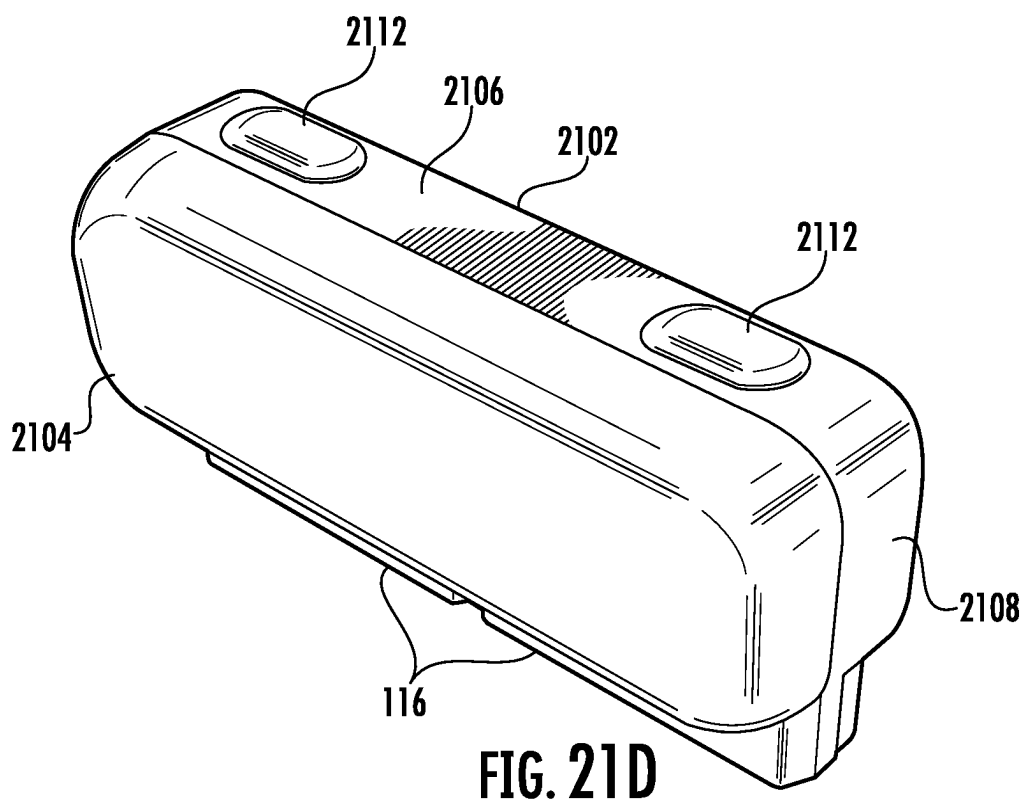
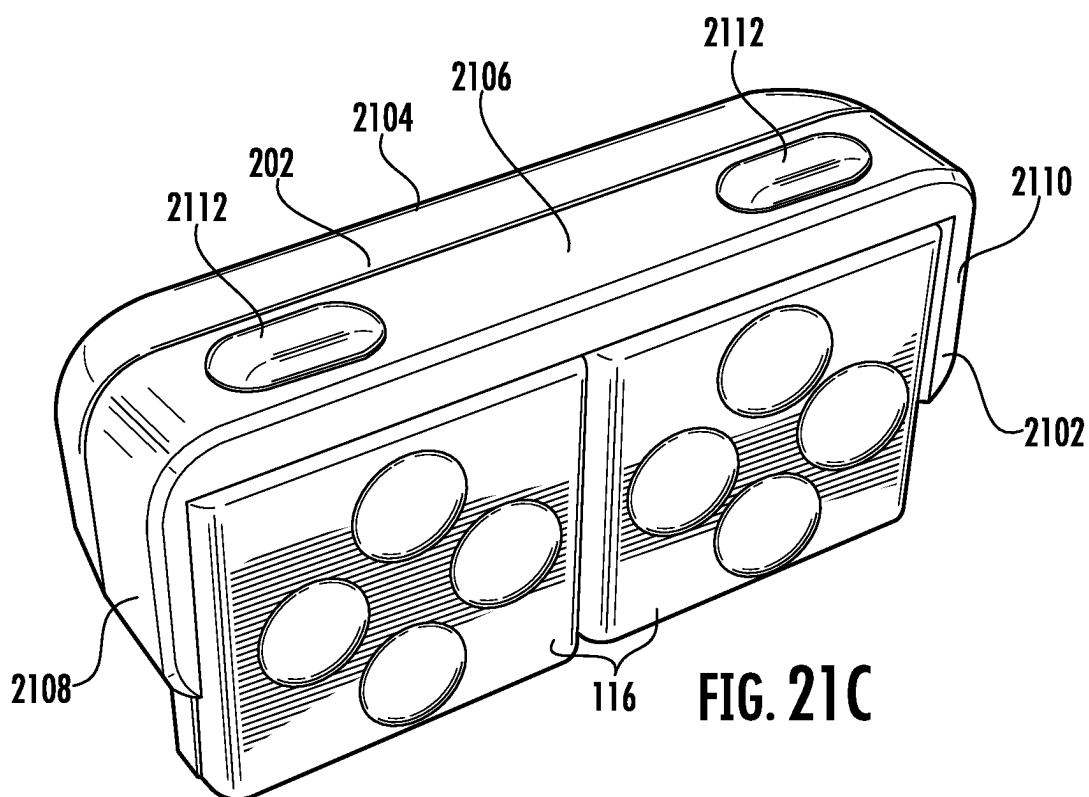


FIG. 17







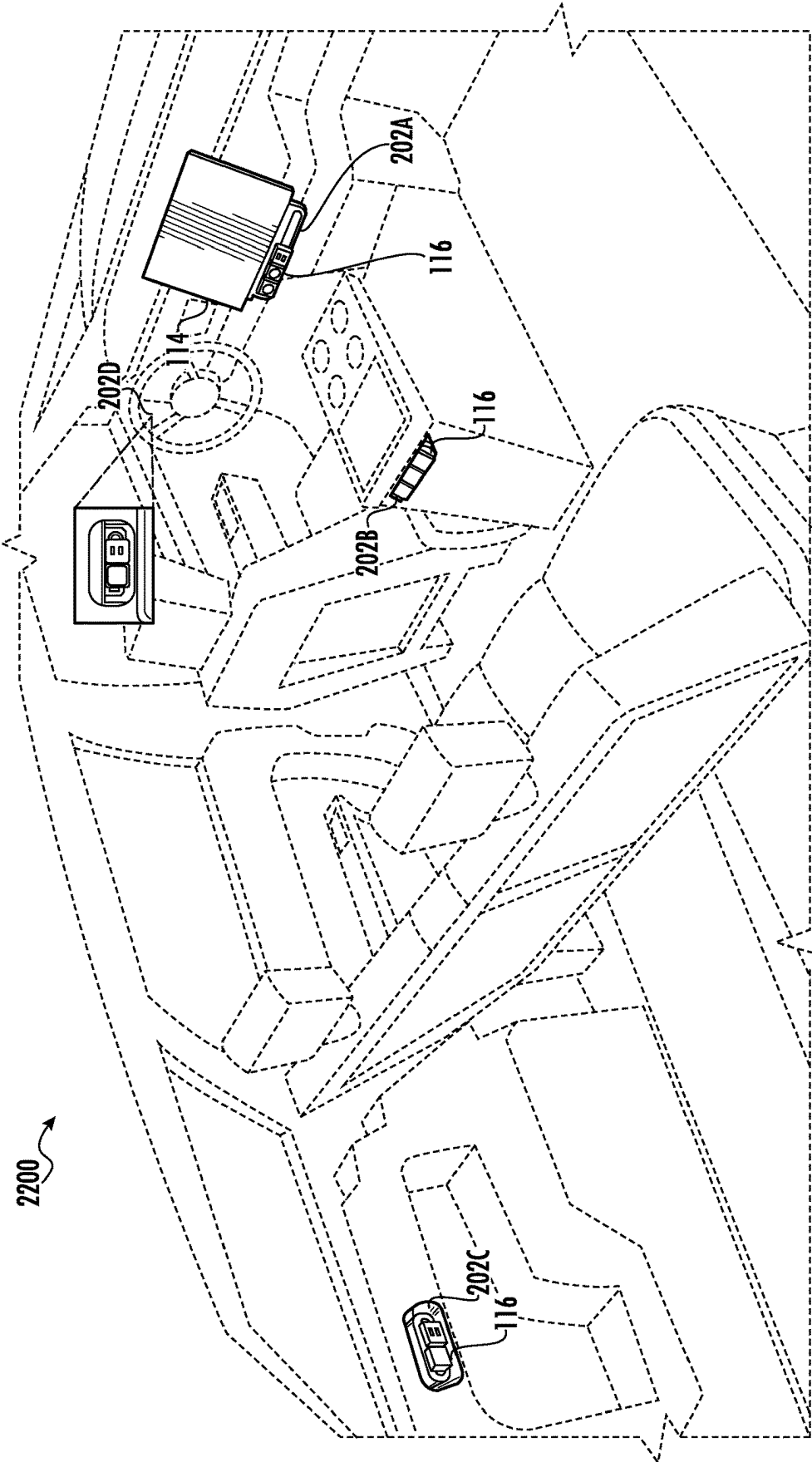


FIG. 22

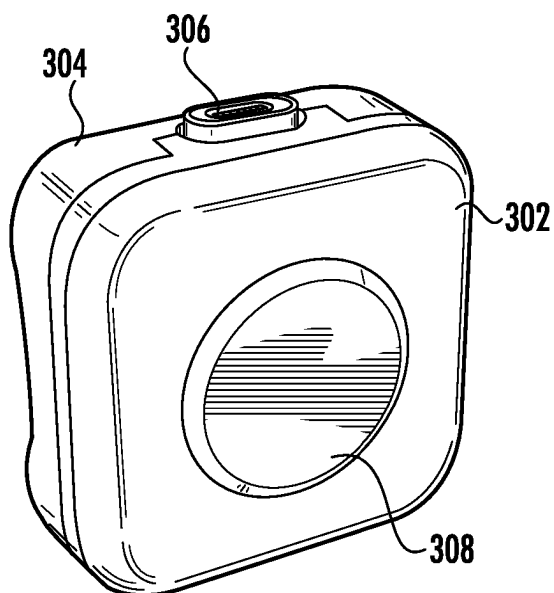


FIG. 23A

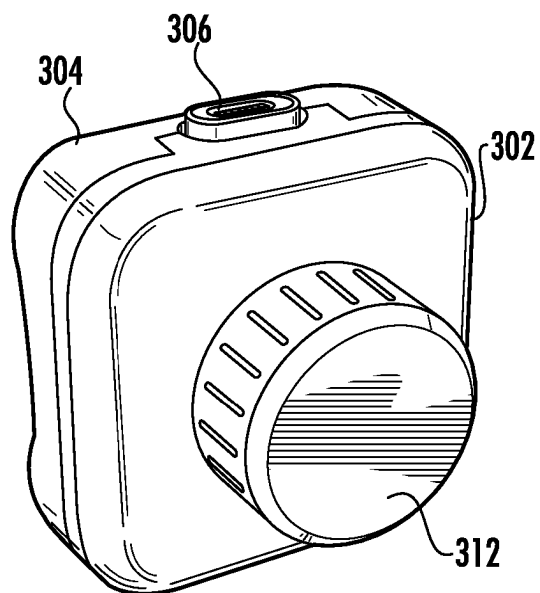


FIG. 23B

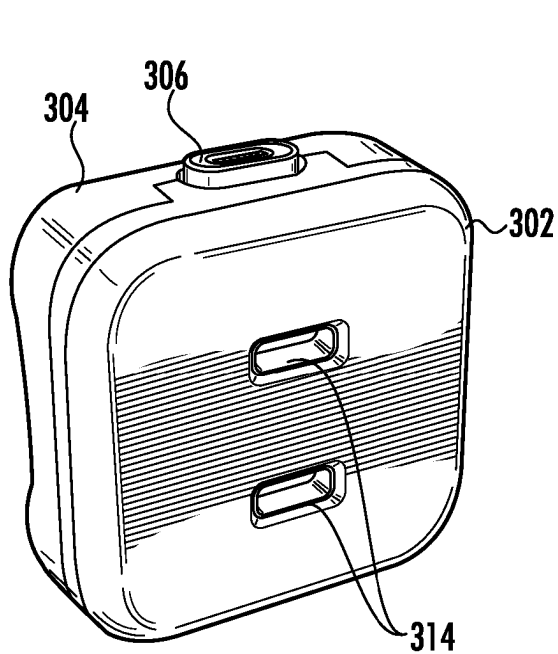


FIG. 23C

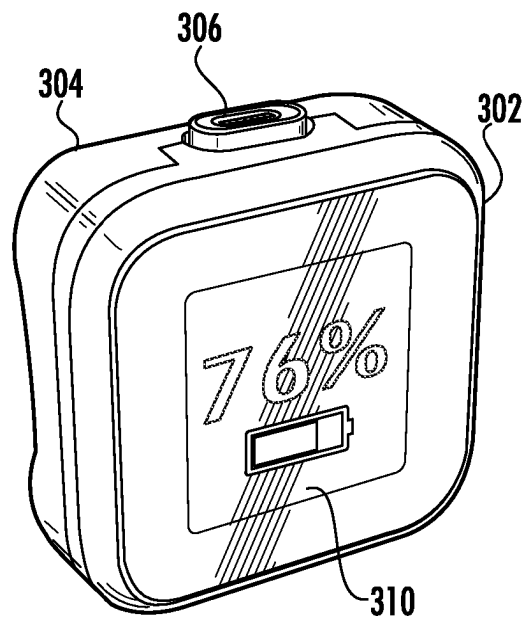


FIG. 23D

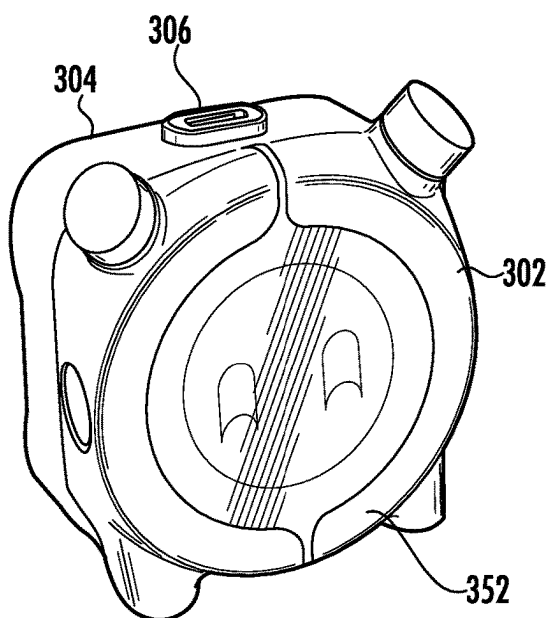


FIG. 23E

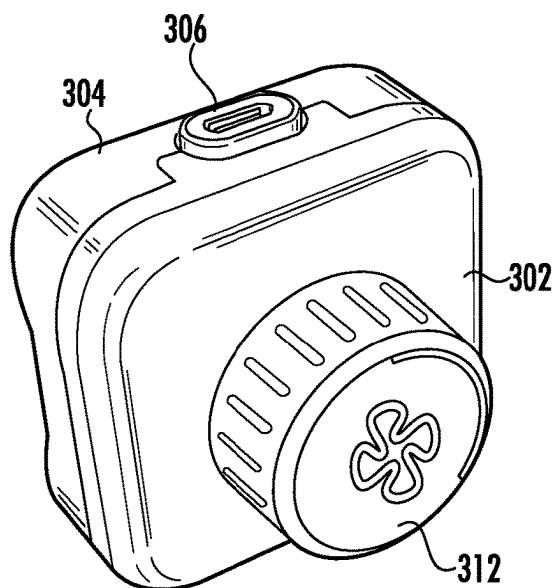


FIG. 23F

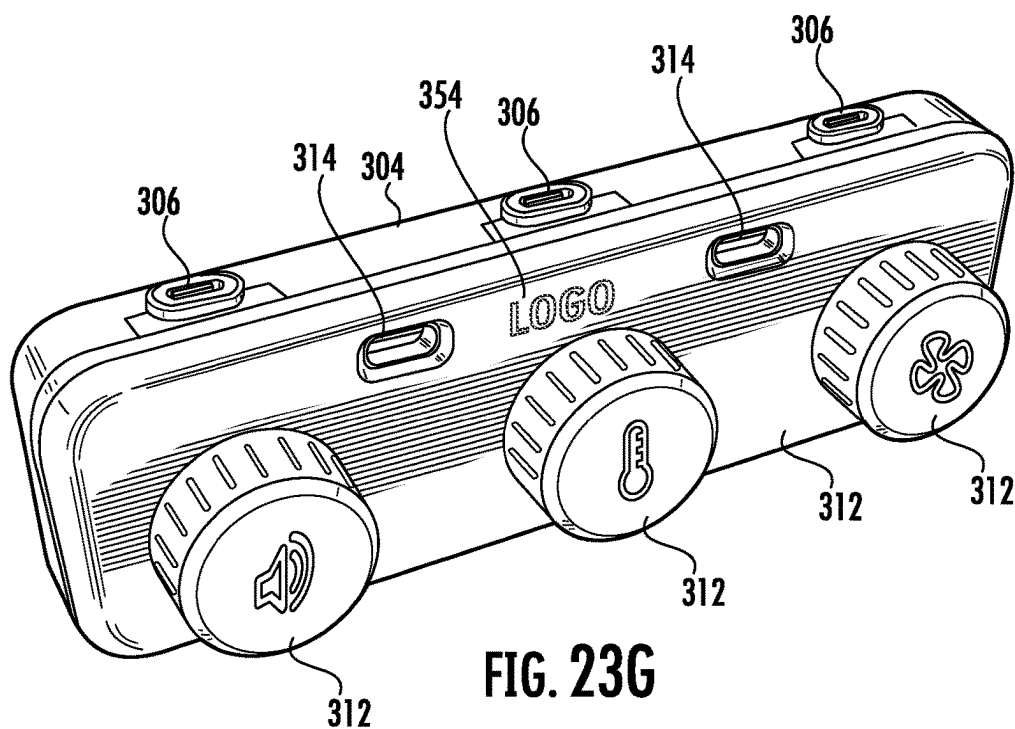


FIG. 23G

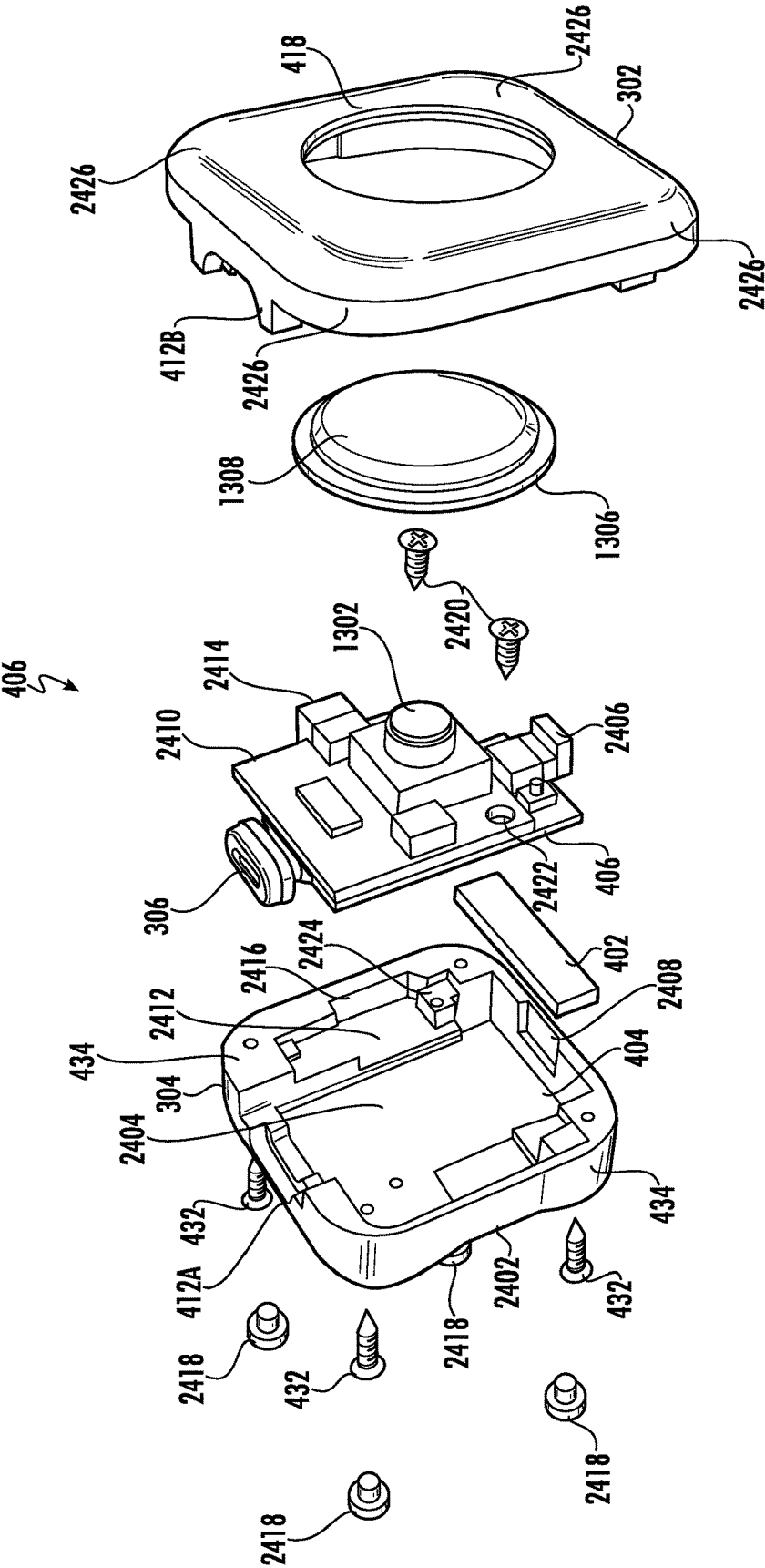
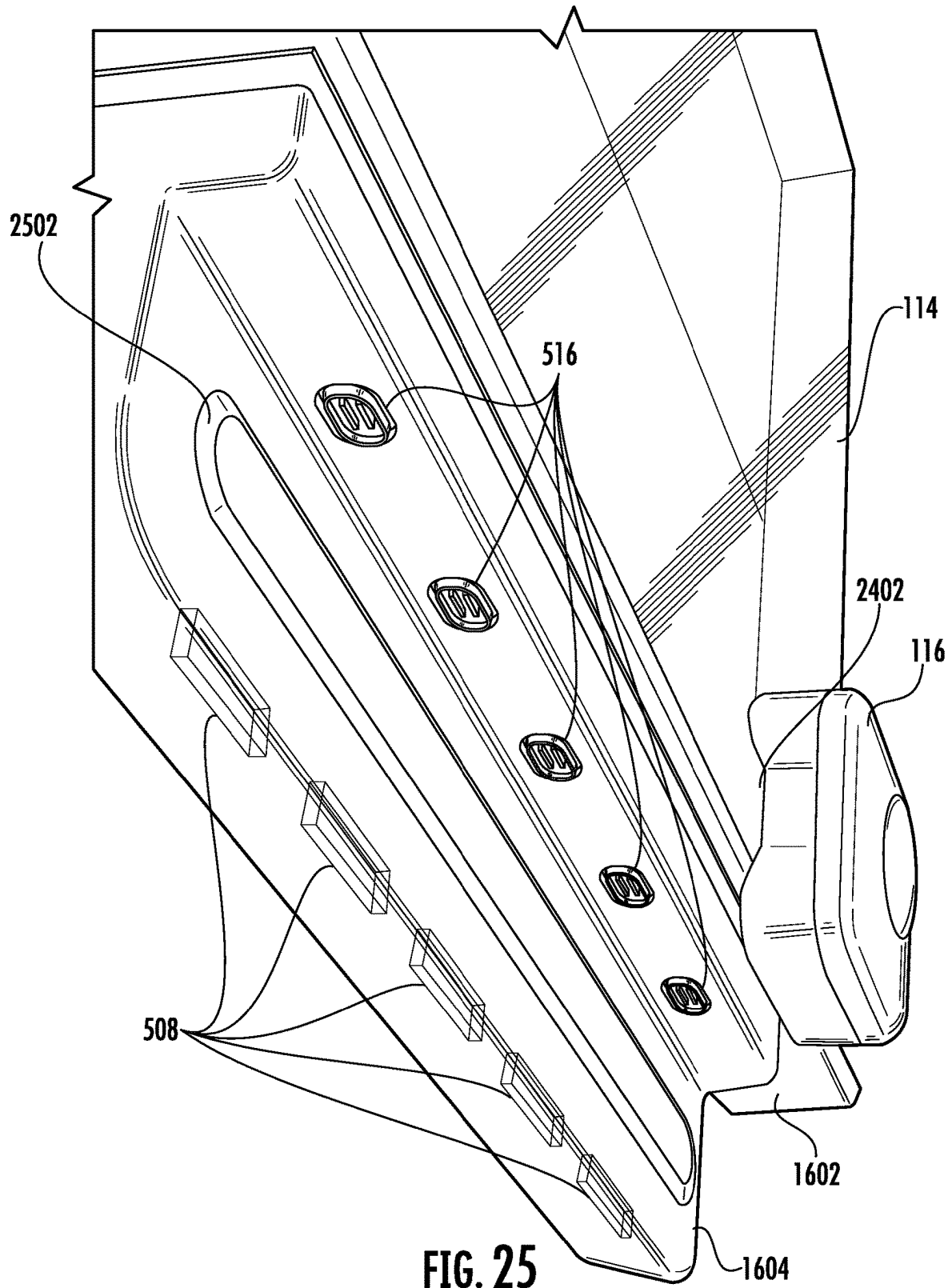


FIG. 24



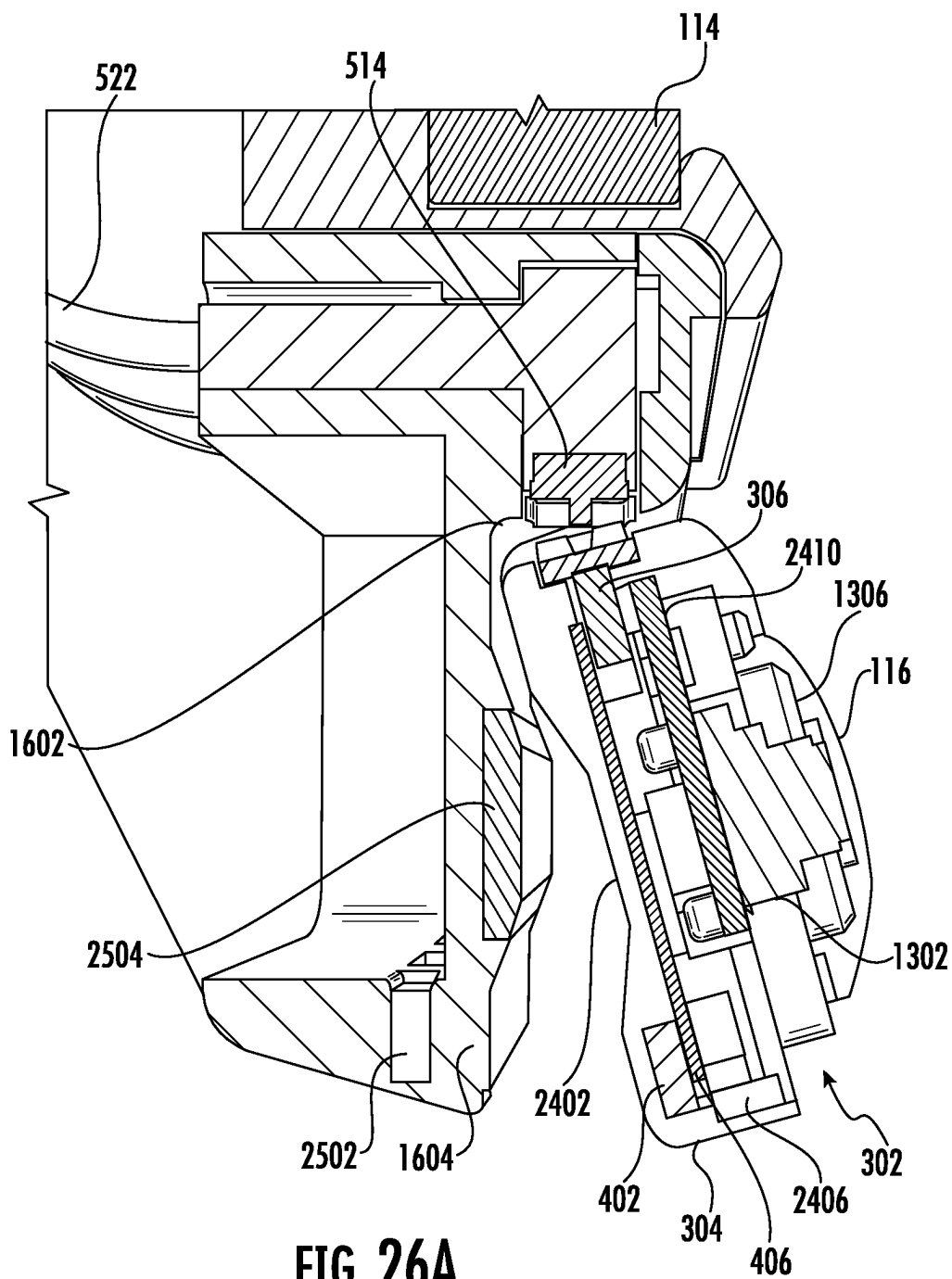
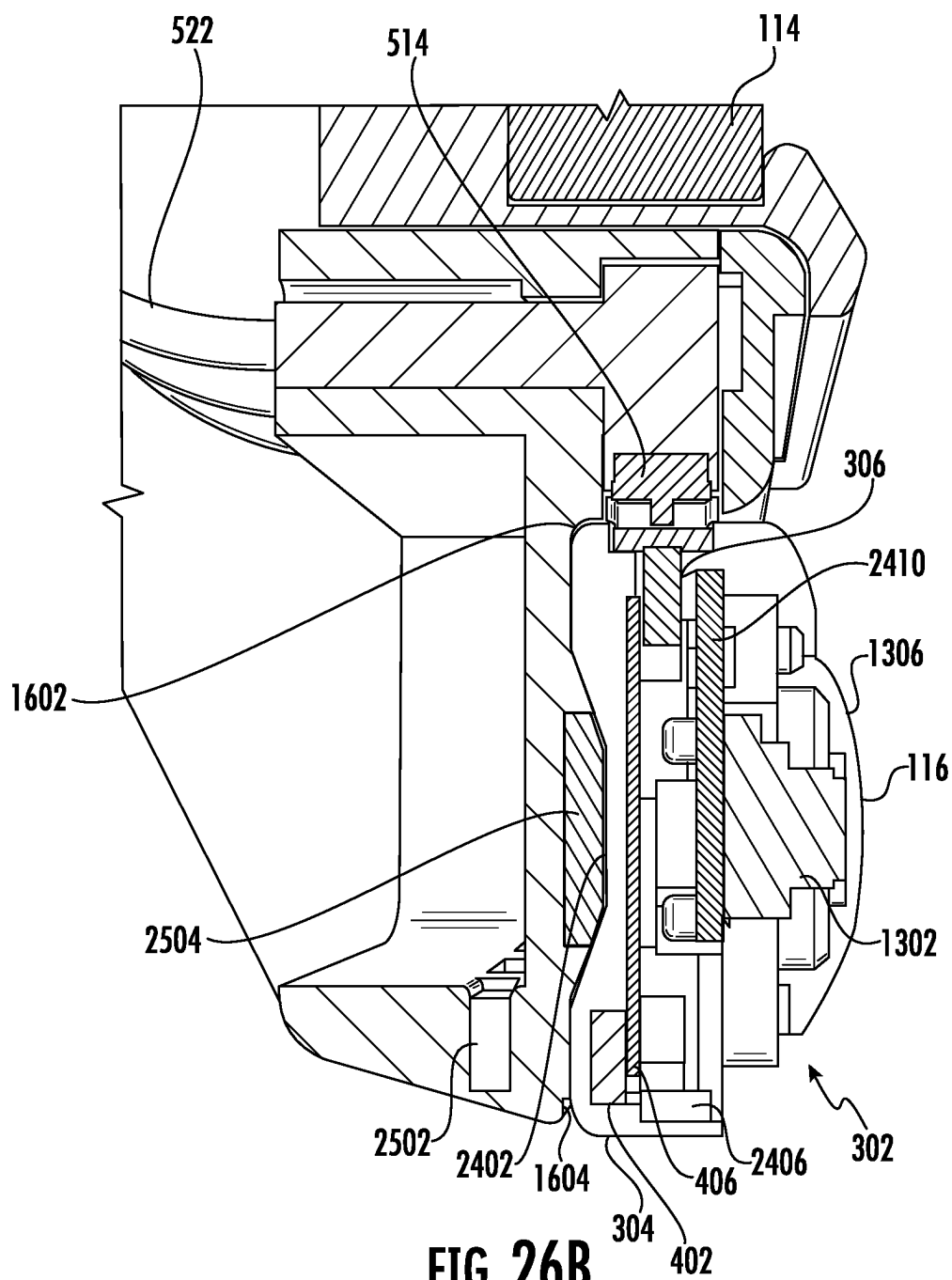


FIG. 26A



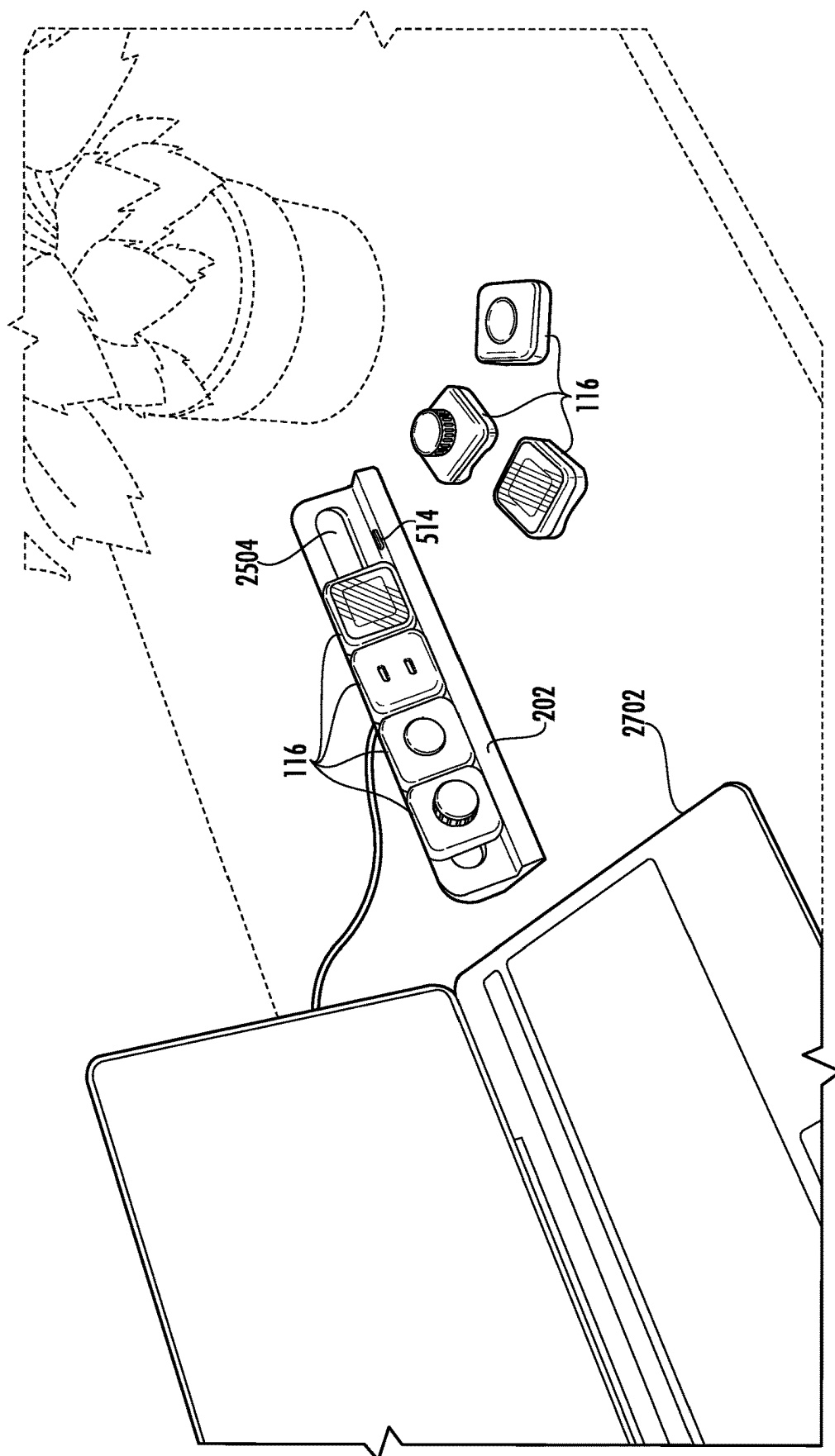
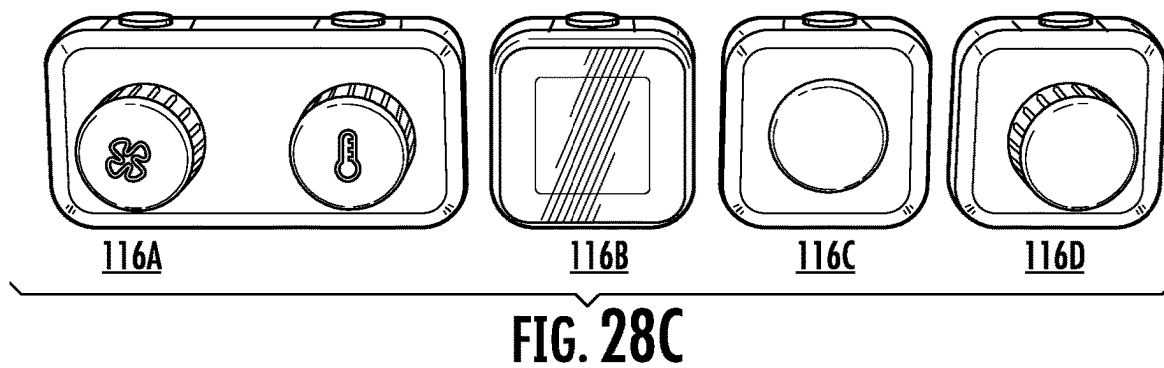
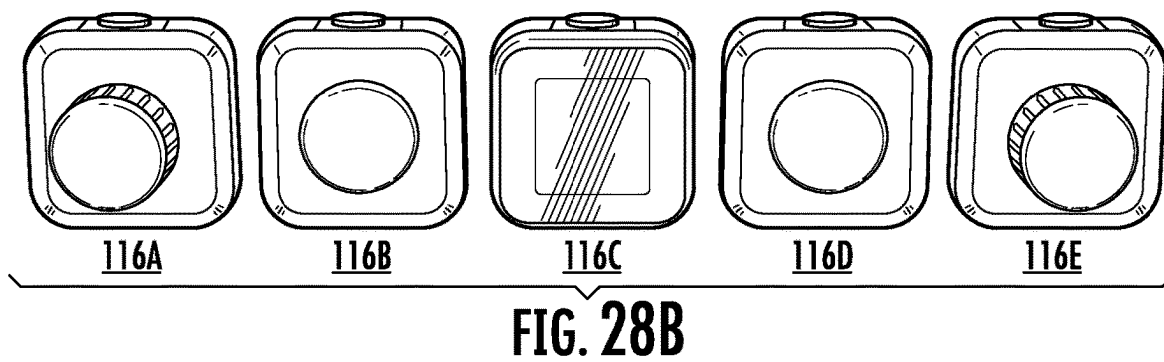
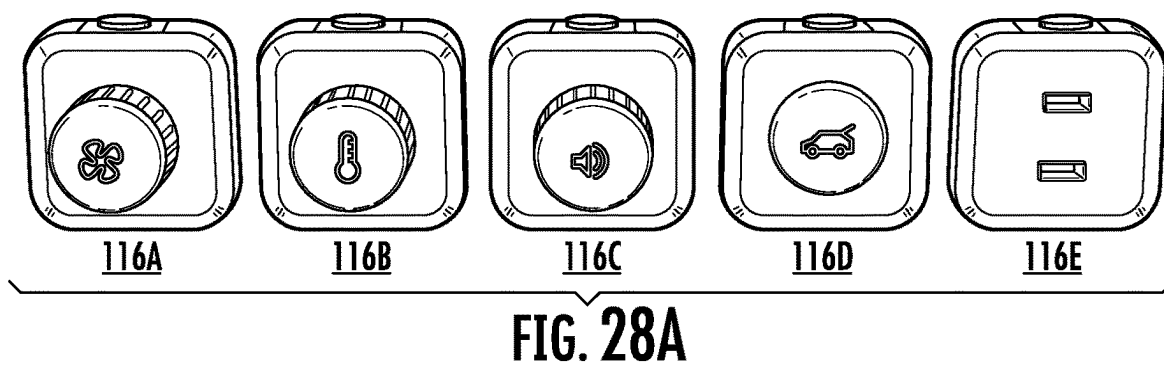
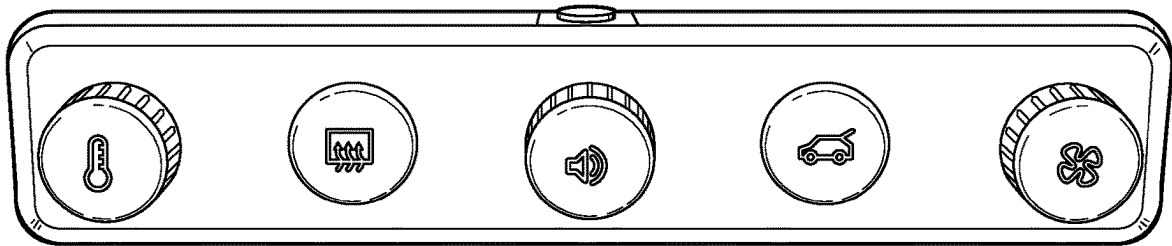
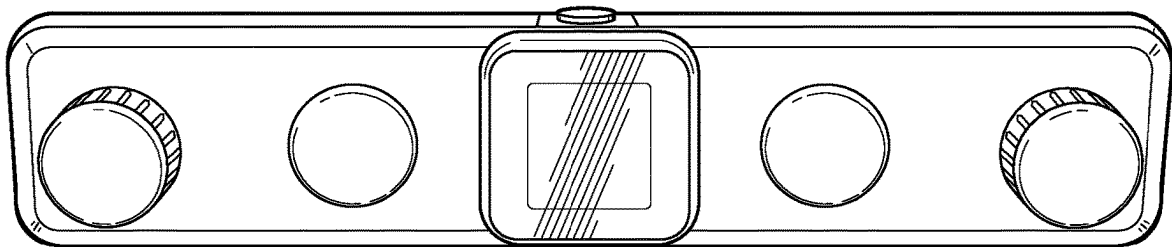


FIG. 27

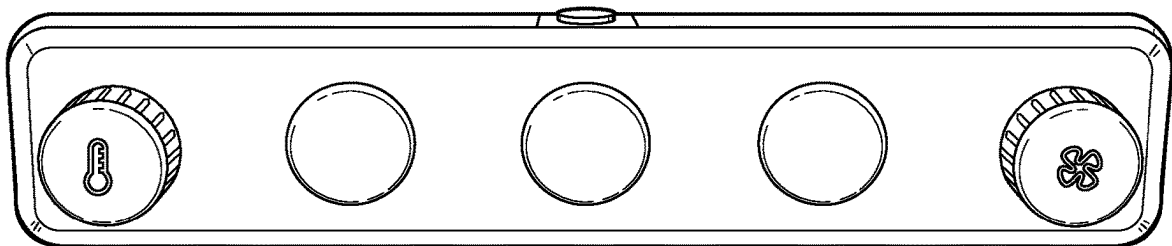




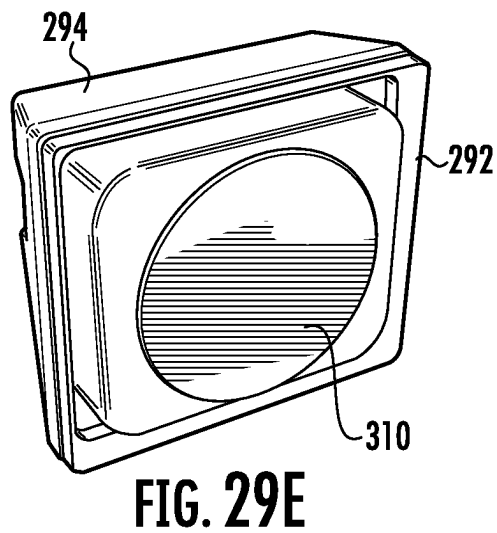
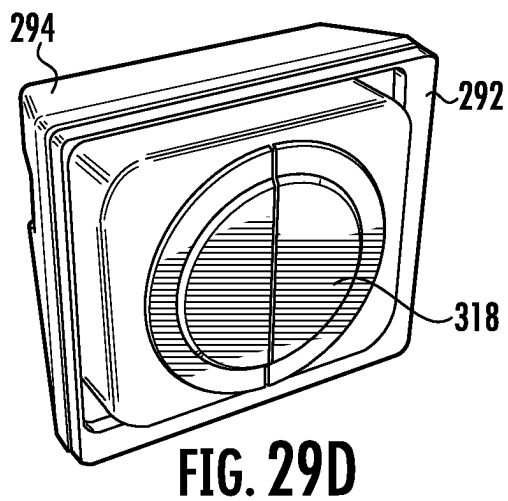
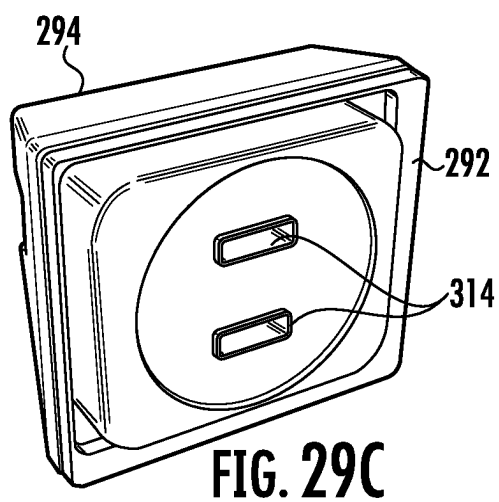
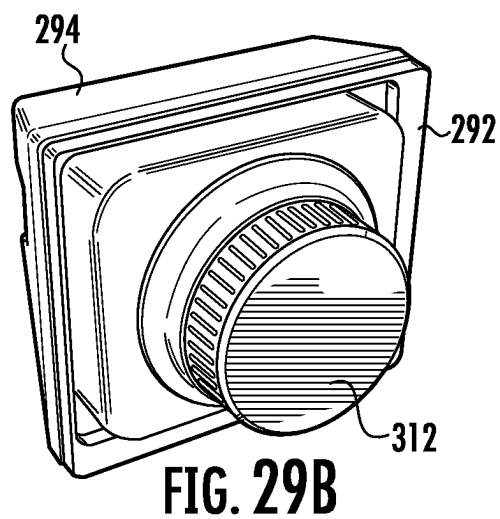
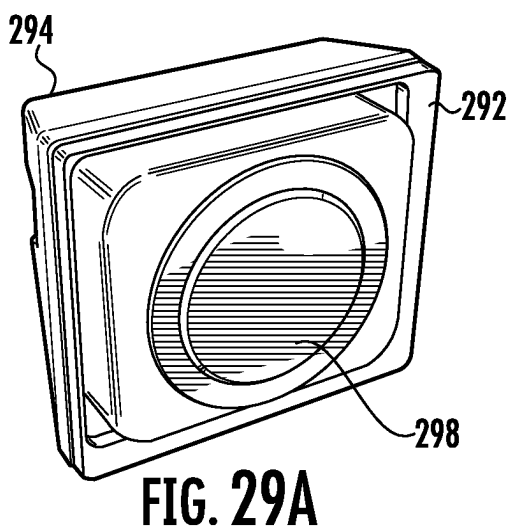
116
FIG. 28D

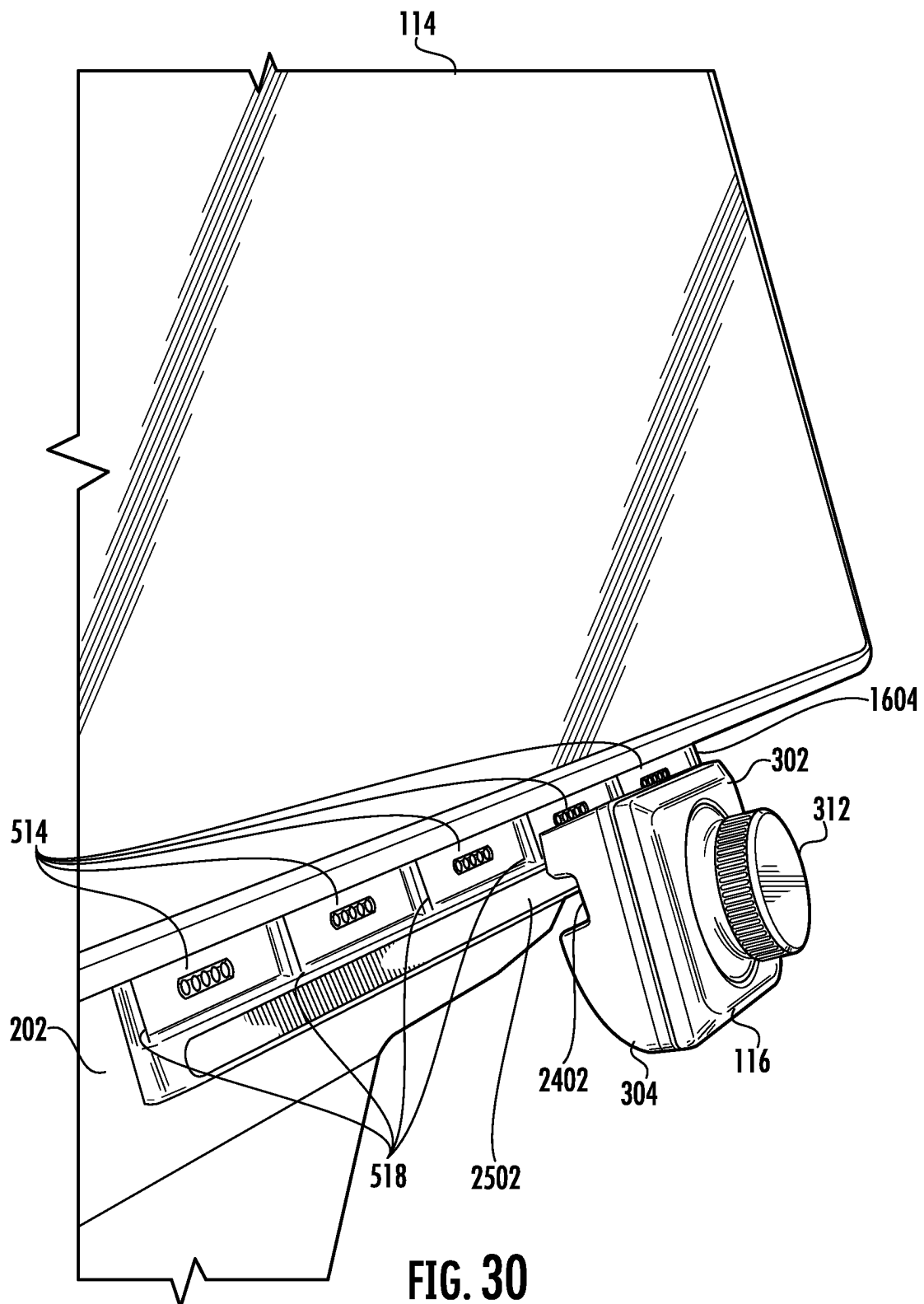


116
FIG. 28E



116
FIG. 28F





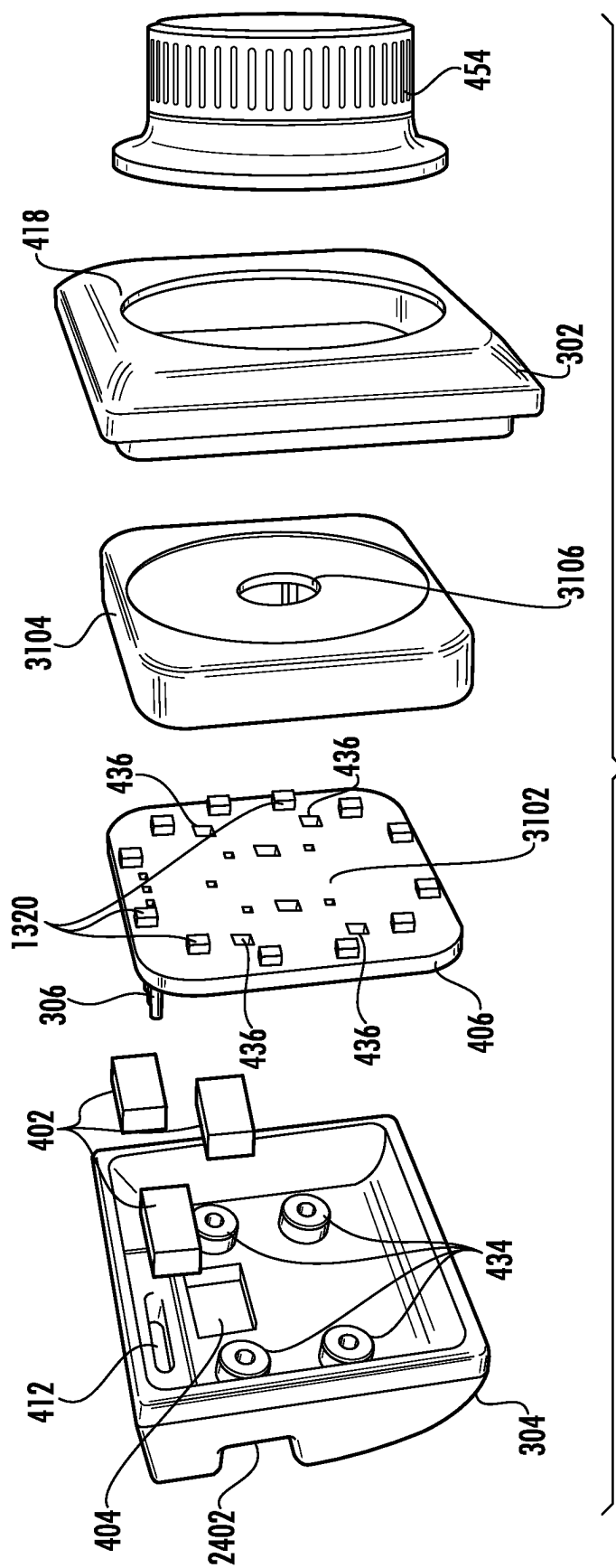
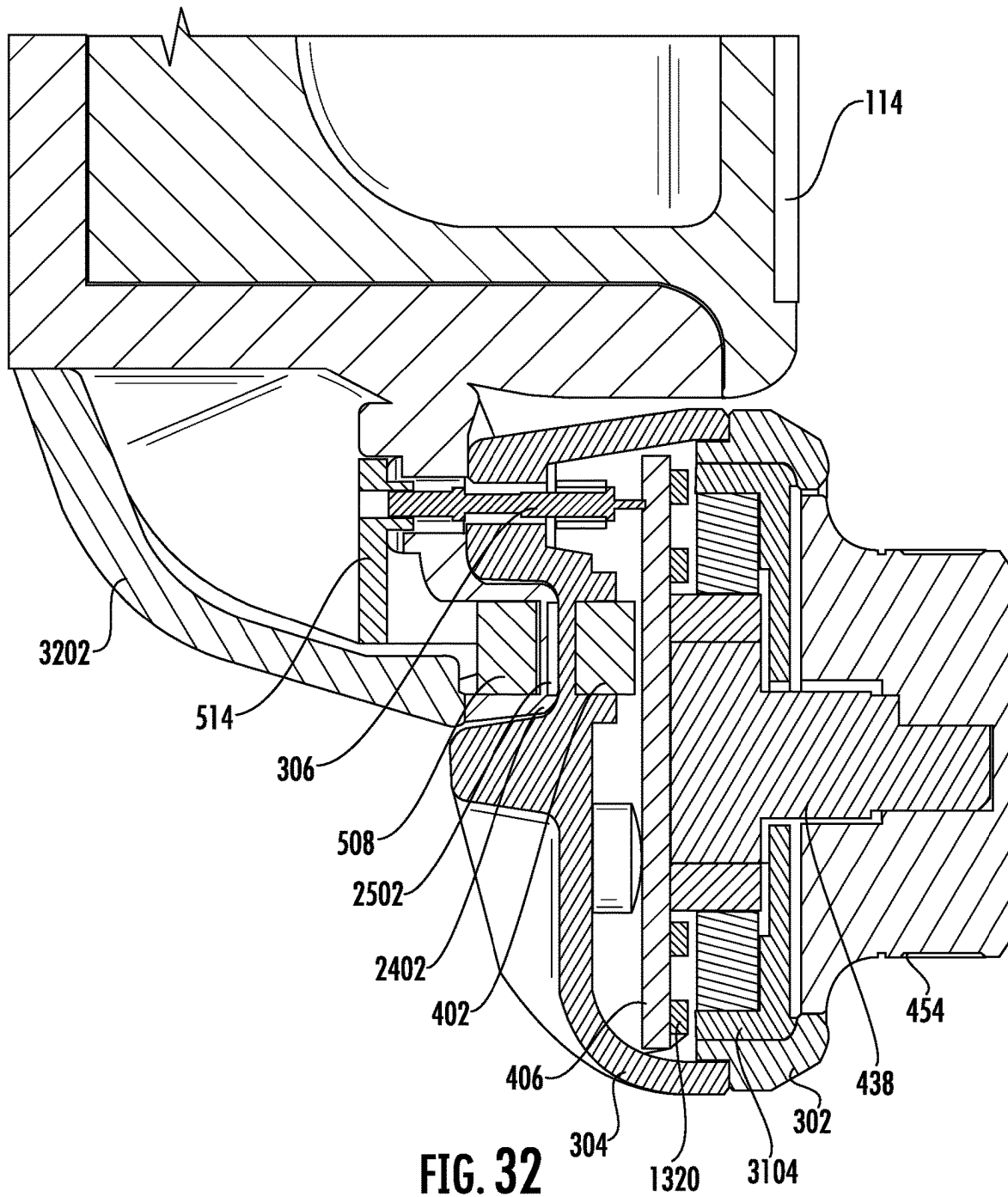


FIG. 31



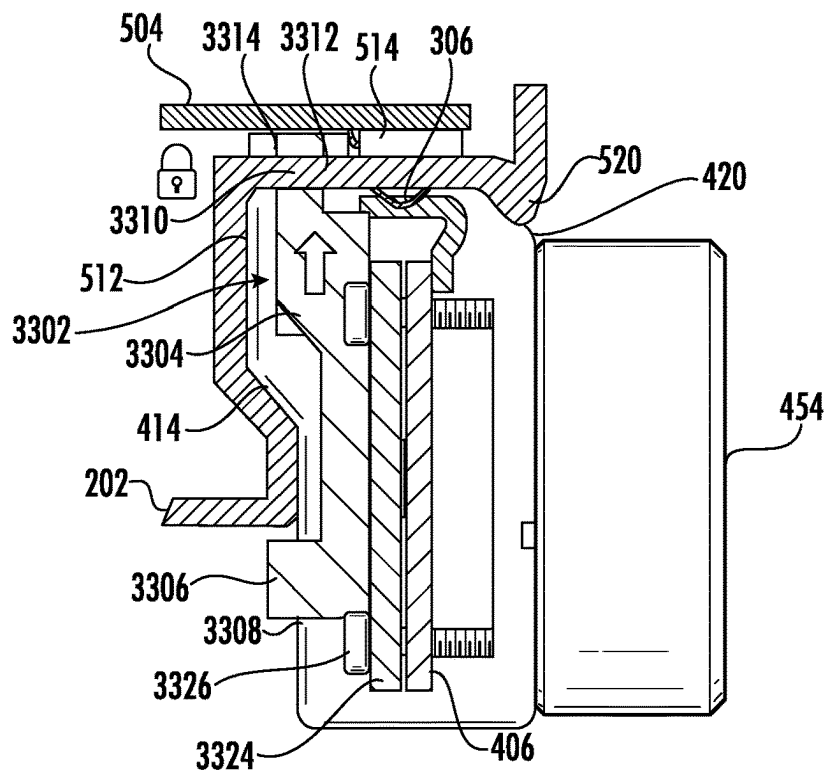


FIG. 33A

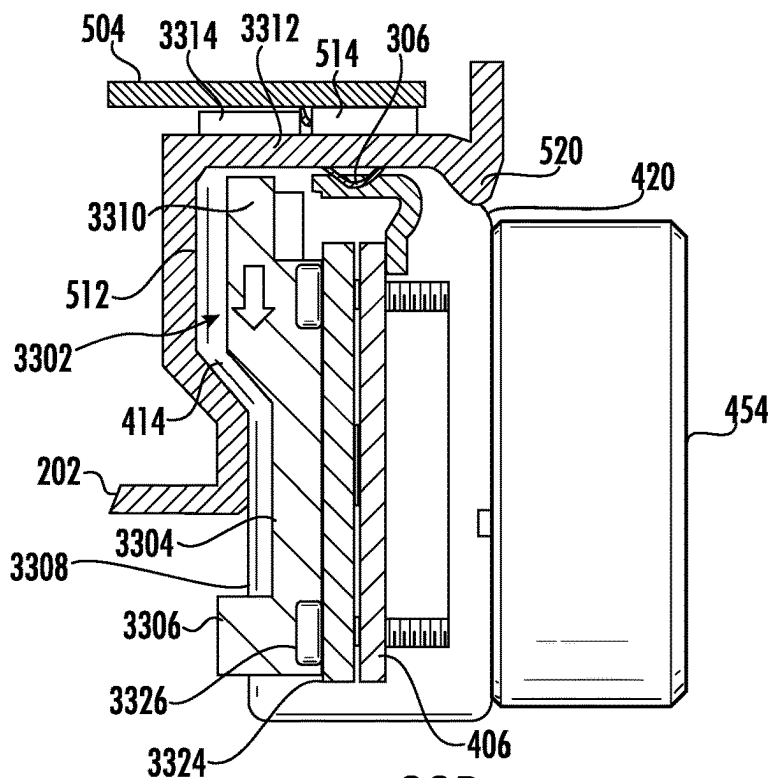
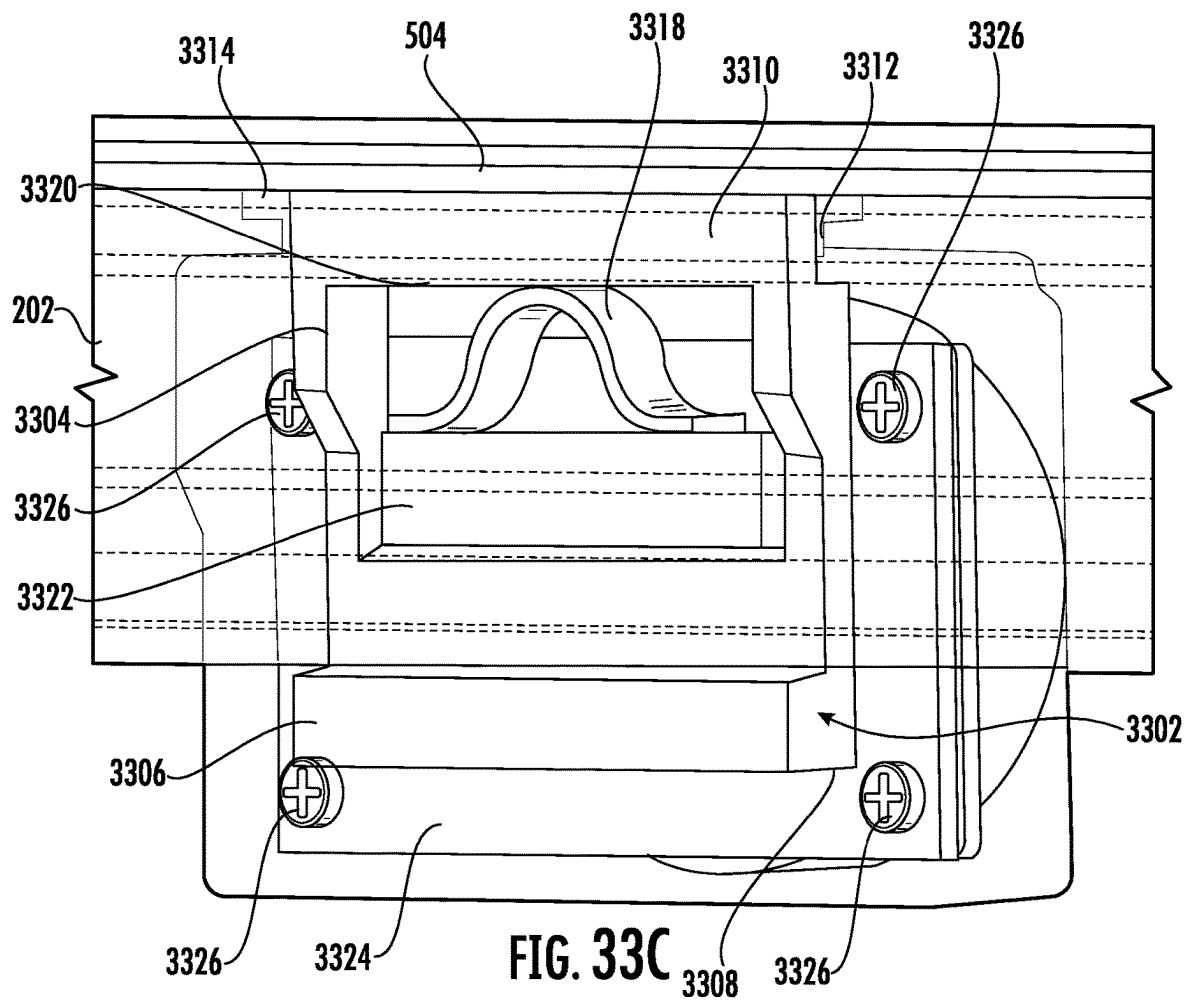


FIG. 33B



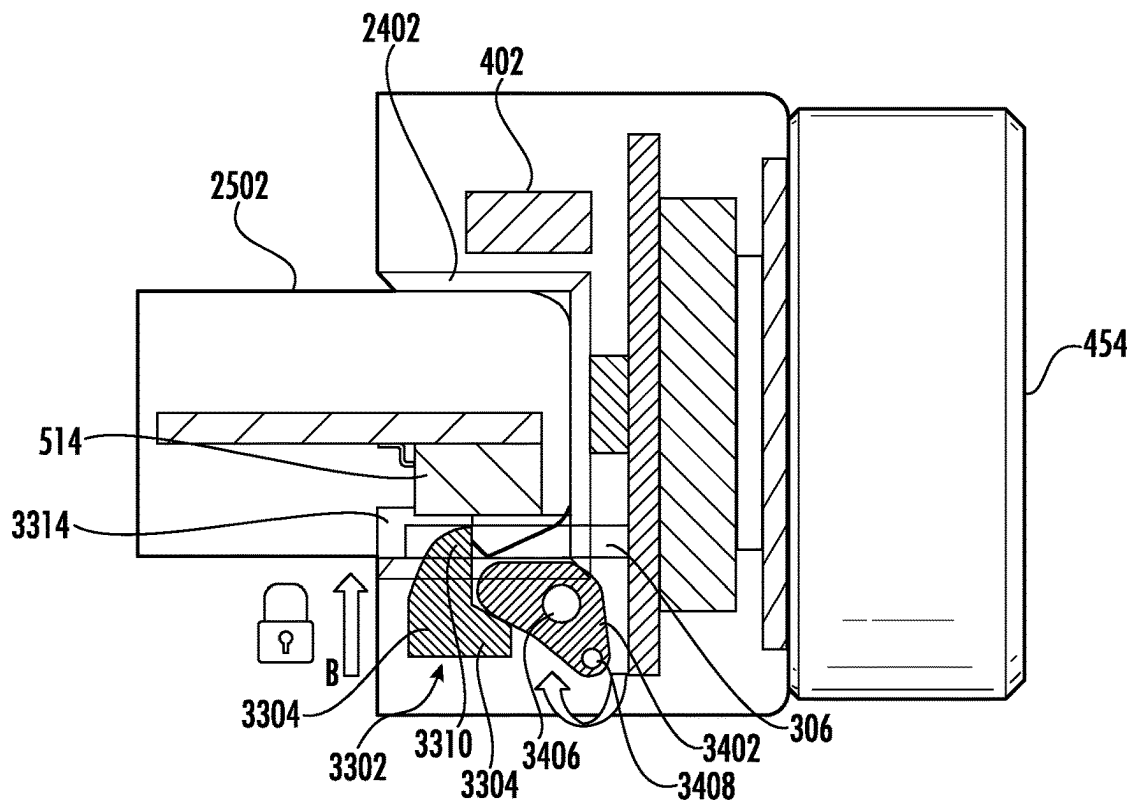


FIG. 34A

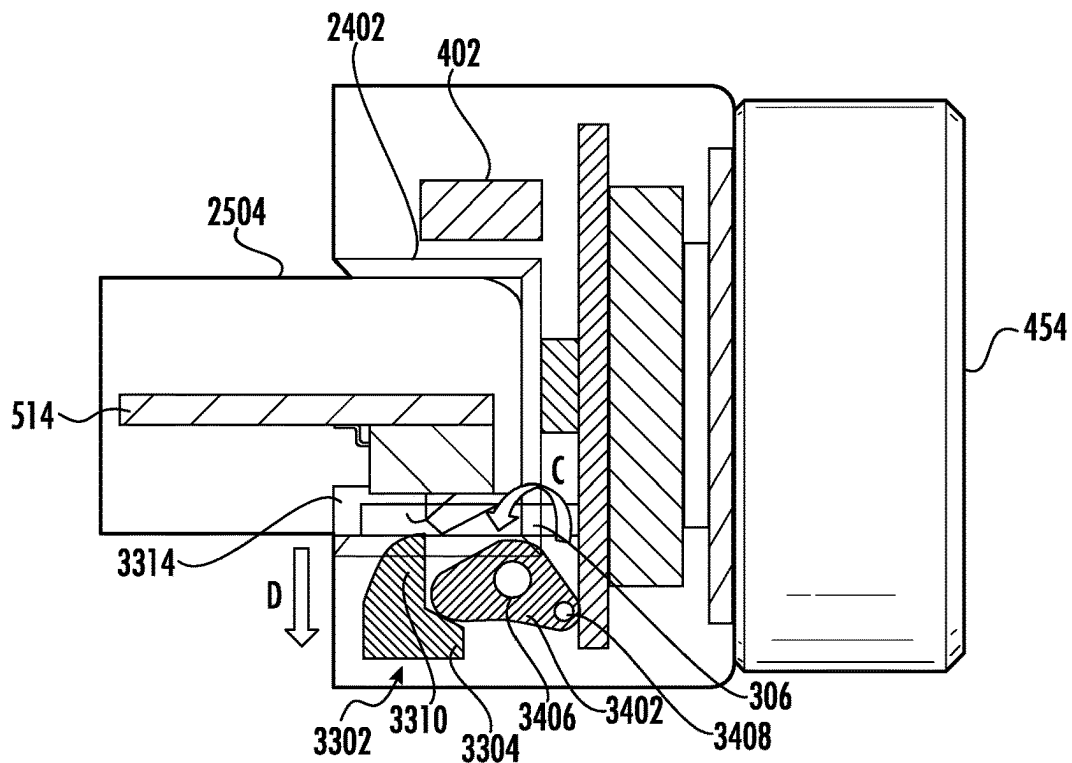
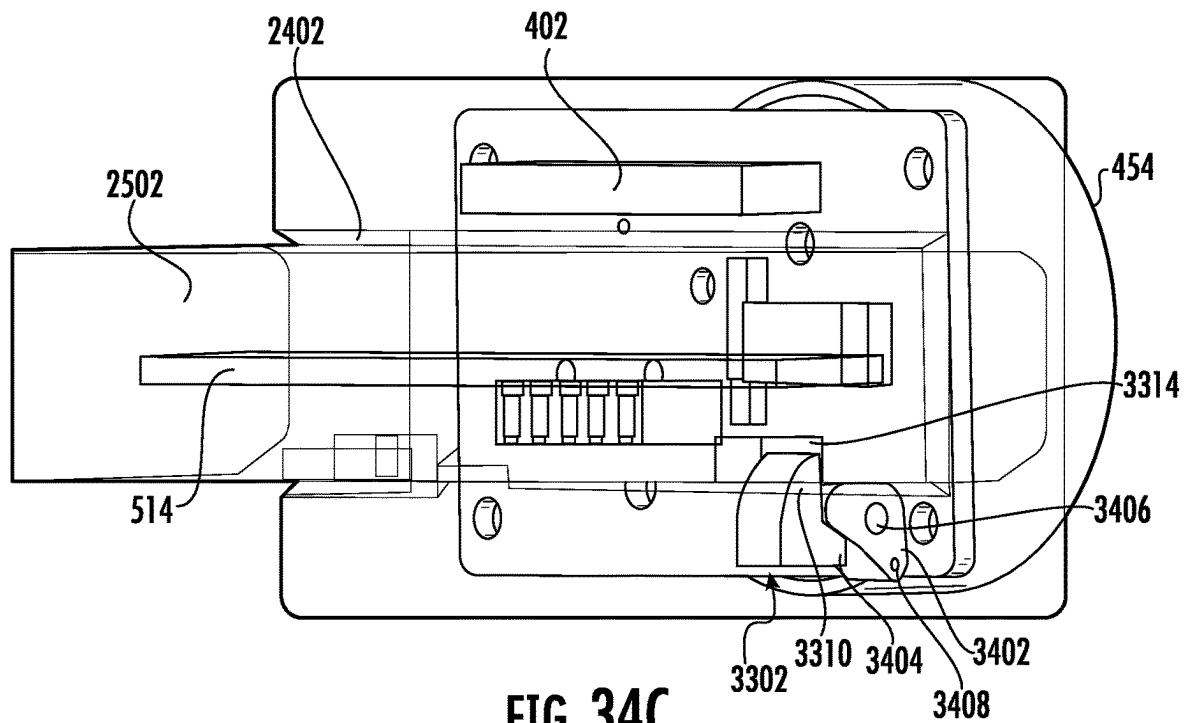
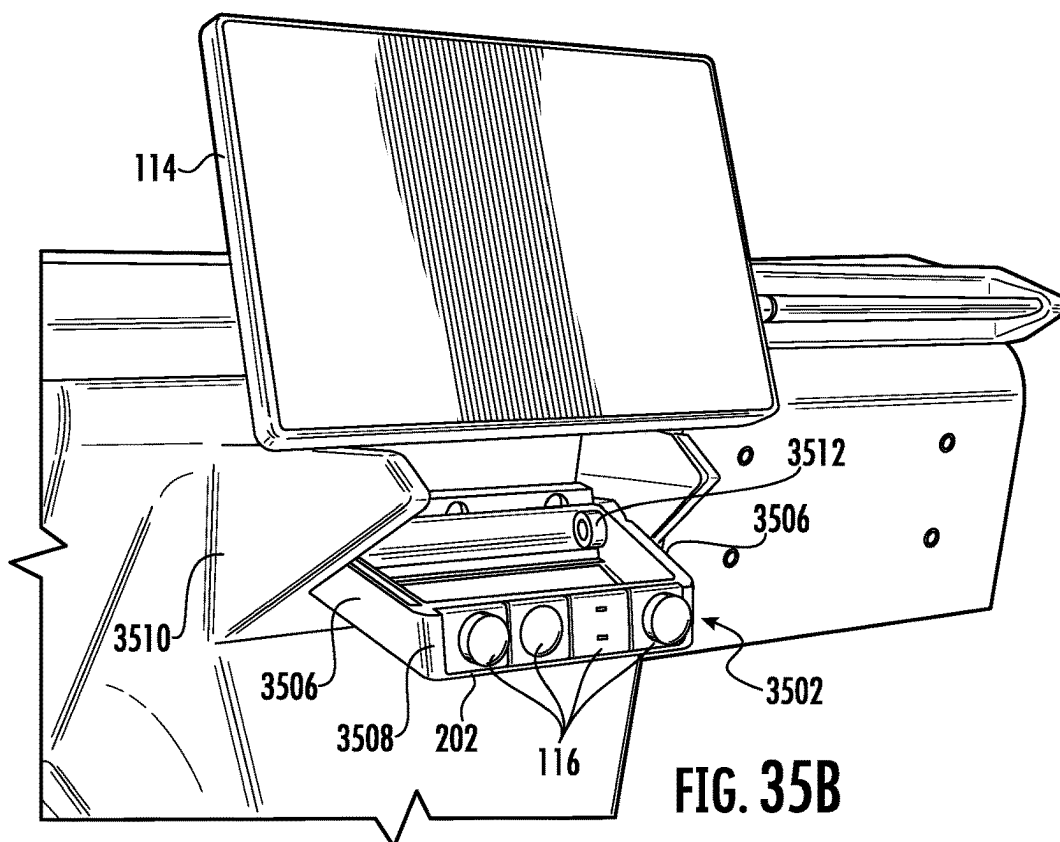
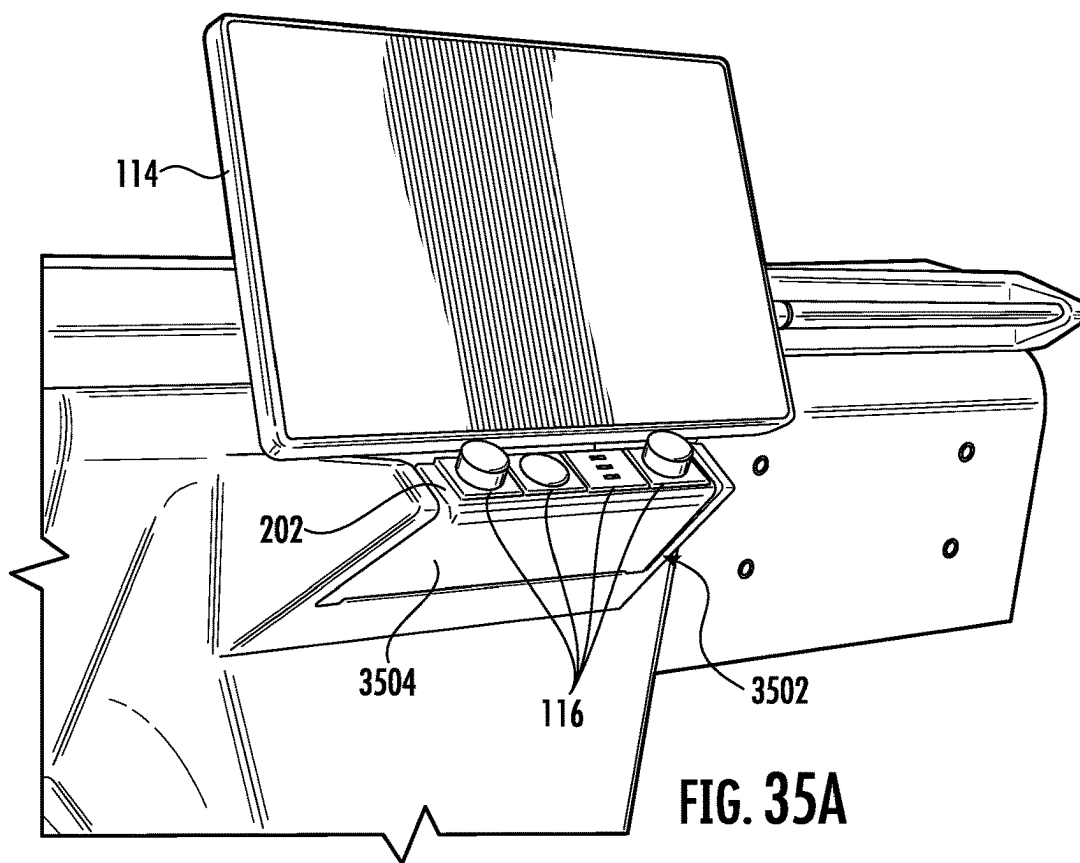
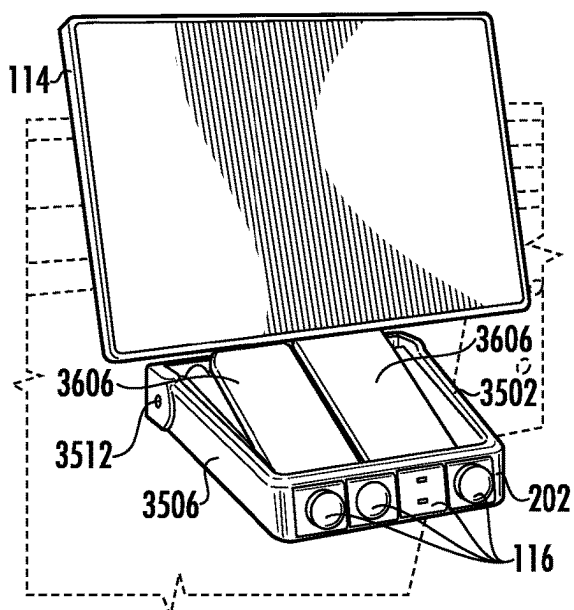
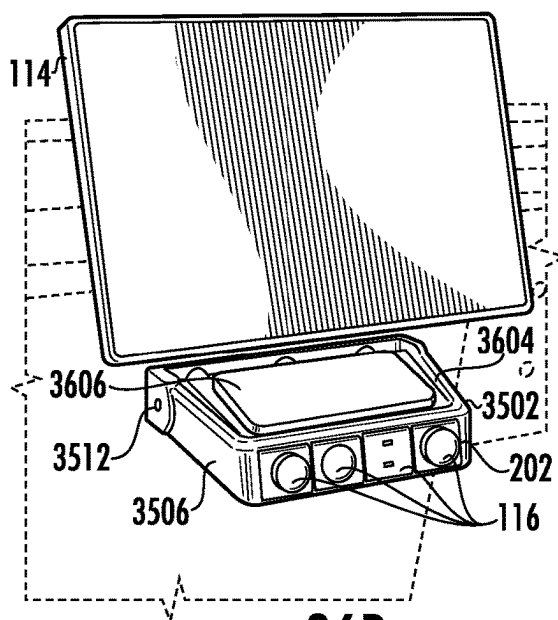
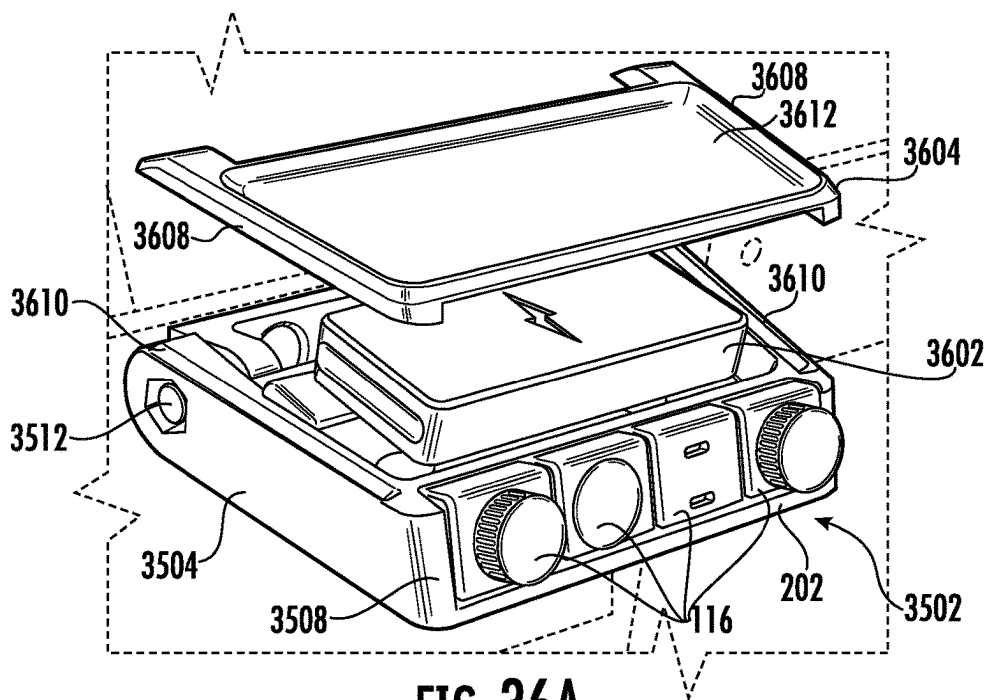


FIG. 34B







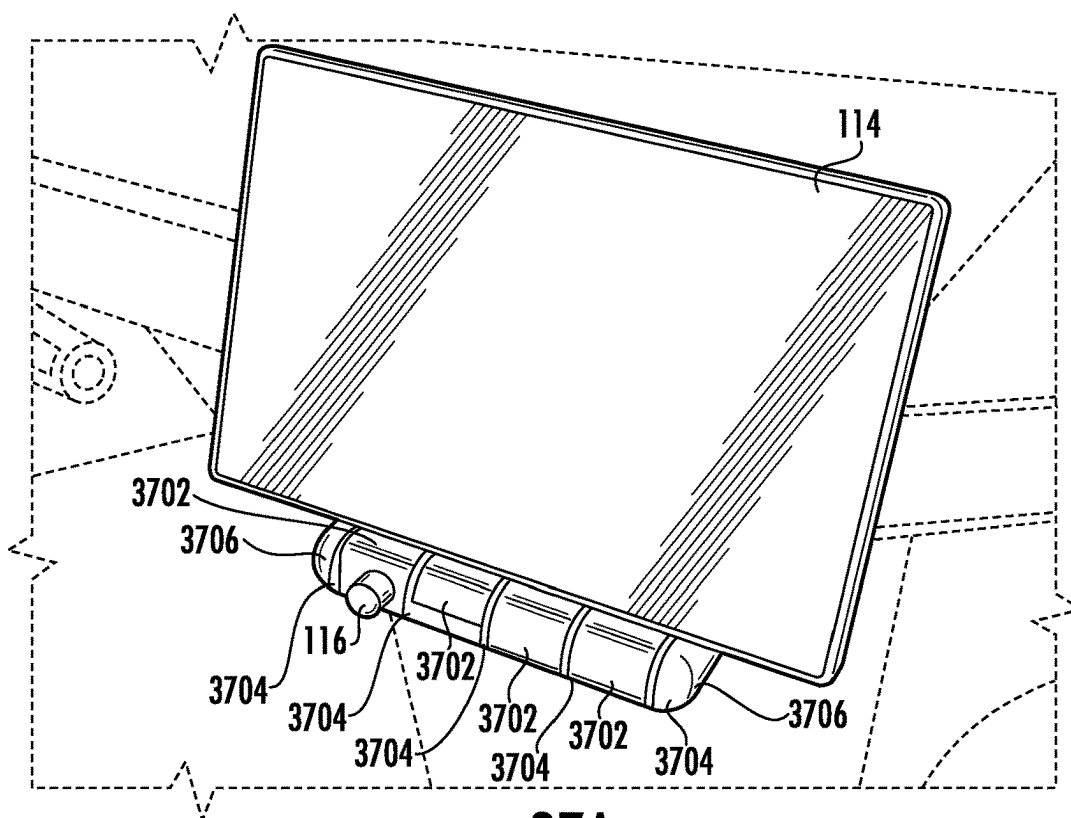


FIG. 37A

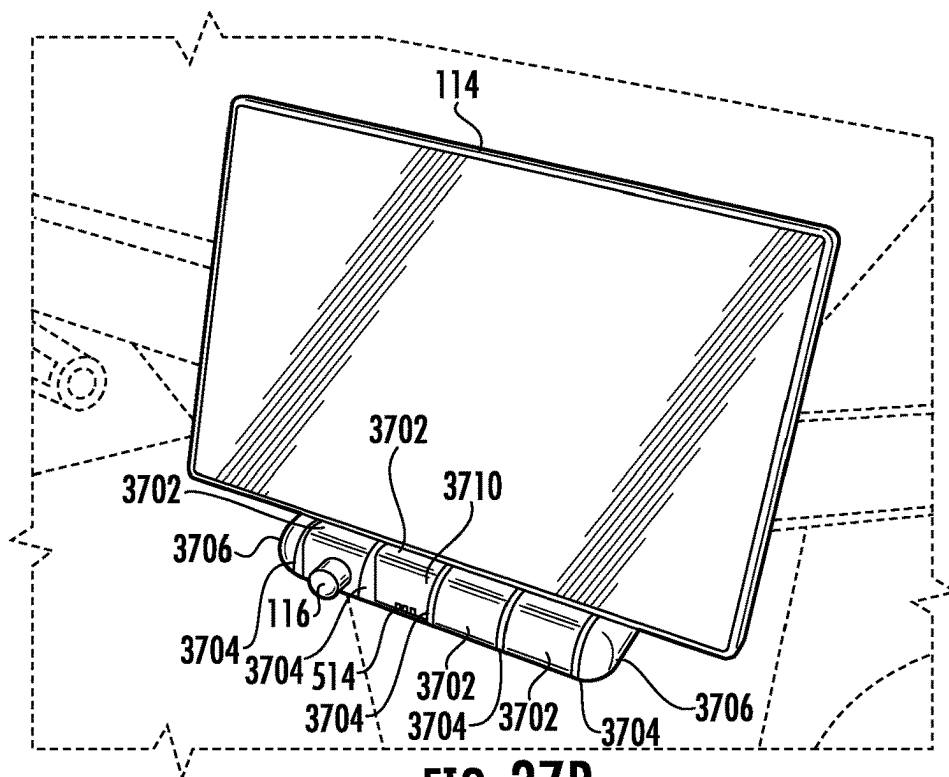


FIG. 37B

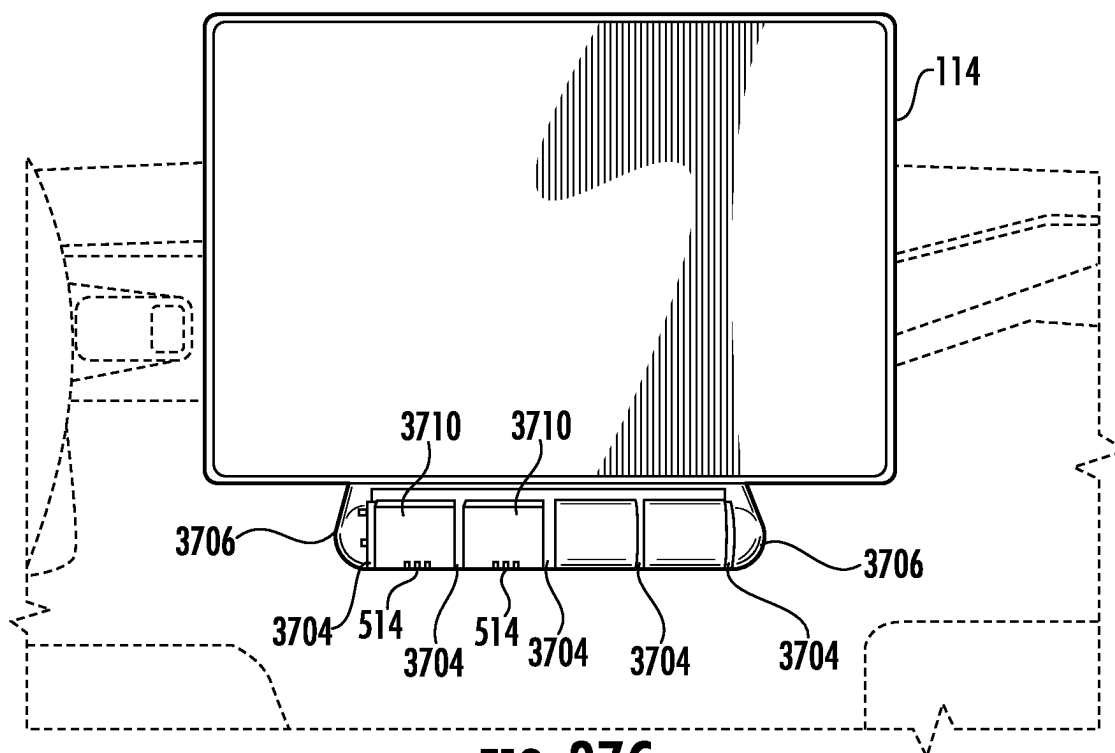


FIG. 37C

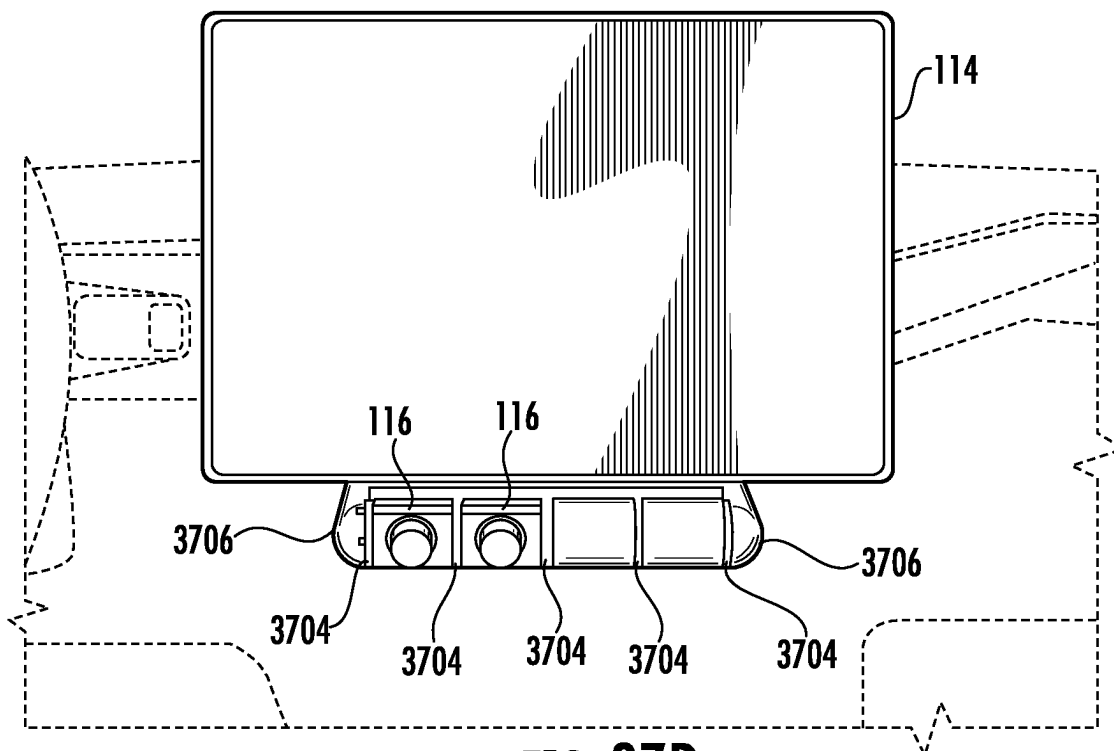


FIG. 37D

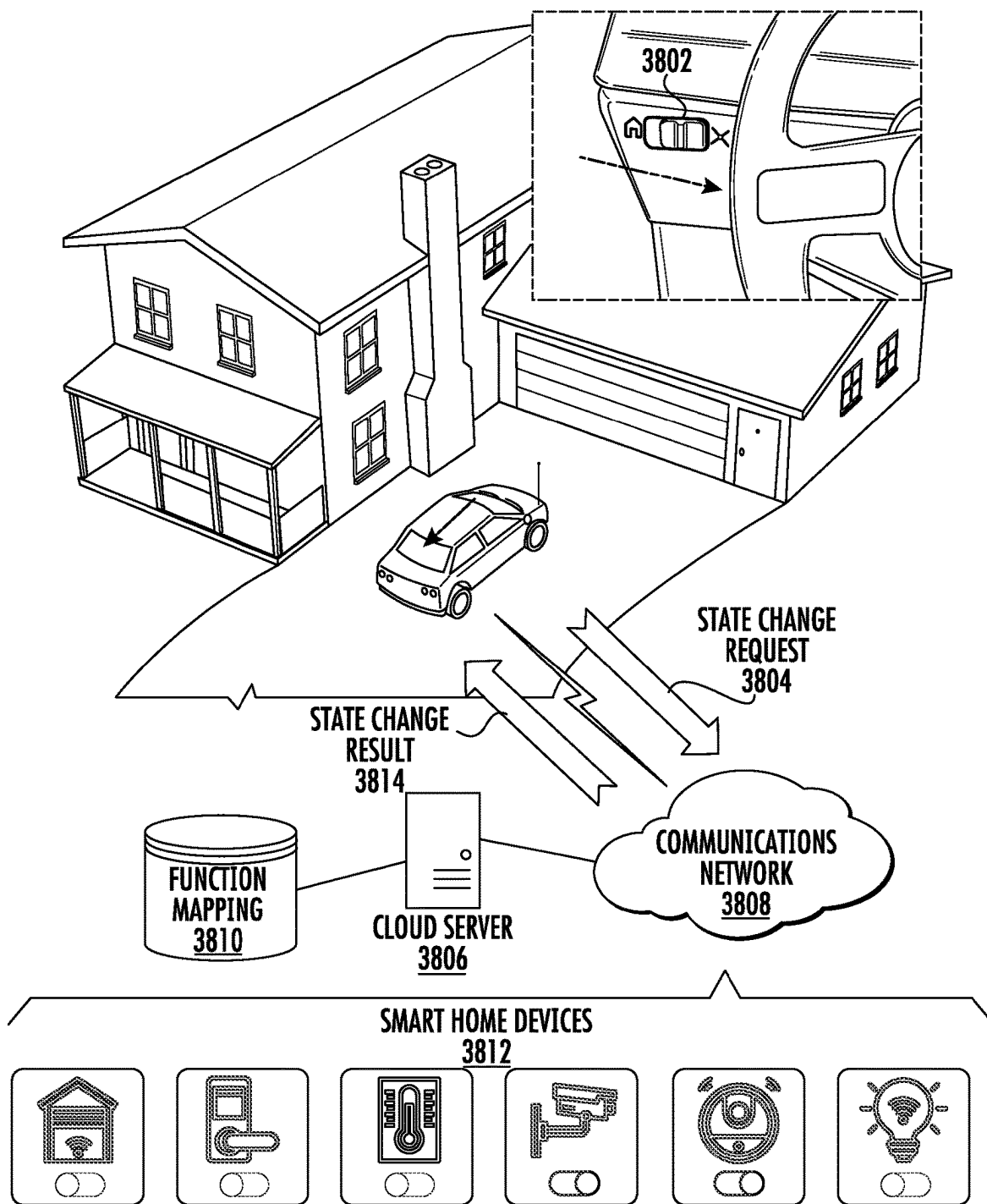


FIG. 38A

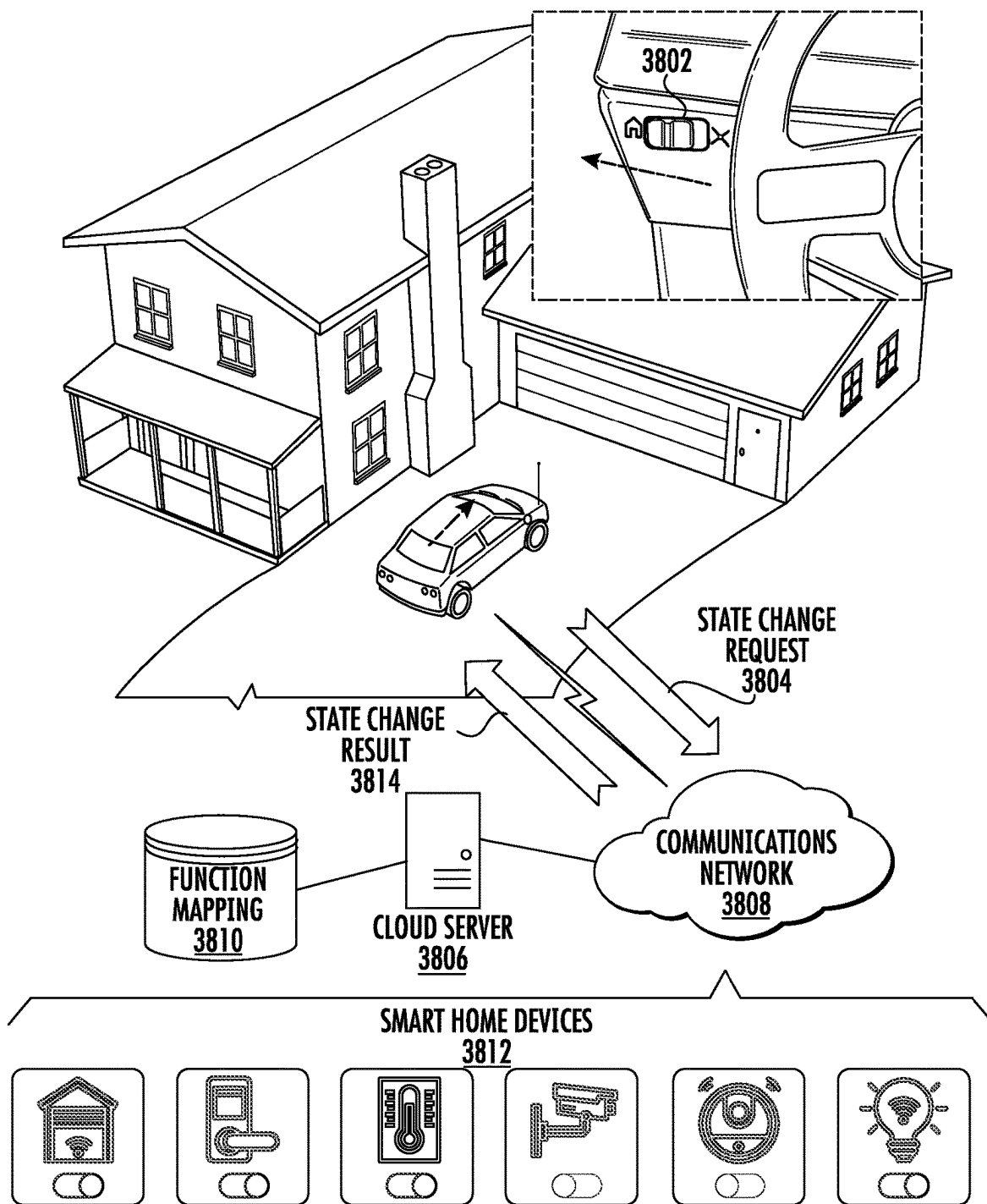


FIG. 38B

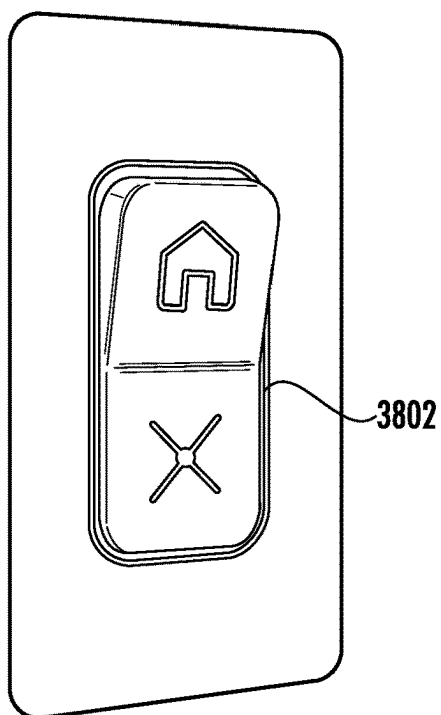


FIG. 39A

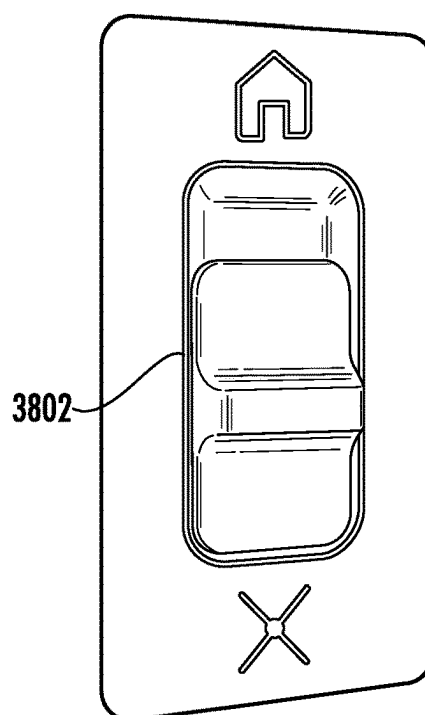


FIG. 39B

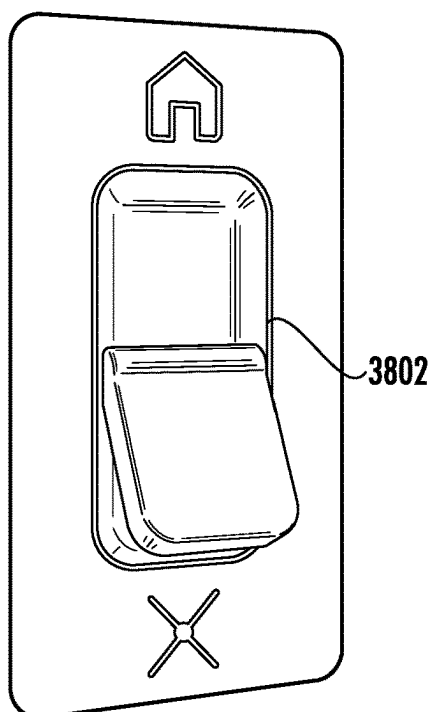


FIG. 39C

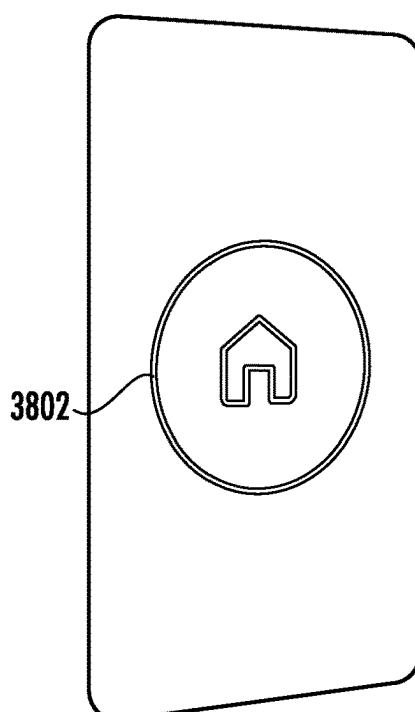
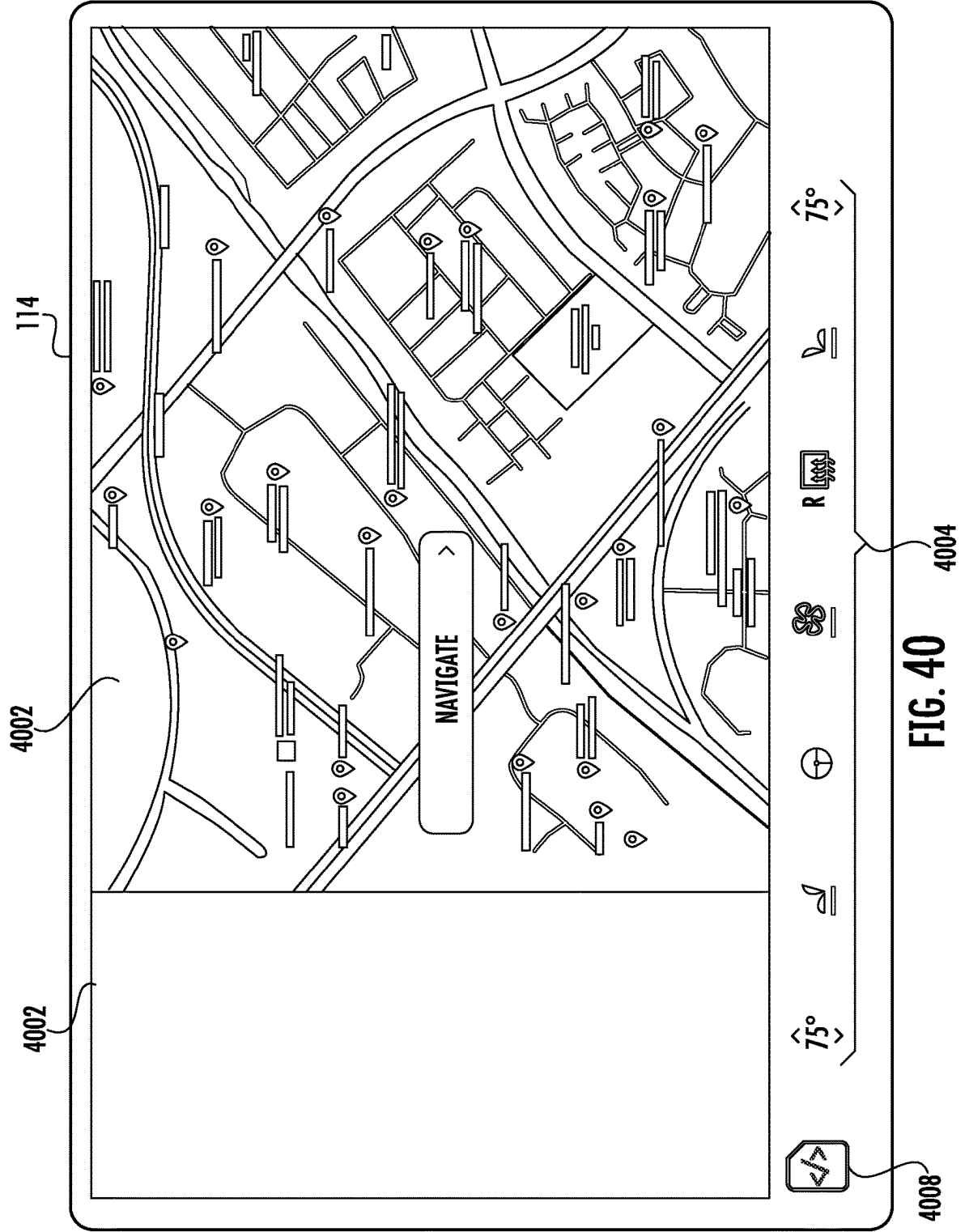


FIG. 39D



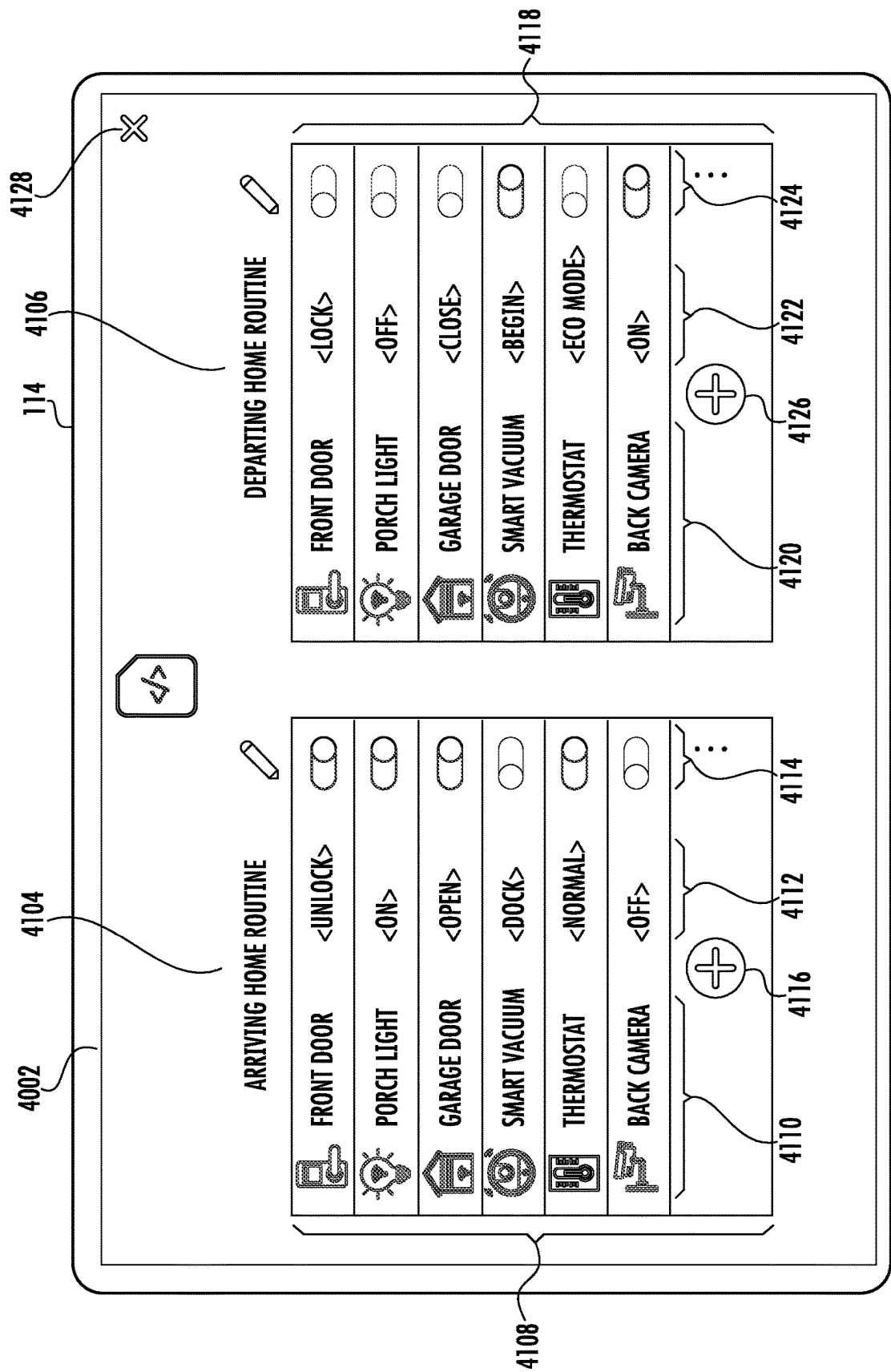
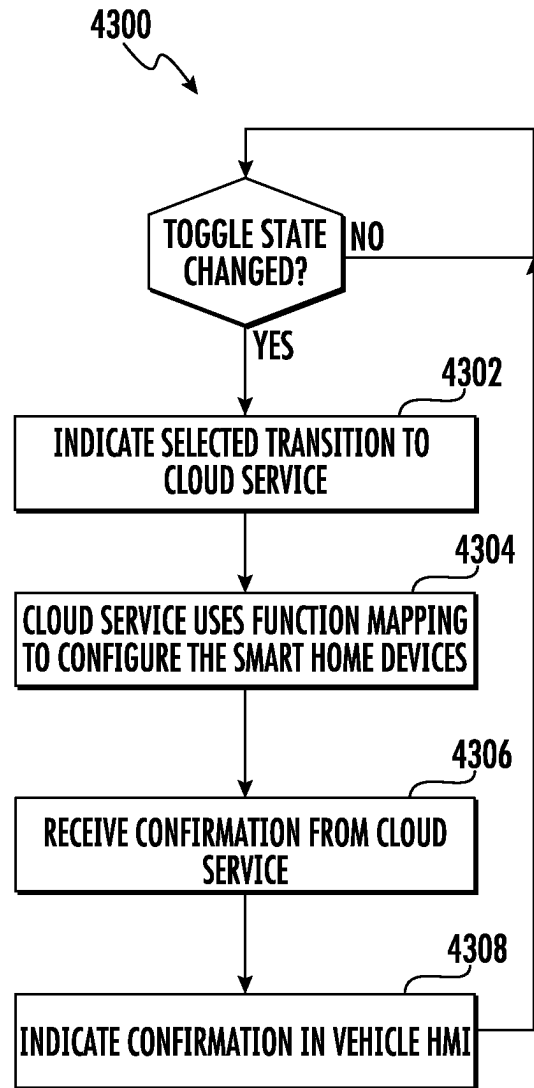
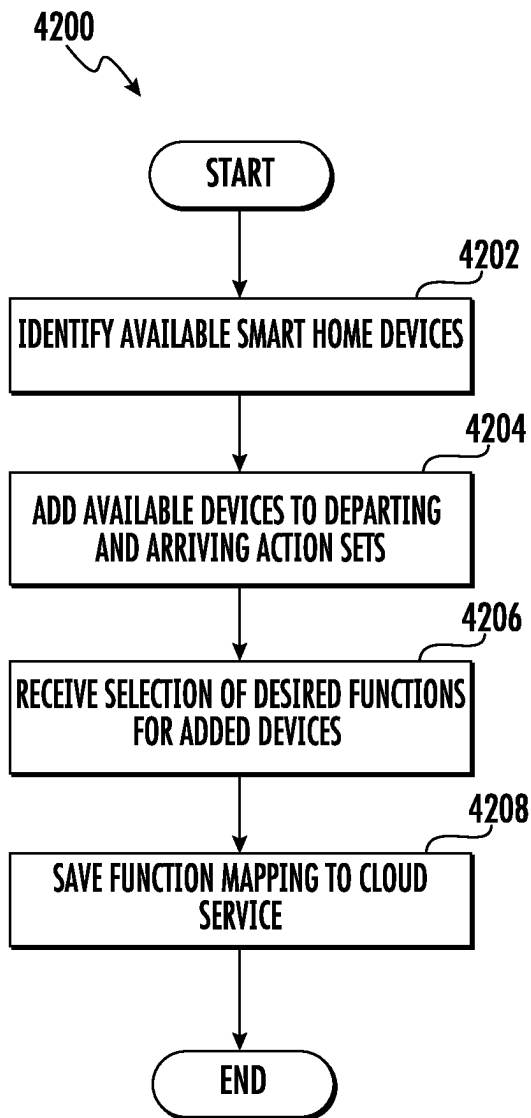


FIG. 41



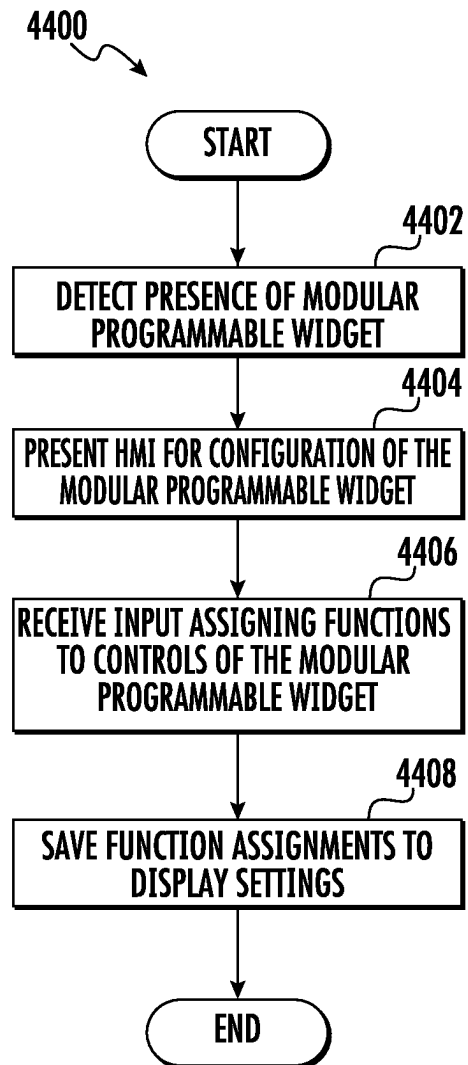


FIG. 44

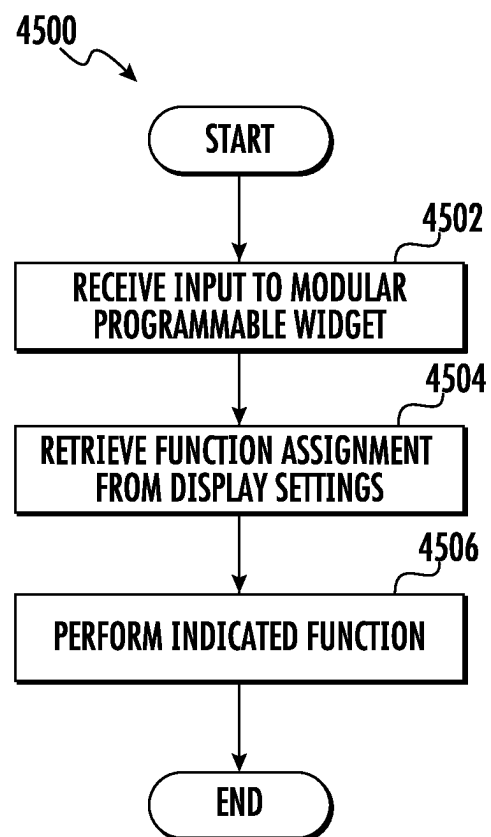


FIG. 45

1

MODULAR VEHICLE HMI**TECHNICAL FIELD**

Aspects of the disclosure generally relate to modular 5 human-machine interfaces (HMIs) for vehicles.

BACKGROUND

Many modern vehicles are equipped with electronic displays. Sometimes referred to as a digital dash, an electronic gauge cluster includes a set of vehicle instrumentation that may be displayed with a configurable digital readout or display panel, rather than with traditional analog gauges. Due to the configurable nature of the digital displays, electronic gauge cluster may be controlled by the user to display different elements of vehicle information.

SUMMARY

In one or more illustrative examples, a modular programmable widget for a vehicle, includes a front housing and a back housing defining an enclosure. One or more physical controls are provided on a front face of the enclosure. A contact pad is exposed by the enclosure for electrical connection to a dock, the contact pad providing for transmission of data and power between the modular programmable widget and the dock. One or more features for mechanical connection and securing of the modular programmable widget to the dock are also provided.

In one or more illustrative examples, a vehicle system for utilizing modular programmable widgets, includes a dock comprising a widget controller; a modular programmable widget, comprising an enclosure, one or more physical controls provided on a front face of the enclosure, a contact pad exposed by the enclosure for electrical connection to the widget controller of the dock, the contact pad providing for transmission of data and power between the modular programmable widget and the dock, and one or more features for mechanical connection and securing of the modular programmable widget to the dock; and a human-machine interface (HMI) display system, in communication with a configurable display, configured to in a configuration mode, present a user interface to the configurable display for the configuration of the one or more physical controls of the modular programmable widget, and in a runtime mode, perform an assigned function responsive to user input to the one or more physical controls.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example block diagram of a vehicle supporting modular programmable widgets;

FIG. 2 illustrates an example cutaway view of a vehicle interior supporting the modular programmable widgets;

FIG. 3A illustrates an example perspective view of a modular programmable widget having a configurable button with a display;

FIG. 3B illustrates an example perspective view modular programmable widget having a configurable dial with a display;

FIG. 3C illustrates an example perspective view modular programmable widget having additive power and data ports;

FIG. 3D illustrates an example perspective view modular programmable widget having a configurable touch screen;

FIG. 3E illustrates an example perspective view modular programmable widget having a toggle button;

2

FIG. 3F illustrates an example perspective view modular programmable widget having a camera;

FIG. 3G illustrates an example perspective view modular programmable widget having a biometric sensor;

FIG. 3H illustrates an example perspective view modular programmable widget having a credit card reader;

FIG. 3I illustrates an example perspective view modular programmable widget having a radio frequency identification (RFID) sensor;

FIG. 3J illustrates an example perspective view modular programmable widget having an inventory tracker;

FIG. 3K illustrates an example perspective view modular programmable widget having an air quality sensor;

FIG. 3L illustrates an example perspective view modular programmable widget having a configurable light emitting diode (LED) light array;

FIG. 3M illustrates an example perspective view modular programmable widget having a configurable joystick;

FIG. 3N illustrates an example perspective view modular programmable widget having a configurable four-way button pad;

FIG. 3O illustrates an example perspective view modular programmable widget having a high-definition multimedia interface (HDMI) port;

FIG. 3P illustrates an example perspective view modular programmable widget having an audio jack;

FIG. 3Q illustrates an example perspective view modular programmable widget having trailer control;

FIG. 3R illustrates an example perspective view modular programmable widget having a configurable multi-directional control;

FIG. 3S illustrates an example perspective view modular programmable widget having an additive magnetic wireless charger;

FIG. 3T illustrates an example perspective view modular programmable widget having a configurable access pass button pad;

FIG. 3U illustrates an example perspective view modular programmable widget having an additive analog clock;

FIG. 3V illustrates an example perspective view modular programmable widget having a companion device;

FIG. 3W illustrates an example perspective view modular programmable widget branded with a logo;

FIG. 3X illustrates an example perspective view modular programmable widget of a multiple of the length of a standard modular programmable widget shown in FIGS. 3A-3W;

FIG. 4A illustrates an example exploded view of the modular programmable widget shown in FIG. 3A having a configurable button with a display;

FIG. 4B illustrates an example exploded view of the modular programmable widget shown in FIG. 3B having a configurable dial with a display;

FIG. 4C illustrates an example exploded view of the modular programmable widget shown in FIG. 3C having additive power and data ports;

FIG. 4D illustrates an example exploded view of the modular programmable widget shown in FIG. 3D having a configurable touch screen;

FIG. 5 illustrates an example exploded view of a dock for receiving the modular programmable widgets;

FIG. 6 illustrates a side cutaway view of the dock of FIG. 5 in an assembled state;

FIG. 7 illustrates a side cutaway view of the modular programmable widget of FIG. 4B in an assembled state;

FIG. 8A illustrates a side view of a first operation in the attachment of an example modular programmable widget to the dock;

FIG. 8B illustrates a side view of a second operation in the attachment of an example modular programmable widget to the dock;

FIG. 8C illustrates a side view of a completed attachment of an example modular programmable widget to the dock;

FIG. 9 illustrates an example cutaway view of a vehicle interior supporting the modular programmable widgets having an alternate packaging design;

FIG. 10A illustrates an example perspective view of a modular programmable widget having a configurable button;

FIG. 10B illustrates an example perspective view of a modular programmable widget having a configurable dial;

FIG. 10C illustrates an example perspective view of a modular programmable widget having additive power/data ports;

FIG. 10D illustrates an example perspective view of a modular programmable widget having a configurable display;

FIG. 10E illustrates an example perspective view of a modular programmable widget having a companion;

FIG. 10F illustrates an example perspective view of a modular programmable widget having a configurable joystick;

FIG. 10G illustrates an example perspective view of a modular programmable widget having trailer control;

FIG. 10H illustrates an example perspective view of a modular programmable widget having a toggle button;

FIG. 10I illustrates an example perspective view of a modular programmable widget that is multiple times the length of a standard modular programmable widget as shown in FIGS. 10A-10H;

FIG. 11 illustrates an example side view of the modular programmable widget of FIG. 10B having a configurable dial;

FIG. 12 illustrates an example rear perspective view of the modular programmable widget of FIG. 8B having a configurable dial;

FIG. 13 illustrates an example exploded view of the modular programmable widget shown in FIG. 8B having a configurable dial;

FIG. 14 illustrates an example front view of the dock for receiving the modular programmable widgets;

FIG. 15A illustrates perspective and side views of a first operation in the attachment of an example alternate design modular programmable widget to the dock;

FIG. 15B illustrates perspective and side views of a second operation in the attachment of an example alternate design modular programmable widget to the dock;

FIG. 15C illustrates perspective and side views of a completed attachment of an example alternate design modular programmable widget to the dock;

FIG. 16 illustrates a perspective view of the dock integrated below a configurable display of the vehicle for attachment of the modular programmable widgets;

FIG. 17 illustrates a side cutaway view of the modular programmable widget shown in FIGS. 8B and 12-13, in assembled form and attached to the dock;

FIG. 18 illustrates the removal of the rotary dial from an assembled modular programmable widget to allow for changing out of the signifier cap;

FIG. 19 illustrates a selection of different signifier caps, each having a different customizable front face;

FIG. 20 illustrates an example selection of different signifier caps installed to various modular programmable widgets;

FIG. 21A illustrates a front perspective view of an empty mobile dock without attachment of any modular programmable widgets;

FIG. 21B illustrates a front perspective view of the mobile dock in the process of attachment of modular programmable widgets;

FIG. 21C illustrates a front perspective view of the mobile dock having modular programmable widgets attached;

FIG. 21D illustrates a rear perspective view of the mobile dock having modular programmable widgets attached;

FIG. 22 illustrates an example cutaway view of a vehicle interior supporting the modular programmable widgets having another alternative alternate packaging design;

FIG. 23A illustrates an example perspective view of a modular programmable widget having a configurable button;

FIG. 23B illustrates an example perspective view of a modular programmable widget having a configurable dial;

FIG. 23C illustrates an example perspective view of a modular programmable widget having additive power/data ports;

FIG. 23D illustrates an example perspective view of a modular programmable widget having a configurable display;

FIG. 23E illustrates an example perspective view of a modular programmable widget having a companion;

FIG. 23F illustrates an example perspective view of a modular programmable widget having a configurable dial with display;

FIG. 23G illustrates an example perspective view of a modular programmable widget that is multiple times the length of a standard modular programmable widget as shown in FIGS. 23A-23F;

FIG. 24 illustrates an example exploded view of the modular programmable widget shown in FIG. 23A having a configurable button;

FIG. 25 illustrates a perspective view of the dock integrated below a configurable display of the vehicle for attachment of the modular programmable widgets;

FIG. 26A illustrates a side view of an operation in the attachment of an example modular programmable widget to the dock;

FIG. 26B illustrates a side view of a completed attachment of an example modular programmable widget to the dock;

FIG. 27 illustrates an example of a home dock configured for receiving the modular programmable widgets;

FIG. 28A illustrates an example of a first set of modular programmable widgets, in a common hardcoded, individual configuration;

FIG. 28B illustrates an example of a second set of modular programmable widgets, in a configurable, individual configuration;

FIG. 28C illustrates an example of a third set of modular programmable widgets, in a common grouped and configurable individual configuration;

FIG. 28D illustrates an example of a fourth set of modular programmable widgets, in a hardcoded grouped configuration;

FIG. 28E illustrates an example of a fifth set of modular programmable widgets in a configurable grouped configuration;

FIG. 28F illustrates an example of a sixth set of modular programmable widgets in a hybrid bar configuration;

5

FIG. 29A illustrates an example perspective view of a modular programmable widget having a configurable button with a display;

FIG. 29B illustrates an example perspective view modular programmable widget having a configurable dial with a display;

FIG. 29C illustrates an example perspective view modular programmable widget having additive power and data ports;

FIG. 29D illustrates an example perspective view modular programmable widget having a toggle button;

FIG. 29E illustrates an example perspective view modular programmable widget having a configurable touch screen;

FIG. 30 illustrates a perspective view of the dock integrated below a configurable display of the vehicle for attachment of the modular programmable widgets;

FIG. 31 illustrates an example exploded view of the modular programmable widget shown in FIG. 29B having a configurable dial;

FIG. 32 illustrates a side cutaway view of an attachment of an example modular programmable widget to the dock;

FIG. 33A illustrates a side cutaway view of a modular programmable widget having a locking slider in a locked position;

FIG. 33B illustrates a side cutaway view of the modular programmable widget of FIG. 33A with the locking slider in an unlocked position;

FIG. 33C illustrates a rear perspective partially transparent view of the modular programmable widget of FIG. 33A having the locking slider in the locked position;

FIG. 34A illustrates a side cutaway view of a modular programmable widget having an alternate locking mechanism using a locking cam in a locked position;

FIG. 34B illustrates a side cutaway view of the modular programmable widget of FIG. 34A having the locking cam in an unlocked position;

FIG. 34C illustrates a rear perspective partially transparent view of a modular programmable widget of FIG. 34A having a locking cam in the locked position;

FIG. 35A illustrates an example of a storage compartment with integrated dock functionality, shown in a closed position;

FIG. 35B illustrates an example of the storage compartment with integrated dock of FIG. 35A in an open position;

FIG. 36A illustrates use of a phone charger and dock plate for use with the storage compartment;

FIG. 36B illustrates a single phone dock with a landscape orientation;

FIG. 36C illustrates a dual phone dock with a portrait orientation;

FIG. 37A illustrates an example of a rotating dock with four rotating dock slots, with the first dock slot in the open position and holding a modular programmable widget, the second in an intermediate position, and the remaining dock slots in the closed position;

FIG. 37B illustrates an example of a rotating dock with four rotating dock slots, with the first dock slot in the open position and modular programmable widget, the second in the open position and empty, and the remaining dock slots in the closed position;

FIG. 37C illustrates an example of a rotating dock with four rotating dock slots, with the first two dock slot open and filled with modular programmable widgets, and the remaining dock slots in the closed position;

FIG. 37D illustrates an example of a rotating dock with four rotating dock slots, with the first two dock open and slot empty, and the remaining dock slots in the closed position;

6

FIG. 38A illustrates an example physical toggle button to facilitate switching between a home mode and an away mode, with the physical toggle button in the away mode;

FIG. 38B illustrates an example physical toggle button to facilitate switching between a home mode and an away mode, with the physical toggle button in the home mode;

FIG. 39A illustrates an example flip style physical toggle for use in switching between the home and away modes;

FIG. 39B illustrates an example slide style physical toggle button for use in switching between the home and away modes;

FIG. 39C illustrates an example toggle style physical toggle button for use in switching between the home and away modes;

FIG. 39D illustrates an example pushbutton style physical toggle button for use in switching between the home and away modes;

FIG. 40 illustrates an example main screen of the configurable display from which a configuration screen for the home and away modes is selectable;

FIG. 41 illustrates an example configuration screen of the configurable display;

FIG. 42 illustrates an example process for the configuration of the physical toggle button;

FIG. 43 illustrates an example process for the execution of a mode change triggered by the physical toggle button;

FIG. 44 illustrates an example process for the configuration of a modular programmable widget; and

FIG. 45 illustrates an example process for the use of a configured modular programmable widget.

DETAILED DESCRIPTION

As required, detailed embodiments of the present disclosure are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the disclosure that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present disclosure.

Some vehicle HMIs provide a primarily touch based interface, with few physical controls. Such an interior may offer simplicity and minimalism. Yet, some users may desire tactile controls. Each user may have different preferences, and the ability to tailor the user interface may be a user expectation.

An enhanced vehicle HMI may support modular programmable widgets. These widgets may include a variety of modular and programmable tactile buttons, knobs, switches, or other physical controls. The modular programmable widgets may be installed to the vehicle and assigned to various functions. Once attached and configured, the physical controls of the modular programmable widgets may offer quick, customized access to the configured functions. Further aspects of the modular programmable widgets are described in detail herein.

FIG. 1 illustrates an example block diagram 100 of a vehicle 102 supporting modular programmable widgets 116. The vehicle 102 may include various types of passenger vehicle, such as crossover utility vehicle (CUV), sport utility vehicle (SUV), truck, recreational vehicle (RV), boat, plane or other mobile machine for transporting people or goods. It

should be noted that the illustrated vehicle **102** is merely an example, and more, fewer, and/or differently located elements may be used.

An HMI display system **104** of the vehicle **102** may include one or more processors **106** configured to perform instructions, commands and other routines in support of the processes described herein. For instance, the HMI display system **104** may be configured to execute instructions of a display application **132** loaded to a memory **108** to provide information display features. Such instructions and other data may be maintained in a non-volatile manner using a variety of types of computer-readable medium **110**. The computer-readable medium **110** (also referred to as a processor-readable medium or storage) includes any non-transitory medium (e.g., a tangible medium) that participates in providing instructions or other data that may be read by the processor **106** of the HMI display system **104**. Computer-executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java, C, C++, C#, Objective C, Fortran, Pascal, Java Script, Python, Perl, hypertext markup language (HTML), etc.

The HMI display system **104** may be provided with various functionality to allow the occupants to interact with the vehicle **102**. For example, the HMI display system **104** may drive or otherwise communicate with one or more configurable displays **114** configured to provide visual output to vehicle **102** occupants and receive touch input from the occupants by way of a display controller **118**. In another example, the HMI display system **104** may include a widget controller **112** configured to receive user input indicative of occupant-vehicle interaction from one or more modular programmable widgets **116** of the vehicle **102**. These modular programmable widgets **116** may include various controls, such as buttons or knobs.

FIG. 2 illustrates an example cutaway view of a vehicle interior **200** supporting the modular programmable widgets **116**. As shown in FIG. 2, the HMI display system **104** for the vehicle **102** may provide a touch-based user interface via one or more configurable displays **114**. To satisfy the requirements to users who wish to use hardware controls, the HMI display system **104** may provide an ability to add, remove, and configure the modular programmable widgets **116** using the configurable display **114**. The modular programmable widgets **116** may offer customizable tactile HMI for various vehicle **102** functions. The controls may be assigned to various functions such as bringing up a book-marked display screen, or allowing for the direct configuration of settings such as utility controls (e.g., towing functions), climate controls, gig work function (e.g., ride-share), media control functions, smart home settings, vehicle settings, etc.

As shown, one or more first modular programmable widgets **116** may be attached to a dock **202A** of the center stack configurable display **114**. This location under the center stack configurable display **114** may be easily accessible for driver and passenger. Also shown, one or more second modular programmable widgets **116** may be attached to a console bin dock **202B**. Here, the second modular programmable widget **116** is attached to the console bin dock **202B** for access by rear passengers, which may be useful for both retail and vehicle-as-a-service (VaaS) use cases. As further shown, one or more third modular programmable widgets **116** may be attached to a cargo dock **202C** in the rear cargo area. It should also be noted that the

illustrated vehicle interior **200** is an example, and interiors having more, fewer, or differently located components may be used.

Referring back to FIG. 1, the HMI display system **104** may be further configured to communicate with other components of the vehicle **102** via one or more in-vehicle buses **120**. The in-vehicle buses **120** may include one or more of a vehicle controller area network (CAN), an Ethernet network, and a media oriented system transfer (MOST), as some non-limiting possibilities. The in-vehicle buses **120** may allow the HMI display system **104** to send or receive information relevant to the HMI to and from other vehicle **102** systems. The exemplary vehicle systems discussed herein may communicate with the HMI display system **104** over a main in-vehicle bus **120**. In other examples, the HMI display system **104** may be connected to more or fewer in-vehicle buses **120**, and one or more modular programmable widgets **116** or other components may be connected to the HMI display system **104** via in-vehicle buses **120** or directly without connection to an in-vehicle bus **120**.

As some non-limiting examples, a powertrain control module **122** may be a component in communication with the HMI display system **104** and may be configured to provide information to the HMI display system **104** regarding control of engine operating components (e.g., idle control, fuel delivery, emissions control, engine diagnostic codes, etc.). A body control module **124** may be configured to manage various power control functions such as exterior lighting, interior lighting, keyless entry, remote start, and provide information to the HMI display system **104** such as point of access status information (e.g., closure status of the hood, doors and/or trunk of the vehicle **102**). A climate control module **126** may be configured to provide control and monitoring of heating and cooling system components, as well as to provide information to the HMI display system **104** regarding the components (e.g., compressor clutch and blower fan control, temperature sensor information, etc.). A stability control module **128** may be configured to perform functions such as active suspension control, traction control, and trailer control, and provide sensed vehicle **102** dynamics information to the HMI display system **104**, such as roll angle, pitch angle, yaw rate, roll rate, pitch rate, lateral and longitudinal velocity, lateral and longitudinal change over time, tire slip, tire slip rate, and an infotainment system module. A telematics control module **130** may include an in-vehicle modem configured to access communications services of a communications network (not shown), and may provide packet-switched network services (e.g., Internet access, voice over Internet protocol (VOIP) communication services) to the HMI display system **104** and to other devices connected over the in-vehicle bus **120**.

A display application **132** may be installed to the HMI display system **104** and utilized to allow the vehicle **102** to provide output to the display controller **118** such that the configurable display **114** conveys the information relating to the operation of the vehicle **102** to the driver. In an example, the display application **132** may be configured to provide a menu structure via the configurable display **114** having set of categories into which information screens are placed, such as media selections, trip/fuel information, navigation, towing information, etc.

The menu structure may further include information screens that may be selected from the categories. For instance, the trip/fuel category may include a trip counter screen for a first trip counter, a trip counter screen for a second trip counter, a fuel economy screen, a fuel history screen, and a navigation/compass screen. When a screen is

selected from the menu structure (e.g., via touch input to the configurable displays **114**, via user manipulation of the modular programmable widgets **116**, via voice commands, etc.), the selected screen or function may be provided on the configurable display **114**.

As shown in greater detail below, the display application **132** may be further configured to allow a user to configure the modular programmable widgets **116** to operate as shortcuts to various functionality of the HMI display system **104**. Accordingly, when driving the user may be able to quickly access the bookmarked functions or screens using the modular programmable widget **116**, to provide for quick selection by the HMI display system **104** on the configurable display **114**.

The display application **132** may maintain display settings **134** indicative of the mapping of modular programmable widgets **116** to the functionality of the HMI display system **104**. For instance, responsive to the connection of a modular programmable widget **116** to the HMI display system **104**, the display application **132** may transition the configurable display **114** to a screen for the configuration of the modular programmable widget **116**. The configurable display **114** may allow the user to select which functions are to be controlled by which controls of the modular programmable widget **116**. The display setting **134** may then be stored to the memory **108** and used to interpret input received to the modular programmable widget **116** by the HMI display system **104**.

Various types of modular programmable widget **116** may be utilized. In a first option, the modular programmable widgets **116** may utilize a toe-in mechanism. Details of the first option are discussed with respect to FIGS. 2-8C. In another option the modular programmable widgets **116** may utilize an alternate form factor for the toe-in mechanism. Details of the second option are discussed with respect to FIGS. 9-21D. In yet another option, the modular programmable widgets **116** may utilize a magnetic attachment port design. Details of the third option are discussed with respect to FIGS. 22-28F. In still another option the modular programmable widgets **116** may utilize a pin connector. Details of the fourth option are discussed with respect to FIGS. 29A-32. Optional locking aspects of the modular programmable widgets **116** are discussed with respect to FIGS. 33A-C and 34A-C.

FIGS. 3A-3D illustrate example modular programmable widgets **116**. Referring collectively to FIGS. 3A-3D, an enclosure having a generally rectangular prism shape may be defined by a front housing **302** and a back housing **304**. The front housing **302** may define a generally flat front surface of the modular programmable widget **116** a top surface of the modular programmable widget **116**, and a front portion of the side walls. The back housing **304** may define the remainder of the enclosure, including the rear wall, and a rear portion of the side, bottom, and top walls. The front housing **302** and back housing **304** may collectively house the internal circuitry and other components of the modular programmable widget **116**. Controls may be provided on front face of the front housing **302**, while connectivity to the vehicle **102** may be provided via contact pads **306** exposed through the top of the back housing **304**. The contact pads **306** may allow for transmission of data and power between the modular programmable widgets **116** and the dock **202**. In an example, the contact pads **306** may be leaf spring contacts.

The front housing **302** may expose various controls that may provide interaction with the user. In the example of FIG. 3A, this may include a configurable button **308**. In

some examples, the configurable button **308** may further include a display **310** integrated into the front face of the configurable button **308**. The display **310** may be programmed to display various content to the user. As a possibility, the display **310** may be used to show an icon or other representation of the functionality assigned to the configurable button **308**.

FIG. 3B illustrates an example modular programmable widget **116** having a configurable dial **312** with a display **310**. The configurable dial **312** may include a rotary encoder configured to receive rotational input from a user, e.g., for selection of a continuous parameter such as volume or temperature, for selection from a menu of choices, etc. The configurable dial **312** may optionally include button functionality, similar to the operation of the configurable button **308**, such that the configurable dial **312** may be pressed for binary parameter control (e.g., mute), preset recall, to move to a next item in a list (e.g., of media inputs, radio stations, etc.). Similar to the configurable button **308**, the configurable dial **312** may further include a display **310**, such the front face of the configurable dial **312** may show an icon or other representation of the functionality assigned to the configurable dial **312**.

FIG. 3C illustrates an example modular programmable widget **116** having additive power/data ports **314**. As shown, the modular programmable widgets **116** in FIG. 3C provides two USB power/data ports **314**, but this is merely an example, and different quantities and types of connectors may be exposed by the modular programmable widgets **116**.

FIG. 3D illustrates an example modular programmable widget **116** having a configurable touch screen **316**. As shown in this example, the modular programmable widget **116** provides a flat rectangular touch screen **316** across the front surface of the modular programmable widget **116**.

It should be noted that the modular programmable widgets **116** shown in FIGS. 3A-3D are merely examples, and modular programmable widgets **116** having more, fewer, or different features may be used. Further examples of possible modular programmable widgets **116** are shown in FIGS. 3E-3W.

FIG. 3E illustrates an example modular programmable widget **116** having a toggle button **318**. The toggle button **318** may be used, for example, to turn a feature or other function on or off. FIG. 3F illustrates an example modular programmable widget **116** having a camera **320**. FIG. 3G illustrates an example modular programmable widget **116** having a biometric sensor **322**. FIG. 3H illustrates an example modular programmable widget **116** having a credit card reader **324**. FIG. 3I illustrates an example modular programmable widget **116** having an RFID sensor **326**. FIG. 3J illustrates an example modular programmable widget **116** having an inventory tracker **328**.

FIG. 3K illustrates an example modular programmable widget **116** having an air quality sensor **330**. FIG. 3L illustrates an example modular programmable widget **116** having a configurable LED light array **332**. FIG. 3M illustrates an example modular programmable widget **116** having a configurable joystick **334**. FIG. 3N illustrates an example modular programmable widget **116** having a configurable four-way button pad **336**. The buttons of the configurable four-way button pad **336** (as well as any other buttons on the various modular programmable widgets **116**) may be individually assignable to functions, and or for adjusting a continuous parameter upwards or downwards, such as volume or temperature. FIG. 3O illustrates an example modular

11

programmable widget 116 having an HDMI port 338. FIG. 3P illustrates an example modular programmable widget 116 having an audio jack 340.

FIG. 3Q illustrates an example modular programmable widget 116 having trailer control 342. FIG. 3R illustrates an example modular programmable widget 116 having a configurable multi-directional control 344. FIG. 3S illustrates an example modular programmable widget 116 having an additive magnetic wireless charger 346. FIG. 3T illustrates an example modular programmable widget 116 having a configurable access pass button pad 348. FIG. 3U illustrates an example modular programmable widget 116 having an additive analog clock 350. FIG. 3V illustrates an example modular programmable widget 116 having a companion 352. FIG. 3W illustrates an example perspective view modular programmable widget branded with a logo 354. This may allow, for example, the modular programmable widget 116 to be offered with hard-coded co-branded functions that relate to a third party.

It should be noted that the modular programmable widgets 116 shown in FIGS. 3A-3W are each of the same exterior dimensions. This standard size allows for easy addition and removal of the modular programmable widgets 116 from the docks 202. In other cases, however, the modular programmable widgets 116 may include modular programmable widgets 116 of a larger physical size. Such modular programmable widgets 116 may be able to include more controls as compared to the default size of the modular programmable widgets 116.

FIG. 3X illustrates an example modular programmable widget 116 that is multiple times the length of a standard modular programmable widget 116 as shown in FIGS. 3A-3W. In this example, the modular programmable widget 116 is three times the length of the standard modular programmable widgets 116. Due to the increased length, the modular programmable widget 116 in FIG. 3X is able to provide additional controls. Here, multiple configurable dials 312 (as shown, three) are spaced apart along the front housing 302, with two additive power/data ports 314 arranged in the spaces between the configurable dials 312. In an example, the configurable dials 312 of this modular programmable widget 116 may provide a set of predefined second row controls, such as volume, temperature, and fan speed controls.

It should also be noted that, in some examples, the modular programmable widgets 116 may be branded with a logo 354 or other mark of a brand. This may allow, for example, the modular programmable widget 116 to be offered with hard-coded co-branded functions that relate to a third party. For instance, when pressed, the modular programmable widget 116 may provide co-branded access to a third-party product, invoke a function of the third-party product, activate a game function of the third-party product, and/or show content from the third-party product, as some non-limiting examples.

FIG. 4A illustrates an example exploded view of the modular programmable widget 116 shown in FIG. 3A having a configurable button 308 with a display 310. As mentioned above, the modular programmable widget 116 includes the front housing 302 and back housing 304. Here, the back housing 304 defines the rear wall and a portion of the side walls of a right rectangular prism shape, while the front housing 302 defines the remaining front face and side wall portions of the rectangular prism.

The modular programmable widget 116 further includes an internal magnet 402 configured to retain the modular programmable widget 116 in position. In an example, the

12

inside rear wall of the back housing 304 defines a magnet recess 404 lengthwise across the rear wall sized to receive the internal magnet 402 and hold the internal magnet 402 in position. As shown, the internal magnet 402 and magnet recess 404 are each of a generally rectangular shape, but variations in shape are possible.

The modular programmable widget 116 further includes a main circuit board 406 configured to physically hold the electronic components of the modular programmable widget 116. For example, the main circuit board 406 may hold the components responsible for data communication with the vehicle 102 and/or receiving power from the vehicle 102. The main circuit board 406 may also hold components supporting the specific functionality of the modular programmable widget 116. The main circuit board 406 may also define circuit traces and/or wires to electrically connect the electrical components to one another.

In one aspect, the main circuit board 406 may be physically and electrically connected to components to allow for the electrical connection of the modular programmable widget 116 to the dock 202. For example, a flexible ribbon 408 may be used to connect the main circuit board 406 to the contact pad 306 that are exposed by the modular programmable widget 116. A first end of the flexible ribbon 408 may terminate in the contact pads 306 allowing for electrical connection of the modular programmable widget 116 to the docks 202. The modular programmable widget 116 may further include a contact mount 410 connected to the second end of the flexible ribbon 408. The contact mount 410 may provide mechanical rigidity to the contact pad 306 and may also slide into and be retained into a contact slot 412 formed into the inside rear wall of the back housing 304.

The back housing 304 of the modular programmable widget 116 may define a generally flat vertical surface of the rear of the modular programmable widget 116. An attachment body 414 may form a frustrum or other shape extending rearwards from the back housing 304. In an example, the attachment body 414 may extend rearwards from the top of the vertical rear wall of the back housing 304, then tapering towards the lower portion of the rear face of the back housing 304.

Due to the greater depth provided at the upper side of the modular programmable widget 116, a contact window 416 may be defined by the top of the attachment body 414, above the contact slot 412, and may be configured to expose the contact pads 306 through the top of the back housing 304 for electrical connection to the dock 202. The attachment body 414 may further serve to lock the modular programmable widget 116 into place in the dock 202, as shown in further detail with respect to FIGS. 8A-8C.

The front housing 302 may define a front window 418 defining a location for mounting of the controls of the modular programmable widget 116. In the example shown in FIG. 4A, the modular programmable widget 116 is a configurable button 308, and the front window 418 is a sized to receive the components of the configurable button 308.

The top face of the front housing 302 may further define a locating groove 420 providing a concave surface along the length of the top face of the front housing 302. The locating groove 420 may aid in aligning, locating, and/or locking the modular programmable widget 116 into the dock 202.

The configurable button 308 may be formed of a stack of components including a display board 422, the display 310 itself, a display lens 424, and a button bezel 426. The display board 422 may be configured to hold the display 310, as well as to provide for mounting of a flexible ribbon 428 to the display 310. The display board 422 may further include the

13

switch components configured to sense whether the configurable button 308 mechanism is being pressed by a user. The flexible ribbon 428 may be connected to the main circuit board 406 and used to communicate information such as the button pressed state and information indicative of what to display to the display 310.

The display lens 424 may be configured to protect the display 310 and may be formed of a flat disk of a material allowing the passage of light, such as a clear plastic or glass. The button bezel 426 may be formed as a disk of material sized to surround the display lens 424. The front window 418 of the front housing 302 may define a flange 430, spaced away from the front face of the front housing 302, of the same thickness as the button bezel 426. Thus, when assembled, the front of the display lens 424 and the button bezel 426 may be flush with the front surface of the front housing 302.

To facilitate assembly of the modular programmable widget 116, one or more fasteners 432, such as screws or pins, may be used to hold the back housing 304 and front housing 302 together. In an example, through holes 434 may be formed through the back of the back housing 304 which may be aligned with through holes 436 through the main circuit board 406, and finally fixed into apertures (e.g., threaded holes, not shown) into the rear face of the front housing 302. It should be noted that other approaches to assembling the modular programmable widgets 116 may be used in other examples, such as snap fit connectors and/or glue.

FIG. 4B illustrates an example exploded view of the modular programmable widget 116 shown in FIG. 3B having a configurable dial 312 with a display 310. Similar to as shown in FIG. 4A, the configurable dial 312 including components such as the front housing 302, back housing 304, internal magnet 402, main circuit board 406, flexible ribbon 428, contact mount 410, display 310, display board 422, display lens 424, and flexible ribbon 428.

To support the functionality of the dial, the modular programmable widget 116 may also include a rotary encoder 438 mounted to the main circuit board 406. The rotary encoder 438 may be configured to convert rotational movement and/or speed of its encoder shaft 440 into an electrical signal, using various mechanical, optical, and/or magnetic techniques. An encoder stop 442 may surround the rotary encoder 438, and may serve to provide spacing between the main circuit board 406 and the front housing 302. The front window 418 may be sized to the diameter of the encoder shaft 440 of the rotary encoder 438. When assembled, the encoder shaft 440 may be configured to be inserted outwards through the front window 418 of the front housing 302 until the rear face of the front housing 302 abuts the front face of the encoder stop 442.

An encoder bearing 444 may be pressed onto the encoder shaft 440 protruding from the front window 418. The encoder bearing 444 may be of a hollow cylindrical shape, such as a washer, defining a bearing aperture 445 through its hollow center of a diameter sufficient to engage the outer cylindrical face of the encoder shaft 440. The encoder bearing 444 may connect to and spin along with the encoder shaft 440. To prevent the encoder bearing 444 from slipping with respect to the encoder shaft 440, in some examples the encoder shaft 440 may provide one or more detents or protrusions configured to be received into corresponding features of the interior of the encoder bearing 444.

A display housing 446 may be configured to hold the stationary components of the modular programmable widget 116 that do not rotate with the rotary encoder 438. For

14

example, the display housing 446 may define a generally cylindrical outer housing 448 configured to hold the display board 422, display 310, and display lens 424. The outer housing 448 may define one or more openings into its circumference to receive corresponding attachment points 450 extending radially outwards from the body of the display board 422. This may allow the display board 422 to be mounted in a fixed position within the outer housing 448.

The display housing 446 may further define a wire holder 452, extending from the outer housing 448 into the interior of the hollow encoder shaft 440. The wire holder 452 may define an opening through its center to retain and guide the flexible ribbon 428 between the display 310 and the main circuit board 406, through the interior of the rotary encoder 438.

A rotary dial 454 may be sized to receive the encoder bearing 444 and to provide a surface for manipulation of the rotary encoder 438 by a user. Once the display board 422, display 310, and display lens 424 components are mounted within the display housing 446, the rotary dial 454 may be snapped onto the exterior of the encoder bearing 444. The rotary dial 454 may be rotatable by the user to allow for rotation of the rotary encoder 438 and may also serve as a bezel surrounding and protecting the display 310 components.

FIG. 4C illustrates an example exploded view of the modular programmable widget 116 shown in FIG. 3C having additive power/data ports 314. Similar to as shown in FIGS. 4A-4B, the modular programmable widget 116 as shown includes components such as the front housing 302, back housing 304, internal magnet 402, main circuit board 406, flexible ribbon 428, and contact mount 410.

Additionally, this modular programmable widget 116 may include one or more power/data ports 314 mounted to the main circuit board 406. The front housing 302 may also define port apertures 456 corresponding to the locations of the power/data ports 314. This accordingly allows for the power/data ports 314 to be accessible for use in the assembled modular programmable widget 116.

FIG. 4D illustrates an example exploded view of the modular programmable widget 116 shown in FIG. 3D having a configurable touch screen 316. Again, similar to as shown in FIGS. 4A-4C, the modular programmable widget 116 having the configurable touch screen 316 includes components such as the front housing 302, back housing 304, internal magnet 402, main circuit board 406, flexible ribbon 428, and contact mount 410.

Additionally, this modular programmable widget 116 includes a display 310 and display lens 424, although in this example of a longer rectangular shape. The front window 418 of the front housing 302 may be sized to a rectangular shape covering the bulk of the front of the modular programmable widget 116. The front window 418 may also define a flange 430 for receiving the display lens 424, similar to as discussed above. The flexible ribbon 428 may, in this example, connect the display 310 to the main circuit board 406, without requiring an intermediate display board 422.

FIG. 5 illustrates an example exploded view of a dock 202 for receiving the modular programmable widgets 116. The illustrated dock 202 may be the dock 202A of the center stack configurable display 114, in an example. As shown, the dock 202 includes a main substrate 502, a main dock board 504, an upper housing 506, a ferrous metal strip 508, and a lower housing 510.

The main substrate 502 may define a securing channel 512 for receiving the modular programmable widgets 116. The securing channel 512 may be sized to conform with the

15

profile of the attachment body **414** of the rear of the modular programmable widgets **116**. The main dock board **504** may expose dock connectors **514** extending downward from its lower surface. The dock connectors **514** may be configured for receiving the contact pads **306** of the modular programmable widgets **116**. These dock connectors **514** may be sized to fit within contact apertures **516** in the upper face of the securing channel **512**.

The securing channel **512** may further define locators **518** extending downwards into the securing channel **512**, on either side of each of the dock connectors **514**. These locators **518** may serve to aid in the placement of the modular programmable widgets **116** in positions where the contact pad **306** of the modular programmable widget **116** is aligned with the dock connector **514**.

The upper end of the securing channel **512** may define a protrusion **520** matching the profile of the concave locating groove **420** in the top of the front housing **302** of the modular programmable widget **116**. Here the protrusion **520** is a rounded shape, with generally the same diameter as the concavity of the locating groove **420**. This protrusion **520** may serve as a boss to guide rotation of the modular programmable widget **116** into position to be snapped into place, as shown in further detail in FIGS. 8A-8C. When installed into place, the contact pads **306** may be electrically connected to the dock connectors **514**, which in turn connect to a cable **522** for connection to the widget controller **112**. The upper housing **506** may be configured to fit over the main substrate **502** and the main dock board **504**, to provide a smooth, finished appearance to the front of the dock **202**.

Below the main substrate **502**, the ferrous metal strip **508** may define an elongated, generically rectangular magnetic surface. The ferrous metal strip **508** may be mounted below the securing channel **512** and may be attracted to by the internal magnets **402** of the modular programmable widget **116** to secure the modular programmable widget **116** into place. The lower housing **510** may be configured to fit below the main substrate **502** and the ferrous metal strip **508**, to provide a smooth, finished appearance to the bottom of the dock **202**.

FIG. 6 illustrates a side cutaway view of the dock **202** of FIG. 6 in an assembled state. FIG. 7 illustrates a side cutaway view of the modular programmable widget **116** of FIG. 4B in an assembled state. As shown, the components of the dock **202** and modular programmable widgets **116** includes aspects to facilitate their selective mechanical and electrical connection.

FIG. 8A illustrates a first operation A in the attachment of an example modular programmable widget **116** to the dock **202**. As shown, the operation A includes moving the modular programmable widget **116** towards the dock **202**, such that the locating groove **420** of the modular programmable widget **116** is aligned with the protrusion **520** of the dock **202**.

FIG. 8B illustrates a second operation B in the attachment of an example modular programmable widget **116** to the dock **202**. As shown, the operation B includes, once the locating groove **420** is mated to the protrusion **520**, pivoting the modular programmable widget **116** around the protrusion **520** such that the attachment body **414** of the modular programmable widget **116** moved into the securing channel **512**. In doing so, the contact pad **306** of the modular programmable widget **116** moves towards electrical contact with the dock connector **514** of the dock **202**.

FIG. 8C illustrates a completed attachment of an example modular programmable widget **116** to the dock **202**. As shown, the attachment body **414** is mated into the inner

16

surface of the securing channel **512**, and the contact pad **306** of the modular programmable widget **116** is in electrical contact with the dock connector **514** of the dock **202**. Also, the internal magnet **402** of the modular programmable widget **116** is attracted to the ferrous metal strip **508** within the dock **202**, providing force to keep the modular programmable widget **116** in place within the dock **202**.

Significantly, despite the differences in controls and size of the modular programmable widgets **116**, the modular programmable widgets **116** each define the same connection components. For instance, each of the modular programmable widgets **116** has the same electrical contact pads **306** and locating groove **420** as discussed above. Thus, any of the modular programmable widgets **116** may be connected to the dock **202**, allowing for a user to choose whichever modular programmable widget **116** has the desired controls and/or appearance.

FIG. 9 illustrates an example cutaway view of a vehicle interior **200** supporting the modular programmable widgets **116** having an alternate packaging design. Similar to as shown before, one or more first modular programmable widgets **116** may be attached to a dock **202A** of the center stack configurable display **114**. This location under the center stack configurable display **114** may be easily accessible for driver and passenger. Also shown, one or more second modular programmable widgets **116** may be attached to a console bin dock **202B**. Here, the second modular programmable widget **116** is attached to the console bin dock **202B** for access by rear passengers. As further shown, one or more third modular programmable widgets **116** may be attached to a cargo dock **202C** in the rear cargo area. It should also be noted that the illustrated vehicle interior **200** is an example, and interiors having more, fewer, or differently located components may be used.

FIGS. 10A-10I illustrate example modular programmable widgets **116** in the alternative packaging. Referring collectively to FIGS. 10A-10I, an enclosure having a generally rectangular prism shape may be defined by a front housing **302** and a back housing **304**. The front housing **302** may define a generally flat front surface of the modular programmable widget **116**, while the back housing **304** may define the remainder of the enclosure, including the rear, side, and top walls that house the internal circuitry and other components of the modular programmable widget **116**. Controls may be provided on the front housing **302**, while connectivity to the vehicle **102** may be provided via contact pads **306** exposed through the top of the back housing **304**. The contact pads **306** may allow for transmission of data and power between the modular programmable widgets **116** and the dock **202**. In an example, the contact pads **306** may be leaf spring contacts.

The front housing **302** may expose various controls that may provide interaction with the user. In the example of FIG. 10A, this may include a configurable button **308**. While not shown, in some examples the configurable button **308** may include a display **310**. FIG. 10B illustrates an example modular programmable widget **116** having a configurable dial **312**. FIG. 10C illustrates an example modular programmable widget **116** having additive power/data ports **314**. As shown, the modular programmable widgets **116** in FIG. 10C provides two USB power/data ports **314**, but this is merely an example, and different quantities and types of connectors may be exposed by the modular programmable widgets **116**. FIG. 10D illustrates an example modular programmable widget **116** having a display **310**. As shown in this example,

17

the modular programmable widget **116** provides a flat display **310** on the front surface of the modular programmable widget **116**.

Further examples are possible, such as discussed above. For instance, FIG. **10E** illustrates an example modular programmable widget **116** having a configurable joystick **334**. FIG. **10F** illustrates an example modular programmable widget **116** having a trailer control **342**. FIG. **10G** illustrates an example modular programmable widget **116** having toggle button **318**. FIG. **10H** illustrates an example modular programmable widget **116** having a companion **352**.

It should be noted that the modular programmable widgets **116** shown in FIGS. **10A-10H** share the same exterior dimensions. This standard size allows for easy addition and removal of the modular programmable widgets **116** from the docks **202**. In other cases, however, the modular programmable widgets **116** may include modular programmable widgets **116** of a larger physical size. Such modular programmable widgets **116** may be able to include more controls as compared to the default size of the modular programmable widgets **116**.

FIG. **10I** illustrates an example modular programmable widget **116** that is multiple times the length of a standard modular programmable widget **116** as shown in FIGS. **10A-10H**. In this example, the modular programmable widget **116** is three times the length of the standard modular programmable widgets **116**. Due to the increased length, the modular programmable widget **116** in FIG. **10I** is able to provide multiple configurable dials **312** (here three) spaced apart along the front housing **302**, with two additive power/data ports **314** arranged in the spaces between the configurable dials **312**. In an example, the configurable dials **312** of this modular programmable widget **116** may provide a set of predefined second row bundle of controls, such as volume, temperature, and fan speed controls. For example, the main circuit board **406** of the modular programmable widget **116** may be preprogrammed to control those aspects by communication with the widget controller **112** when the modular programmable widget **116** is electrically connected to the dock **202**.

FIG. **11** illustrates an example side view of the modular programmable widget **116** of FIG. **10B** having a configurable dial **312**. FIG. **12** illustrates an example rear perspective view of the modular programmable widget **116** of FIG. **10B** having a configurable dial **312**. As collectively shown in FIGS. **11-12**, the back housing **304** of the modular programmable widget **116** may define a generally flat vertical surface of the rear of the modular programmable widget **116**.

The back housing **304** may further define the attachment body **414** extending rearward from the vertical surface of the rear wall. In an example the attachment body **414** may form a frustrum or other shape extending rearwards from the back housing **304**. In an example, the attachment body **414** may extend rearwards in the upper portion of the back housing **304** and may then taper towards the flat back of the rear face of the back housing **304**.

Due to the greater depth provided by the attachment body **414** at the upper portion of the modular programmable widget **116**, a contact window **416** may be defined by the top of the attachment body **414**, above the contact slot **412**, and may be configured to expose the contact pads **306** through the top of the back housing **304** for electrical connection to the dock **202**. The attachment body **414** may further serve to lock the modular programmable widget **116** into place in the dock **202**, as shown in further detail with respect to FIGS. **14-16**.

18

FIG. **13** illustrates an example exploded view of the modular programmable widget **116** shown in FIG. **10B** having a configurable dial **312**. As mentioned above, the modular programmable widget **116** includes the front housing **302** and back housing **304**. The back housing **304** of the modular programmable widget **116** may define a generally flat vertical surface of the rear of the modular programmable widget **116**. An attachment body **414** may form a frustrum or other shape extending rearwards from the back housing **304**. In an example, the attachment body **414** may extend rearwards from the top of the vertical rear wall of the back housing **304**, then tapering towards the lower portion of the rear face of the back housing **304**.

The modular programmable widget **116** further includes an internal magnet **402** configured to retain the modular programmable widget **116** in position. In an example, the inside rear wall of the back housing **304** defines a magnet recess **404** lengthwise across the rear wall, sized to receive the internal magnet **402** and hold the internal magnet **402** in position. As shown, the internal magnet **402** and magnet recess **404** are each of a generally rectangular shape.

The modular programmable widget **116** further includes a main circuit board **406** configured to physically hold the electronic components of the modular programmable widget **116**. For example, the main circuit board **406** may hold the components responsible for data communication with the vehicle **102** and/or receiving power from the vehicle **102**. The main circuit board **406** may also hold components supporting the specific functionality of the modular programmable widget **116**. The main circuit board **406** may also define circuit traces and/or wires to electrically connect the electrical components to one another.

In one aspect, the main circuit board **406** may be physically and electrically connected to components to allow for the electrical connection of the modular programmable widget **116** to the dock **202**. In an example, the top of the main circuit board **406** may hold the contact pads **306** configured to provide electrical connection to the dock **202**. The contact pads **306** may slide into a contact slot **412** formed into the top of the attachment body **414** of the back housing **304**.

As opposed to being defined by the front housing **302** as shown earlier, in the alternate packaging, the top face of the back housing **304** defines the locating groove **420** providing a concave surface along the length of its top face. Despite the different assembly, this locating groove **420** functions similarly, however to may aid in locating and locking the modular programmable widget **116** into the dock **202**.

To support the functionality of the dial, the modular programmable widget **116** may also include a rotary encoder **438** mounted to the main circuit board **406**. The rotary encoder **438** may be configured to convert rotational movement and/or speed of its encoder shaft **440** into an electrical signal, using various mechanical, optical, and/or magnetic techniques.

The front housing **302** may define a front window **418** defining a location for mounting of the controls of the modular programmable widget **116**. In the example shown in FIG. **11**, the modular programmable widget **116** is a configurable dial **312**, and the front window **418** is sized to the diameter of assembled rotary encoder **438**. The encoder shaft **440** may be configured to be inserted outwards through the front window **418** of the front housing **302**.

A switch **1302** may be mounted onto the main circuit board **406** (better seen in FIG. **16**) within a cylindrical aperture **3312** defined by the rotary encoder **438**. The switch **1302** may have a front-facing button mounted to the main

19

circuit board **406** that, when pressed, activates the switch **1302** and that when unpressed or released, deactivates the switch **1302**.

A signifier cap **1306** may define a front center surface of the rotary encoder **438** and may be pressed by the user to activate the switch **1302**. The signifier cap **1306** may define a generally flat, circular shape, having a customizable front face **1308**. The customizable front face **1308** may include various content, such as an icon indicating a function to be performed by pressing the signifier cap **1306**.

To allow the signifier cap **1306** to reach the switch **1302**, a switch stem **1310** may be configured to transfer pressure provided by a user to the signifier cap **1306** onto the switch **1302** mounted within the rotary encoder **438**. The switch stem **1310** may define an elongate body with an inner end aimed towards the button to selectively transfer force onto the button. The outer end of the switch stem **1310** may define a stem receptacle **1312** for receiving a corresponding pin **1314** extending towards the switch stem **1310** from the inner face of the signifier cap **1306** opposite the customizable front face **1308**. In one non-limiting example, the stem receptacle **1312** and pin **1314** may be of a triangular cross-sectional shape.

A switch guide **1316** may be installed within the aperture **3312**, to limit the switch stem **1310** to linear motion towards or away from the switch **1302**. The switch guide **1316** may be defined by a circular outer surface sized to allow for insertion into the aperture **3312** as well as a guide aperture **1318** open to receive the elongate body of the switch stem **1310**. The switch guide **1316** may remain stationary, while the aperture **3312** may slide inwards responsive to the user pressing the signifier cap **1306**. In an example, the switch stem **1310** and the guide aperture **1318** may each define a generally square cross-section sized to allow the switch stem **1310** to move linearly in and out, but not rotate or otherwise move. Thus, when the signifier cap **1306** is pressed, the signifier cap **1306** pushes the switch stem **1310** inwards, activating the switch **1302**.

An encoder bearing **444** may be pressed onto the encoder shaft **440** protruding from the front window **418**. The encoder bearing **444** may spin with the encoder shaft **440**, although the signifier cap **1306** may not, allowing the signifier cap **1306** to preserve its upright orientation.

A rotary dial **454** (sometime referred to as a rotary cap) may be sized to receive the encoder bearing **444** and signifier cap **1306** and to provide a surface for manipulation of the rotary encoder **438** by a user. The rotary dial **454** may be snapped onto the exterior of the encoder bearing **444**. The rotary dial **454** may be rotatable by the user to allow for rotation of the rotary encoder **438** and may also serve as a bezel surrounding and protecting the rotary encoder **438** components.

The modular programmable widget **116** may also include lighting features. For example, the main circuit board **406** may include one or more light modules **1320**. These light modules **1320** may be LEDs, in an example, that may be configured to display different colors and/or intensities of light. The front housing **302** may include a light ring **1322** around the front window **418** to act as a light guide to direct the light from the light modules **1320** for providing illumination around the perimeter of the front window **418**.

FIG. 14 illustrates an example front view of the dock **202** for receiving the modular programmable widgets **116**. A modular programmable widget **116** having a design as shown in FIG. 13 shown along with the dock **202**. To facilitate the attachment, the contact pad **306** exposed by the

20

attachment body **414** is aligned with the middle dock connector **514** of the dock **202**.

FIG. 15A illustrates perspective and side views of a first operation in the attachment of an example alternate design modular programmable widget **116** to the dock **202**. As shown, the first operation includes moving the modular programmable widget **116** towards the dock **202**, such that the locating groove **420** of the modular programmable widget **116** is aligned with the protrusion **520** of the dock **202**.

FIG. 15B illustrates perspective and side views of a second operation in the attachment of an example alternate design modular programmable widget **116** to the dock **202**. As shown, the second operation includes, once the locating groove **420** is mated to the protrusion **520**, pivoting the modular programmable widget **116** around the protrusion **520** such that the attachment body **414** of the modular programmable widget **116** moved into the securing channel **512**. In doing so, the contact pad **306** of the modular programmable widget **116** moves towards electrical contact with the dock connector **514** of the dock **202**.

FIG. 15C illustrates perspective and side views of a completed attachment of an example alternate design modular programmable widget **116** to the dock **202**. Once installed, the attachment body **414** is mated into the inner surface of the securing channel **512**, and the contact pad **306** of the modular programmable widget **116** is in electrical contact with the dock connector **514** of the dock **202**. Also, the internal magnet **402** of the modular programmable widget **116** is attracted to a ferrous metal strip **508** within the dock **202**, providing force to keep the modular programmable widget **116** in place within the dock **202**.

FIG. 16 illustrates a perspective view of the dock **202A** integrated below a configurable display **114** of the vehicle **102** for attachment of the modular programmable widgets **116**. In this example, the securing channel **512** of the dock **202** is collectively defined by a housing surface **1602** of the configurable display **114**, a rear surface **1604** extending downwards from the housing surface **1602** spaced away from the front of the configurable display **114**, and a lower dock housing **1606** completing the lower portion of the securing channel **512**. The dock connectors **514** may accordingly be exposed through contact apertures **516** defined by housing surface **1602**.

The housing surface **1602** may further define locators **518** spaced on either side of each of the contact apertures **516**. These locators **518** may extend downwards into the securing channel **512** from the housing surface **1602**, on either side of each of the dock connectors **514**. These locators **518** may serve to aid in the placement of the modular programmable widgets **116** in positions where the contact pad **306** of the modular programmable widget **116** is aligned with the dock connector **514**. As shown, the locators **518** define a triangular shape that widens into the depth of the securing channel **512**. This triangular shape may conform to the tapering of the frustrum shape of the attachment body **414**, serving to guide the modular programmable widget **116** into position when inserted (such as shown in FIGS. 15A-15C).

Thus, despite the differences in controls and size of the modular programmable widgets **116**, the modular programmable widgets **116** each define the same connection components. For instance, each of the modular programmable widgets **116** has the same electrical contact pads **306** and locating groove **420** as discussed above. Thus, any of the modular programmable widgets **116** may be connected to

21

the dock **202**, allowing for a user to choose whichever modular programmable widget **116** has the desired controls and/or appearance.

FIG. **17** illustrates a side cutaway view of the modular programmable widget **116** shown in FIGS. **8B** and **12-13**, in assembled form and attached to the dock **202**. As shown, the components of the dock **202** and modular programmable widgets **116** includes aspects to facilitate their selective mechanical and electrical connection.

FIGS. **18-20** collectively illustrate further aspects of the customizable front face **1308** of the signifier cap **1306**. Referring to FIG. **18**, the figure illustrates the removal of the rotary dial **454** from an assembled modular programmable widget **116** to allow for changing out of the signifier cap **1306**. As shown, the rotary dial **454** has been slid off the encoder bearing **444**, allowing the signifier cap **1306** to be pulled off of the stem receptacle **1312**.

FIG. **19** illustrates a selection of different signifier caps **1306**, each having a different customizable front face **1308**. For instance, the signifier cap **1306A** has a customizable front face **1308A** having an icon representation of a function (here a fan). The signifier cap **1306B** has a customizable front face **1308B** of a customized color. The signifier cap **1306C** has a customizable front face **1308C** of a customized material or texture. These signifier caps **1306** may be freely changeable by the user, allowing for customization of the look of the modular programmable widgets **116**.

FIG. **20** illustrates an example selection of different signifier caps **1306** installed to various modular programmable widgets **116**. As shown, the different signifier caps **1306** expose different customizable front faces **1308**.

FIGS. **21A-21D** collectively illustrates aspects of a mobile dock **202**. FIG. **21A** illustrates a front perspective view of an empty mobile dock **202** without attachment of any modular programmable widgets **116**. FIG. **21B** illustrates a front perspective view of the mobile dock **202** in the process of attachment of modular programmable widgets **116**. FIG. **21C** illustrates a front perspective view of the mobile dock **202** having modular programmable widgets **116** attached. FIG. **21D** illustrates a rear perspective view of the mobile dock **202** having modular programmable widgets **116** attached.

In general, the mobile dock **202** may be configured to receive modular programmable widgets **116**, using a mating geometry of the attachment body **414** with the securing channel **512** and the internal magnets **402**, consistent with attachment of the modular programmable widgets **116** to the in-vehicle docks **202**. The mobile dock **202** may include wireless communications functionality (e.g., BLUETOOTH), and an internal battery, which may allow the mobile dock **202** to be used for remote control and/or game controller use cases with the vehicle **102** (or with other wireless devices such as smartphones, tablets, video game consoles, etc.). For instance, mobile dock **202** may sync with the configurable display **114** of the vehicle **102** for gaming and entertainment (although the main display may be restricted unless the vehicle **102** is in Park).

The mobile dock **202** may comprise a front housing **2102** and a rear housing **2104**. The front housing **2102** and rear housing **2104** may collectively define an enclosure of a size suitable for hand-held operation. The enclosure may have rounded corners and/or faces to allow for comfortable holding in the user's hands. As shown, the length of the dock **202** allows for a quantity of modular programmable widgets **116** to be received. The illustrated dock **202** has length for receiving two modular programmable widgets **116** but docks

22

202 of different sizes may allow for the insertion of more or fewer modular programmable widgets **116**.

The front housing **2102** may define a top wall **2106**, opposing side walls **2108**, **2110**. With the area enclosed by the inner faces of the walls **2106**, **2108**, and **2110**, the front housing **2102** may define the securing channel **512** for attachment of the modular programmable widgets **116**.

Consistent with the other docks **202**, the locating groove **420** may be sized to conform with the profile of the attachment body **414** of the rear of the modular programmable widgets **116**. The mobile dock **202** may include dock connectors **514** in the locating groove **420** configured for receiving the contact pads **306** of the modular programmable widgets **116**, sized to fit within contact apertures **516** in the upper face of the locating groove **420**. The securing channel **512** may also define locators **518** extending downwards into the securing channel **512**, on either side of each of the dock connectors **514**, again to aid in the placement of the modular programmable widgets **116** in positions where the contact pad **306** of the modular programmable widget **116** is aligned with the dock connector **514**. The upper end of the securing channel **512** may define a protrusion **520** matching the profile of the concave locating groove **420** in the top of the front housing **302** of the modular programmable widget **116**. Additionally, a ferrous metal strip **508** may be mounted within the mobile dock **202**, below the securing channel **512**, and may be attracted to by the internal magnets **402** of the modular programmable widget **116** to secure the modular programmable widget **116** into place.

As the mobile dock **202** is portable and not connected to the vehicle **102**, the exterior faces of the front housing **2102** and/or rear housing **2104** may be able to provide fixed controls, independent of the removable controls of the modular programmable widgets **116**. As shown, the top of the front housing **2102** defines two control buttons **2112** which may always be available, regardless of whether modular programmable widgets **116** are installed to the mobile dock **202**.

FIG. **22** illustrates an example cutaway view of a vehicle interior **200** supporting the modular programmable widgets **116** having another alternative alternate packaging design. Similar to as shown before, one or more first modular programmable widgets **116** may be attached to a dock **202A** of the center stack configurable display **114**. This location under the center stack configurable display **114** may be easily accessible for driver and passenger. Also shown, one or more second modular programmable widgets **116** may be attached to a console bin dock **202B**. Here, the second modular programmable widget **116** is attached to the console bin dock **202B** for access by rear passengers. As further shown, one or more third modular programmable widgets **116** may be attached to a cargo dock **202C** in the rear cargo area. Also shown, in this variation one or more fourth modular programmable widgets **116** may be attached to a gauge cluster dock **202D** in the front dash area adjacent the steering wheel. It should also be noted that the illustrated vehicle interior **200** is an example, and interiors having more, fewer, or differently located components may be used.

FIGS. **23A-23F** illustrate example modular programmable widgets **116** in the second alternative packaging. Referring collectively to FIGS. **23A-23F**, an enclosure having a generally rectangular prism shape may be defined by a front housing **302** and a back housing **304**. The front housing **302** may define a generally flat front surface of the modular programmable widget **116**, while the back housing **304** may define the remainder of the enclosure, including the rear, side, and top walls that house the internal circuitry and

23

other components of the modular programmable widget 116. Controls may be provided on the front housing 302, while connectivity to the vehicle 102 may be provided via a magnetic contact pad 306 exposed through the top of the back housing 304. The magnetic contact pad 306 may allow for transmission of data and power between the modular programmable widgets 116 and the dock 202. In this example, the contact pad 306 may be a magnetic universal serial bus (USB) jack.

The front housing 302 may expose various controls that may provide interaction with the user. In the example of FIG. 23A, this may include a configurable button 308. While not shown in FIG. 10A, in some examples the configurable button 308 may include a display 310. FIG. 23B illustrates an example modular programmable widget 116 having a configurable dial 312. FIG. 10C illustrates an example modular programmable widget 116 having additive power/data ports 314. As shown, the modular programmable widgets 116 in FIG. 23C provides two USB power/data ports 314, but this is merely an example, and different quantities and types of connectors may be exposed by the modular programmable widgets 116. FIG. 23D illustrates an example modular programmable widget 116 having a display 310. As shown in this example, the modular programmable widget 116 provides a flat display 310 on the front surface of the modular programmable widget 116. FIG. 23E illustrates an example modular programmable widget 116 having a companion 352. FIG. 23F illustrates an example perspective view of a modular programmable widget 116 having a configurable dial 312 with configurable display 114.

Similar to as noted previously, the modular programmable widgets 116 shown in FIGS. 23A-23F share the same exterior dimensions. This standard size allows for easy addition and removal of the modular programmable widgets 116 from the docks 202. In other cases, however, the modular programmable widgets 116 may include modular programmable widgets 116 of a larger physical size. Such modular programmable widgets 116 may be able to include more controls as compared to the default size of the modular programmable widgets 116.

FIG. 23G illustrates an example modular programmable widget 116 that is multiple times the length of a standard modular programmable widget 116 as shown in FIGS. 23A-23F. In this example, the modular programmable widget 116 is three times the length of the standard modular programmable widgets 116. Due to the increased length, the modular programmable widget 116 in FIG. 23G is able to provide multiple configurable dials 312 (here three) spaced apart along the front housing 302, with two additive power/data ports 314 arranged in the spaces between the configurable dials 312. In an example, the configurable dials 312 of this modular programmable widget 116 may provide a set of predefined second row bundle of controls, such as volume, temperature, and fan speed controls. For example, the main circuit board 406 of the modular programmable widget 116 may be preprogrammed to control those aspects by communication with the widget controller 112 when the modular programmable widget 116 is electrically connected to the dock 202.

It should also be noted that, as this example modular programmable widget 116 mechanically and electrically connects via magnetic USB, the modular programmable widget 116 includes multiple contact pads 306 for connection to the dock 202 (e.g., three in this example). This use of multiple magnetic connections may allow for better securing of the modular programmable widget 116 to the dock 202.

24

FIG. 24 illustrates an example exploded view of the modular programmable widget 116 shown in FIG. 23A having a configurable button 308. Similar to as mentioned above, these modular programmable widgets 116 includes the front housing 302 and back housing 304. The back housing 304 of the modular programmable widget 116 may define a generally flat vertical surface of the rear of the modular programmable widget 116. Here, instead of an attachment body 414, the back housing 304 instead defines a horizontal channel 2402 extending along the length of the back housing 304. As shown, the horizontal channel 2402 is placed along the middle of the back housing 304.

The modular programmable widget 116 further includes an internal magnet 402 configured to retain the modular programmable widget 116 in position. In an example, the inside rear wall of the back housing 304 defines a magnet recess 404 lengthwise across the rear wall, sized to receive the internal magnet 402 and hold the internal magnet 402 in position. As shown, the internal magnet 402 and magnet recess 404 are each of a generally rectangular shape.

The modular programmable widget 116 further includes a main circuit board 406 configured to physically hold the electronic components of the modular programmable widget 116. For example, the main circuit board 406 may hold the components responsible for data communication with the vehicle 102 and/or receiving power from the vehicle 102. The main circuit board 406 may also hold components supporting the specific functionality of the modular programmable widget 116. The main circuit board 406 may also define circuit traces and/or wires to electrically connect the electrical components to one another.

In one aspect, the main circuit board 406 may be physically and electrically connected to components to allow for the electrical connection of the modular programmable widget 116 to the dock 202. In an example, the top of the main circuit board 406 may hold the contact pad 306 configured to provide electrical connection to the dock 202. In this illustrated example, the contact pad 306 is a magnetic USB jack. The magnetic USB jack may have a rounded rectangular profile, and as shown is mounted lengthwise along the top of the modular programmable widget 116.

The contact pad 306 may slide into a contact slot 412 formed into the top of the attachment body 414 of the back housing 304. In this example, the contact slot 412 may be formed in two portions—a rear contact slot portion 412A defined by the upper face of the back housing 304 and conforming to the contour of a rear portion of the magnetic USB jack, and a front contact slot portion 412B defined by the upper face of the front housing 302 and conforming to the contour of a front portion of the magnetic USB jack. Collectively, the rear contact slot portion 412A and the front contact slot portion 412B may serve to hold the contact pad 306 in place when the modular programmable widget 116 is assembled.

The back housing 304 may define a main board cavity 2404 sized to receive the main circuit board 406 of the modular programmable widget 116. To ease placement of the main circuit board 406, the main circuit board 406 may hold a board locator 2406 extending from the main circuit board 406 and configured to be received into a corresponding indent 2408 in the back housing 304. Once located, the main circuit board 406 may be inserted into the main board cavity 2404.

The main circuit board 406 may contain the components in common to the different modular programmable widgets 116, such as the components supporting connectivity with the vehicle 102. The modular programmable widget 116

25

may further include a sister board **2410** configured to hold the controls that are specific to the type of the modular programmable widgets **116** and that may be exposed through the front of the modular programmable widget **116**.

The back housing **304** may further define a sister board cavity **2412** sized to receive the sister board **2410**. To ease placement of the sister board cavity **2412**, the sister board cavity **2412** may similarly hold a board locator **2414** extending from the sister board cavity **2412** and configured to be received into a corresponding indent **2416** in the back housing **304**. By use of the board locators **2406**, **2414**, the main circuit board **406** and the sister board **2410** may be secured in place in a spaced apart relationship. One or more fasteners **2420** (such as screws or rivets) placed through attachment through holes **2422** in the sister board **2410** and secured into connectors **2424** in the front side of the back housing **304**.

A switch **1302** may be mounted onto the front side of the sister board **2410**. The switch **1302** may have a front-facing button that, when pressed, activates the switch **1302** and that when unpressed or released, deactivates the switch **1302**.

A signifier cap **1306** may define a front center surface of the rotary encoder **438** and may be pressed by the user to activate the switch **1302**. The signifier cap **1306** may define a generally flat, circular shape, having a customizable front face **1308**. The customizable front face **1308** may include various content, such as an icon indicating a function to be performed by pressing the signifier cap **1306**.

The front housing **302** may define a front window **418** defining a location for mounting of the controls of the modular programmable widget **116**. In the example, the modular programmable widget **116** is a configurable button **308**, and the front window **418** is a sized to receive the signifier cap **1306** of the configurable button **308**. For instance, the signifier cap **1306** may define a flange extending radially outwards from the customizable front face **1308**, such that the customizable front face **1308** is visible through the front window **418** but the flange of the signifier cap **1306** is retained behind the front window **418**.

To facilitate assembly of the modular programmable widget **116**, one or more fasteners **432**, such as screws or pins, may be used to hold the back housing **304** and front housing **302** together. In an example, through holes **2422** may be formed through the back of the back housing **304** which may be aligned with apertures **2426** (e.g., threaded holes, not shown) into the rear face of the front housing **302**. It should be noted that other approaches to assembling the modular programmable widgets **116** may be used in other examples, such as snap fit connectors and/or glue. Dampening feet **2428** may be applied to the fasteners **432** (e.g., that fit into the screw heads of the fasteners **432**). The dampening feet **2428** may be a rubber material may serve to maintain the modular programmable widgets **116** in place when installed to the dock **202**.

FIG. 25 illustrates a perspective view of the dock **202A** integrated below a configurable display **114** of the vehicle **102** for attachment of the modular programmable widgets **116**. In this example, the dock **202** is collectively defined by a lower housing surface **1602** of the configurable display **114**, and a rear surface **1604** extending downwards from the housing surface **1602** spaced away from the front of the configurable display **114**. The dock connectors **514** may accordingly be exposed through contact apertures **516** defined through lower housing surface **1602**.

The dock **202** may define additional features for the alignment and securing of the modular programmable widgets **116** to the dock **202**. For instance, one or more dock

26

magnets **2502** may be mounted behind the rear surface **1604**, positioned to be attracted to the internal magnets **402** of the modular programmable widget **116** to secure the modular programmable widget **116** into place. The dock magnets **2502** may be placed in positions corresponding to the locations of the USB dock connectors **514**, such that the modular programmable widgets **116** are attracted into the docking positions.

Additionally, a mounting bar **2504** may be provided along the length of the front of the rear surface **1604**. This mounting bar **2504** may define a contour corresponding to contour of the horizontal channel **2402** and may serve to guide the modular programmable widget **116** to the proper height for connection of the magnetic USB contact pad **306** to the USB dock connector **514** in the docking position being used by the modular programmable widgets **116**.

FIG. 26A illustrates a side view of an operation in the attachment of an example modular programmable widget **116** to the dock **202**. As shown, the operation includes moving the modular programmable widget **116** towards the dock **202**, such that the horizontal channel **2402** of the modular programmable widget **116** is aligned with the mounting bar **2504** of the dock **202**. Additionally, the magnetic USB contact pad **306** of the modular programmable widget **116** is aligned with one of the dock connectors **514**.

FIG. 26B illustrates a side view of a completed attachment of an example modular programmable widget **116** to the dock **202**. As shown, the horizontal channel **2402** is mated with the mounting bar **2504**, and the contact pad **306** of the modular programmable widget **116** is in electrical and physical contact with the dock connector **514** of the dock **202**. Also, the internal magnet **402** of the modular programmable widget **116** is attracted to a dock magnet **2502** within the dock **202**, providing force to keep the modular programmable widget **116** in place within the dock **202**.

Thus, despite the differences in controls and size of the modular programmable widgets **116**, the modular programmable widgets **116** each define the same connection components. For instance, each of the modular programmable widgets **116** has the same electrical contact pads **306** and horizontal channel **2402** as discussed above. Thus, any of the modular programmable widgets **116** may be connected to the dock **202**, allowing for a user to choose whichever modular programmable widget **116** has the desired controls and/or appearance.

FIG. 27 illustrates an example of a home dock **202** configured for receiving the modular programmable widgets **116**. As shown, the home dock **202** may provide for connection of the modular programmable widgets **116** similar to the in-vehicle dock **202** discussed above with respect to FIGS. 25 and 26A-26B. The home dock **202** may similarly have dock magnets **2502**, a mounting bar **2504**, and dock connectors **514** for receiving the modular programmable widgets **116**. Here it can be seen that the dock connectors **514** allow for attachment of the modular programmable widgets **116** with the contact pads **306** on the bottom as opposed to the top, but this is only an example and home docks **202** having different designs are possible.

In an example use case, the home dock **202** may allow the modular programmable widgets **116** to be used for controlling the vehicle **102** from outside the vehicle **102**. For instance, the home dock **202** may include wireless connectivity configured to allow the home dock **202** to communicate with the vehicle **102**, e.g., via the telematics control module **130** of the vehicle **102**. This may allow the dock **202** to send commands to the widget controller **112** via the

27

telematics control module **130**, allowing the user to utilize the modular programmable widgets **116** to control the vehicle **102** as if the modular programmable widget **116** were connected to the in-vehicle dock **202**.

In another example use case, the home dock **202** may allow the modular programmable widgets **116** to be used for controlling smart home features independent of the vehicle **102**. In such an example, the home dock **202** may be connected to a computing device **2702**, such as a laptop or other computer, and may be used as additional input devices to the computing device **2702**. This may allow for the easy control of home features from the tactile user interface provided by the modular programmable widgets **116**.

FIGS. **28A-F** illustrates examples of various configurations of modular programmable widgets **116** for a dock **202** supporting five modular programmable widgets **116**. These examples illustrate some possibilities of how the modular programmable widgets **116** can evolve with user need or desire. Moreover, it should be noted that the modular programmable widgets **116** may also offer partnership opportunities with third parties and personalization opportunities for the user of the vehicle **102**.

FIG. **28A** illustrates an example of a first set of modular programmable widgets **116**, in a common hardcoded, individual configuration. In this example, the set includes five separate modular programmable widgets **116** having hardcoded functions. As shown, the modular programmable widgets **116** include a first modular programmable widget **116A** in the form of a dial **312** for operating a fan function; a second modular programmable widget **116B** in the form of a dial **312** for operating a temperature function; a third modular programmable widget **116C** in the form of a dial **312** for operating a volume function; a fourth modular programmable widget **116D** in the form of a button **308** for operating a lift gate function; and a fifth modular programmable widget **116E** in the form of additive power/data ports **314**.

FIG. **28B** illustrates an example of a second set of modular programmable widgets **116**, in a configurable, individual configuration. In this example, the set includes five separate modular programmable widgets **116** having configurable functions. As shown, the modular programmable widgets **116** include a first modular programmable widget **116A** in the form of a configurable dial **312**; a second modular programmable widget **116B** in the form of a configurable button **308**; a third modular programmable widget **116C** in the form of a configurable touch screen **316**; a fourth modular programmable widget **116D** in the form of a configurable button **308**; and a fifth modular programmable widget **116E** in the form of a configurable dial **312**.

FIG. **28C** illustrates an example of a third set of modular programmable widgets **116**, in a common grouped and configurable individual configuration. As shown, the modular programmable widgets **116** include a first modular programmable widget **116A** in the form of a combined double-length module including a dial **312** for operating a fan function and also a dial **312** for operating a temperature function; a second modular programmable widget **116B** in the form of a configurable touch screen **316**; a third modular programmable widget **116C** in the form of a configurable button **308**; and a fourth modular programmable widget **116D** in the form of a configurable dial **312**.

FIG. **28D** illustrates an example of a fourth set of modular programmable widgets **116**, in a hardcoded grouped configuration. As shown, a single modular programmable widget **116** having a width of five modular programmable widgets **116** includes a dial **312** for operating a fan function,

28

a button **308** for operating a defrost function, a dial **312** for operating a volume function; a button **308** for operating a lift gate function; and dial **312** for operating a fan function.

FIG. **28E** illustrates an example of a fifth set of modular programmable widgets **116** in a configurable grouped configuration. As shown, a single modular programmable widget **116** having a width of five modular programmable widgets **116** includes a configurable dial **312**, a configurable button **308**, a configurable touch screen **316**, a configurable button **308**, and a configurable dial **312**.

FIG. **28F** illustrates an example of a sixth set of modular programmable widgets **116** in a hybrid bar configuration. In this context, hybrid refers to a modular programmable widget **116** having both preprogrammed and configurable controls. As shown, the single modular programmable widget **116** has a width of five modular programmable widgets **116** includes a dial **312** for operating a temperature function, three configurable buttons **308**, and a dial **312** for operating a fan function.

FIGS. **29A-29E** illustrate example modular programmable widgets **116** in yet another alternative packaging. Referring collectively to FIGS. **29A-29E**, an enclosure having a generally rectangular prism shape may be defined by a front housing **302** and a back housing **304**. The front housing **302** may define a generally flat front surface of the modular programmable widget **116**, while the back housing **304** may define the remainder of the enclosure, including the rear, side, and top walls that house the internal circuitry and other components of the modular programmable widget **116**. Controls may be provided on the front housing **302**, while connectivity to the vehicle **102** may be provided via contact pads **306** exposed through the rear of the back housing **304**. The contact pads **306** may allow for transmission of data and power between the modular programmable widgets **116** and the dock **202**. In an example, the contact pads **306** may include a pin-based connector (as best seen in FIG. **31**).

The front housing **302** may expose various controls that may provide interaction with the user. In the example of FIG. **29A**, this may include a configurable button **308**. While not shown, in some examples the configurable button **308** may include a display **310**. FIG. **29B** illustrates an example modular programmable widget **116** having a configurable dial **312**. FIG. **29C** illustrates an example modular programmable widget **116** having additive power/data ports **314**. As shown, the modular programmable widgets **116** in FIG. **10C** provides two USB power/data ports **314**, but this is merely an example, and different quantities and types of connectors may be exposed by the modular programmable widgets **116**. FIG. **29D** illustrates an example modular programmable widget **116** having a toggle button **318**. FIG. **29E** illustrates an example modular programmable widget **116** having a display **310**. As shown in this example, the modular programmable widget **116** provides a flat display **310** on the front surface of the modular programmable widget **116**.

FIG. **30** illustrates a perspective view of the dock **202A** integrated below a configurable display **114** of the vehicle **102** for attachment of the modular programmable widgets **116**. Similar to the previous packaging example, the back housing **304** of the modular programmable widgets **116** may define a horizontal channel **2402** configured to mate with a corresponding mounting bar **2504** disposed on the rear surface **1604** of the dock **202**. Also similar to as discussed, the rear surface **1604** may expose dock connectors **514** for connection to the contact pads **306** of the modular programmable widgets **116**. The rear surface **1604** may also define locators **518** to aid in the placement of the modular programmable widgets **116** with respect to the dock connectors

29

514. In this example the locators 518 may define vertical bars extending forwards between the locations for receiving the modular programmable widgets 116, serving to guide the modular programmable widget 116 into position when inserted (such as shown in FIG. 32).

FIG. 31 illustrates an example exploded view of the modular programmable widget 116 shown in FIG. 29B having a configurable dial 312. As mentioned above, the modular programmable widget 116 includes the front housing 302 and back housing 304. The back housing 304 of the modular programmable widget 116 may define a generally flat vertical surface of the rear of the modular programmable widget 116. In this example, the back housing 304 defines a horizontal channel 2402 extending along the length of the back housing 304. As shown, the horizontal channel 2402 is defined along length of the back housing 304 between about one quarter from the top through about one half from the top of the modular programmable widget 116, forming about one quarter of the vertical height of the modular programmable widget 116. Below the horizontal channel 2402, the back housing 304 of the modular programmable widget 116 curves down to the front housing 302 of the modular programmable widget 116.

The modular programmable widget 116 further includes one or more internal magnets 402 configured to retain the modular programmable widget 116 in position. In the illustrated example, the modular programmable widget 116 includes three internal magnets 402 at different positions within the modular programmable widget 116. The inside rear wall of the back housing 304 defines magnet recesses 404 lengthwise across the rear wall, sized to receive the internal magnets 402 and to hold the internal magnet 402 in position. As shown, the internal magnets 402 and magnet recesses 404 are each of a generally rectangular shape.

The modular programmable widget 116 further includes a main circuit board 406 configured to physically hold the electronic components of the modular programmable widget 116. For example, the main circuit board 406 may hold the components responsible for data communication with the vehicle 102 and/or receiving power from the vehicle 102. The main circuit board 406 may also hold components supporting the specific functionality of the modular programmable widget 116. In an example, the main circuit board 406 may define predefined connection points 3102 for receiving component such as rotary encoders 438 and/or switches 1302. The main circuit board 406 may also include one or more light modules 1320. These light modules 1320 may be LEDs, in an example, that may be configured to display different colors and/or intensities of light. The main circuit board 406 may also define circuit traces and/or wires to electrically connect the electrical components to one another.

The main circuit board 406 may also be physically and electrically connected to components to allow for the electrical connection of the modular programmable widget 116 to the dock 202. In an example, the top of the main circuit board 406 may hold the contact pad 306 configured to provide electrical connection to the dock 202. In this illustrated example, the contact pad 306 is a set of pins extending rearwards towards the back of the modular programmable widget 116. The contact pad 306 may slide into a contact slot 412 formed into the rear of the attachment body 414 of the back housing 304. In this example, the contact slot 412 may be formed as an aperture allowing the pins of the contact pad 306 to pass through the back housing 304.

An inner housing 3104 may define a cover having a front face and peripheral walls extending towards the rear of the

30

modular programmable widgets 116. The inner housing 3104 may be sized to enclose the main circuit board 406 against the front face of the back housing 304. To facilitate assembly of the modular programmable widget 116, one or more fasteners 432, such as screws or pins, may be used to hold the back housing 304 and inner housing 3104 together. In an example, through holes 434 may be formed through the back of the back housing 304 which may be aligned with through holes through the main circuit board 406, and finally fixed into apertures (e.g., threaded holes, not shown) into the rear face of the inner housing 3104. It should be noted that other approaches to assembling the modular programmable widgets 116 may be used in other examples, such as snap fit connectors and/or glue. The inner housing 3104 may also define one or more inner windows 3106 to allow the user-manipulable controls, such as configurable buttons 308, configurable dials 312, configurable displays 114, etc., to be accessible through the inner housings 3104. As shown, the inner housing 3104 includes an inner window 3106 sized to allow passage of the rotary encoder 438 outside of the inner housing 3104.

The front housing 302 may define a front window 418 defining a location for mounting of the controls of the modular programmable widget 116. In this example, the modular programmable widget 116 is a configurable dial 312 and the front window 418 is a sized to receive the rotary dial 454 of the configurable dial 312 onto the rotary encoder 438. The front housing 302 may be configured to snap fit into the walls of the back housing 304, over the inner housing 3104, to form the finished front exterior of the modular programmable widget 116.

FIG. 32 illustrates a side cutaway view of an attachment of an example modular programmable widget 116 to the dock 202. As shown, the horizontal channel 2402 is mated with the mounting bar 2504, and the pins of the contact pad 306 of the modular programmable widget 116 are electrical and physical contact with the dock connector 514 of the dock 202. Also, the internal magnet 402 of the modular programmable widget 116 is attracted to a dock magnet 2502 within the dock 202, providing force to keep the modular programmable widget 116 in place within the dock 202.

Thus, despite the differences in controls and size of the modular programmable widgets 116, the modular programmable widgets 116 each define the same connection components. For instance, each of the modular programmable widgets 116 has the same electrical contact pads 306 and locating groove 420 as discussed above. Thus, any of the modular programmable widgets 116 may be connected to the dock 202, allowing for a user to choose whichever modular programmable widget 116 has the desired controls and/or appearance.

The modular programmable widgets 116 are designed to be removable and changeable by the user. However, in some cases it may be desirable to implement locking features to secure the modular programmable widgets 116 into place.

FIGS. 33A-33C collectively illustrates aspects of an approach to locking the modular programmable widgets 116. FIG. 33A illustrates a side cutaway view of a modular programmable widget 116 having a slidable lock 3302 in a locked position. FIG. 33B illustrates a side cutaway view of the modular programmable widget 116 showing the slidable lock 3302 in an unlocked position. FIG. 33C illustrates a rear perspective partially transparent view of the modular programmable widget 116 having the lock 3302 in the locked position.

31

The lock 3302 may define a lock body 3304 provided within the interior of the modular programmable widget 116, such that the lock body 3304 is nested between the front housing 302 and back housing 304 of the modular programmable widget 116. The lock body 3304 may be movable within and relative to the modular programmable widgets 116, such that the lock body 3304 is vertically slidable within and relative to the door modular programmable widget 116 between a lower, unlocked position and a raised, locked position.

The lock 3302 may further define a grip portion 3306, illustrated herein at a lower end of the lock 3302 extending outwards from an opening 3308 through the back housing 304. The grip portion 3306 may facilitate operation of the lock 3302 by giving the user a convenient handle to grip onto and lower the lock 3302 as desired. The grip portion 3306 that is exposed to a user's hand is hidden on the back side of the modular programmable widget 116. Advantageously, this may prevent affecting the outer look or aesthetics of the front of the modular programmable widget 116.

The lock body 3304 may further define a locking pin 3310 protruding upwards from an upper end of the lock body 3304. The housing of the modular programmable widget 116 may expose an aperture 3312 through which the locking pin 3310 may extend upwards above the top face of the housing. The locking pin 3310 may be configured to engage with a receptacle 3314 defined by a portion of the housing below the configurable display 114, such as through the securing channel 512. The receptacle 3314 may be sized and located in a position to allow the locking pin 3310 to pass upwards into the dock 202 when the lock 3302 is in the raised, locked position.

A biasing element 3318 may be provided within the modular programmable widget 116 to bias the locking pin 3310 upwards when not being lowered by the user. In an example, the biasing element 3318 may be a spring element placed between a biasing surface 3320 of the lock body 3304 and a stop 3322, such that in a default state the biasing element 3318 presses the biasing surface 3320 upwards and therefore the lock body 3304 upwards into the raised, locked position.

The stop 3322 may be fixed to a mounting plate 3324, which may be installed inside the modular programmable widget 116 parallel to the main circuit board 406. As shown, fasteners 3326 secure the mounting plate 3324 to the main circuit board 406. Thus, the mounting plate 3324 and the inside of the rear wall of the modular programmable widget 116 server to collectively form a channel into which the lock 3302 may slide up and down, in a controlled linear motion.

When installing the modular programmable widget 116 into the dock 202, the locating groove 420 of the modular programmable widget 116 may be aligned with the protrusion 520 of the dock 202. As the locating groove 420 is mated to the protrusion 520, the modular programmable widget 116 may be around the protrusion 520 such that the attachment body 414 of the modular programmable widget 116 moves into the securing channel 512. In doing so, the contact pad 306 of the modular programmable widget 116 moves towards electrical contact with the dock connector 514 of the dock 202. As placement of the modular programmable widget 116 is completed, the biasing element 3318 urges the locking pin 3310 of the lock 3302 through the aperture 3312 and into the receptacle 3314, locking the modular programmable widget 116 into place in the dock 202.

32

To remove the modular programmable widget 116, the user may grab and push down on the grip portion 3306 of the lock 3302 on the back of the modular programmable widget 116. This may accordingly move the locking post upwards, in turn lifting the locking pin 3310 out of the receptacle 3314 and into the lowered, unlocked state. Once moved, the user may then pull the modular programmable widget 116 from the dock 202.

Variations on the lock 3302 are possible. In an example, the operation of the lock 3302 may be automated by use of a solenoid, nitinol wire, or other device that may be electrically signaled by the widget controller 112 to allow for the controlled locking and unlocking of the modular programmable widget 116. Such a lock feature may provide a level of security, e.g., if there are modular programmable widgets 116 in a second row dock 202 of a vehicle 102 being used for ride-share. For instance, the user may be required to enter a credential (e.g., a PIN, another code, a voice command, a voice print, presence of an authorized key fob, presence of an authorized mobile device, etc.) to lock (or unlock) the modular programmable widgets 116 from the dock 202. Responsive to receipt of the credential, the vehicle 102 may perform the locking (or unlocking) operation.

In another example, the vehicle 102 may implement a feature that allows an owner to restrict access by the modular programmable widgets 116 to certain vehicles 102. In an example, the modular programmable widget 116 may be keyed to a credential e.g., a PIN, another code, a voice command, a voice print, presence of an authorized key fob, presence of an authorized mobile device, etc.), and that credential may be required to be received to the configurable display 114 or other HMI of the vehicle 102 to enable the modular programmable widget 116 to operate.

FIGS. 34A-34C collectively illustrate aspects of an alternate approach to locking the modular programmable widgets 116. FIG. 34A illustrates a side cutaway view of a modular programmable widget 116 having the slidable lock 3302 in a locked position. FIG. 33B illustrates a side cutaway view of the modular programmable widget 116 showing the slidable lock 3302 in an unlocked position. FIG. 33C illustrates a rear perspective partially transparent view of the modular programmable widget 116 having the lock 3302 in the locked position.

As compared to the approach shown in FIGS. 33A-33C, in FIGS. 34A-34C the lock 3302 may be driven by an asymmetrical cam 3402. In this example, the lock 3302 may define a follower 3404 angling outwards from the generally vertical lock body 3304.

Similar to as discussed above, the lock body 3304 may further define a locking pin 3310 protruding upwards from an upper end of the lock body 3304. The housing of the modular programmable widget 116 may expose an aperture 3312 through which the locking pin 3310 may extend upwards above the top face of the housing. In this example, the locking pin 3310 may be configured to engage with a receptacle 3314 defined into the lower face of the mounting bar 2504. The receptacle 3314 may be sized and located in a position to allow the locking pin 3310 to pass upwards into the mounting bar 2504 when the lock 3302 is in the raised, locked position.

The cam 3402 may be mounted to an axle via through hole 3406 to provide for rotational movement of the asymmetrical cam 3402 about the axis defined by the through hole 3406. Additionally, the cam 3402 may define an attachment 3408 for receiving a connection to a release actuator. For instance, the attachment 3408 may operably connect to a nitinol wire driven by a release actuator, such that the release

33

actuator may pull the cam **3402** counterclockwise, forcing the ovoid cam **3402** to push the follower **3404** downwards, and moving the locking pin **3310** downwards out of the receptacle **3314**. Although not shown in FIGS. **34A-34C**, a biasing element **3318** may be provided to bias the locking pin **3310** upwards when the release actuator is disengaged.

When installing the modular programmable widget **116** into the dock **202**, the horizontal channel **2402** of the modular programmable widget **116** may be inserted onto the mounting bar **2504**. In doing so, the contact pad **306** of the modular programmable widget **116** moves towards electrical contact with the dock connector **514** of the dock **202**. As placement of the modular programmable widget **116** is completed, the biasing element **3318** urges the locking pin **3310** of the lock **3302** through the aperture **3312** and into the receptacle **3314**, locking the modular programmable widget **116** into place on the mounting bar **2504**.

To remove the modular programmable widget **116**, the user may select a release option from the HMI display system **104**. This may accordingly drive the release actuator to pull the cam **3402**, in turn pulling the locking pin **3310** out of the receptacle **3314** and into the lowered, unlocked state. Once moved, the user may then pull the modular programmable widget **116** from the mounting bar **2504**.

FIGS. **35A-35B** collectively illustrate an example of a storage compartment **3502** with integrated dock **202** functionality. FIG. **35A** illustrate the storage compartment **3502** in a closed position while FIG. **35B** illustrate the storage compartment **3502** in an opened position. In general, the storage compartment **3502** may provide a glove box style of hidden storage to the vehicle **102**.

The storage compartment **3502** may define a generally rectangular tray with a flat tray bottom **3504**, left and right side faces **3506**, and a front face **3508**. The rear ends of the left and right side faces **3506** may be hingedly attached to the dash **3510** of the vehicle **102** at pivot points **3512**. This allows the storage compartment **3502** to rotate downwards via the pivot points **3512** into the open position as shown in FIG. **35B**, and upwards into the closed position as shown in FIG. **35A**. In some examples, the storage compartment **3502** may be lockable in the closed position with entry of an on-screen pin, e.g., into the configurable display **114**.

The front face **3508** of the storage compartment **3502** may provide integrated dock **202** functionality. As shown, a selection of modular programmable widgets **116** are retained in the dock **202** located on the front face **3508** of the storage compartment **3502**. The dock **202** and modular programmable widget **116** may be implemented using any of the various designs discussed in detail herein. A configurable dial **312**, a configurable button **308**, additive power/data ports **314**, and another configurable dial **312** are connected in the illustration, but this is only one example and different quantities and types of modular programmable widgets **116** may be used.

FIGS. **36A-36C** collectively illustrate examples of adding a phone dock **3602** to the storage compartment **3502** in addition to the with integrated dock **202** functionality. FIG. **36A** illustrates use of a phone dock **3602** and dock plate **3604** for use with the storage compartment **3502**. FIG. **36B** illustrates a single phone dock **3602** with a landscape orientation. FIG. **36B** illustrates a dual phone dock **3602** with a portrait orientation.

The phone dock **3602** may be a device configured to wirelessly charge a user's mobile device **3606**. The phone dock **3602** may also provide a data connection between the mobile device **3606** and the vehicle **102**, for example, to

34

allow for applications executed by the mobile device **3606** to utilize the configurable display **114** or other features of the HMI display system **104**.

The dock plate **3604** may define a surface above the phone dock **3602** onto which the mobile device **3606** may be placed for charging and/or connection to the vehicle **102**. As shown, the phone dock **3602** may then be placed into the storage compartment **3502**, and then the dock plate **3604** may be attached above the storage compartment **3502** to act as a cover or top over the storage compartment **3502**. In an example, the left and right sides of the dock plate **3604** may define connectors **3608** (such as indentations) configured to secure onto corresponding protrusions extending upwards from the left and right side faces **3506** of the storage compartment **3502**.

The upper face of the dock plate **3604** may include a rubberized material or other material to provide friction between the mobile device **3606** and the storage compartment **3502**, to prevent the mobile device **3606** from moving while being charged. In some examples, the dock plate **3604** may define a recessed area **3612** into which the mobile device **3606** may be retained into position for charging and/or connection to the vehicle **102**. In the example of FIG. **36B**, a single mobile device **3606** is placed on top of the dock plate **3604**, while in the example of FIG. **36C**, two mobile devices mobile device **3606** are placed onto the larger dock plate **3604**.

FIGS. **37A-37D** collectively illustrate aspects of an alternative dock **202** having rotating dock slots **3702**. As shown, the dock **202** may extend for a length below the configurable display **114** between left and right dock endcaps **3704**. Each of the rotating dock slots **3702** may define a generally horizontal cylindrical shape with a length sufficient to hold a modular programmable widget **116**. Each of dock spacers **3706** may also define a generally horizontal cylindrical shape, as shown with the same radius as that of the rotating dock slots **3702**. The dock **202** may define a series of alternating dock spacers **3706** and rotating dock slots **3702** aligned end-to-end along their flat surfaces. Each of the dock endcaps **3704** may also provide a smooth rounded appearance for the ends of the dock **202**, with generally the same radius as the rotating dock slots **3702**. The dock spacers **3706** may be fixed in rotation, but the rotating dock slots **3702** may be mounted to rotate along their cylindrical axis.

The rotating dock slots **3702** may each also define a flat face **3708**, cut or otherwise formed into the surface of the cylinder. The flat face **3708** may expose the dock connector **514** for providing electrical connection and/or mechanical connection to the modular programmable widgets **116** when attached to the dock **202**. Each of the rotating dock slots **3702** may be opened to reveal an individual dock connector **514** on the flat face **3708** for receiving a modular programmable widget **116**. When not being used, the rotating dock slots **3702** may be rotated closed to hide the flat face **3708** and the dock connector **514**. When closed, the dock slots **3702** may give the appearance of a decorative trim piece, without showing the details of the disconnected electrical dock connectors **514**.

FIG. **37A** illustrates an example of the rotating dock **202** with four rotating dock slots **3702**. From left to right, the first of the rotating dock slots **3702** is in the opened position, and the second of the rotating dock slots **3702** is in an intermediate rotation between the opened and closed position. The first of the rotating dock slots **3702** is holding a modular programmable widget **116**. The remaining two rotating dock slots **3702** are in the closed position.

35

FIG. 37B illustrates an example of a rotating dock 202 with four rotating dock slots 3702, with the first rotating dock slot 3702 in the open position and holding a modular programmable widget 116, the second rotating dock slot 3702 in the open position and empty, and the remaining rotating dock slot 3702 in the closed position.

FIG. 37C illustrates an example of a rotating dock 202 with four rotating dock slots 3702, with the first two rotating dock slots 3702 open and filled with modular programmable widgets 116, and the remaining rotating dock slot 3702 in the closed position.

FIG. 37D illustrates an example of a rotating dock 202 with four rotating dock slots 3702, with the first two rotating dock slots 3702 open and empty, and the remaining two rotating dock slots 3702 in the closed position.

A user may typically perform a set of activities or actions upon arriving home. These actions may include opening the garage door, unlocking the front door to the house, setting the home thermostat to a desired temperature, deactivating security cameras, recalling a smart vacuum device to its dock, and turning on the kitchen lights or other entry lights. The performance of these actions is referred to herein as transitioning to the home mode.

The user may also typically perform a second set of activities or actions in anticipation of departing the home. These actions may include closing the garage door, locking the front door, setting the home thermostat to an eco mode, activating security cameras, setting the smart vacuum device to be on patrol, and turning off the kitchen lights or other lights. The performance of these actions is referred to herein as transitioning to the away mode.

FIGS. 38A-B illustrate an example system 3800 for automatically switching between the home mode and the away mode. The system 3800 may include a physical toggle button 3802 included within the cabin of the vehicle 102. The physical toggle button 3802 may be a two-state physical device, allowing the vehicle 102 to be easily transitioned between the home and away modes. FIG. 38A illustrates the physical toggle button 3802 in the away mode, while FIG. 38B illustrates the physical toggle button 3802 in the home mode.

As shown, the physical toggle button 3802 is located outboard of the steering wheel, but this is only one example position. In another example, the physical toggle button 3802 may be located below the center screen (e.g., below the configurable display 114). In yet another example, the physical toggle button 3802 may be located on the center console. In still another example, the physical toggle button 3802 may be located on the header above the windshield, on the rear view mirror, etc.

FIGS. 39A-39D illustrate example designs for the physical toggle button 3802. FIG. 39A illustrates an example flip style physical toggle button 3802. This example physical toggle button 3802 may be toggled by pressing in the desired side. FIG. 39B illustrates an example slide style physical toggle button 3802. This example physical toggle button 3802 may be toggled by sliding the switch to the desired side. FIG. 39C illustrates an example toggle style physical toggle button 3802. This example physical toggle button 3802 may be toggled by moving the toggle to the desired side. FIG. 39D illustrates an example pushbutton style physical toggle button 3802. This example physical toggle button 3802 may be toggled by pressing the button.

Instead of or in addition to the physical toggle button 3802, it should be noted that one or more of the modular programmable widgets 116 may be programmed to perform the toggle function as well. For instance, a configurable

36

button 308, configurable display 114, configurable four-way button pad 336, configurable multi-directional control 344, etc. may be programmed by the user to invoke the toggle between the home and away modes.

Returning to FIGS. 38A-38B, the system 3800 further includes one or more smart home devices 3804. The smart home devices 3804 may include various devices that may be remotely controlled to perform one or more home automation functions. An example smart home device 3804 may be a garage door opener. The garage door opener smart home device 3804 may support functions such as open garage door and close garage door. Another example smart home device 3804 may be a front door lock. The front door opener smart home device 3804 may support functions such as unlock front door and lock front door. Yet another example smart home device 3804 may be a home thermostat. The home thermostat door opener smart home device 3804 may support functions such as set temperature, turn on air conditioning, turn off air conditioning, turn on heat, turn off heat, set to eco mode, set to occupied mode, etc. In another example, the smart home device 3804 may be a security camera. The security camera smart home devices 3804 may support function such as activate camera and deactivate camera. As still another example, the smart home device 3804 may be a robot vacuum. The robot vacuum smart home device 3804 may support function such as begin cleaning patrol and return to charger base. In another example, the smart home device 3804 may be a light. The light smart home device 3804 may support functions such as turn on light, turn off light, set light brightness, and set light color.

When the state of the physical toggle button 3802 is changed by the user, the arriving or departing behavior may be automatically performed. For example, the set of arriving actions may be automatically performed by the vehicle 102 responsive to transitioning the physical toggle button 3802 from the away mode to the home mode. In addition, the set of departing actions may be automatically performed by the vehicle 102 responsive to transitioning the physical toggle button 3802 from the home mode to the away mode.

To perform the actions, the vehicle 102 may send a state change request 3806 to a cloud server 3808 over a communications network 3810. The cloud server 3808 may use a function mapping 3812 stored to or otherwise accessible to the cloud server 3808 to identify the actions to be performed. For instance, the function mapping 3812 may indicate network addresses of the smart home devices 3804, account/login details of the smart home devices 3804, and/or the functions to activate or deactivate for the smart home devices 3804. The cloud server 3808 may then communicate with the smart home devices 3804 to implement the identified actions. By sending the state change request 3806 to the cloud server 3808 and having the cloud server 3808 forward the commands to the smart home devices 3804, the vehicle 102 may be able to configure the mode of the smart home devices 3804 without having to be in Wi-Fi or other communications with the home itself.

The cloud server 3808 may further provide a state change result 3814 as a confirmation to the vehicle 102 indicating the result of the configuration of the smart home devices 3804. In an example, the state change result 3814 may indicate whether or not the state change is completed. In another example, the state change result 3814 may indicate which functions from the function mapping 3812 could or could not be completed.

In some examples, the HMI display system 104 may provide a chime and/or an on screen icon to confirm the change in status. In another example, the physical toggle

button **3802** may include a light, and the light may be activated responsive to the physical toggle button **3802** being confirmed by the state change result **3814** as being in sync with the home. In other words, when the physical toggle button **3802** is toggled, the light may be deactivated until the state change result **3814** is received, and if the state change result **3814** indicates success, then the light may be reactivated. This accordingly allows the user to easily understand that the state change has been performed.

FIG. **40** illustrates an example main screen **4002** of the configurable display **114** from which a configuration screen for the home and away modes is selectable. As shown, the configurable display **114** include a climate screen area **4004** offering touch controls to manipulate various climate control functions of the vehicle **102**. The configurable display **114** may include a main screen area **4006** in which the currently active application may be displayed. As shown a navigation map application is currently being shown. Additionally, the main screen **4002** also provides a mode configuration control **4008** that, when selected causes the configuration screen for the home and away modes to be displayed.

FIG. **41** illustrates an example mode configuration screen **4102** of the configurable display **114**. In an example, the mode configuration screen **4102** may be displayed responsive to selection of the mode configuration control **4008** from the main screen **4002**.

The mode configuration screen **4102** may allow the user to specify the functions to be performed for both arriving and departing the home. As shown, the mode configuration screen **4102** includes an arriving home routine **4104** label indicating a set of controls for configuring the set of arriving actions and a departing home routine **4106** indicating a set of controls for configuring the set of departing actions.

To configure the set of arriving actions, the mode configuration screen **4102** includes an arriving actions list **4108**. The list includes a set of items that correspond to the different actions to be automatically performed when transitioning to the home mode. Each action may be shown in the arriving actions list **4108**. For instance, for each action a smart device description **4110** may be shown. Here the smart device description **4110** includes a name and icon indicative of the smart home device **3804** to be configured are shown, but different examples are possible, such as a name only, icon only, a user customizable name, etc.

Additionally, a function selector **4112** including a list of the functions supported by the smart home device **3804** may be shown. For instance, the user may select the function selector **4112**, which may open a dropdown of the available functions to be controlled. The available functions may be based on the device discovery or may be predefined based on the type of smart home device **3804**. The user may select the function to be performed by the smart home devices **3804** from dropdown list. The arriving actions list **4108** may also include a toggle **4114** for each list item. This toggle **4114** allows the user to select how to set the chosen function when the user arrives at the home. For instance, the toggle **4114** allows the user to either turn on or off the chosen function upon arrival.

In some cases, the user may wish to add an additional device to the arriving actions list **4108**. To do so, the user may select the add devices control **4116**. Once selected, the vehicle **102** may initiate a device discovery function. In an example, the vehicle **102** may use the telematics control module **130** to wirelessly communicate with the smart home device **3804** local to the vehicle **102**. For instance, the telematics control module **130** may use any of various discovery protocols to broadcast and/or receive messages

indicative of the presence and/or available features of the smart home devices **3804**. The vehicle **102** may perform the discovery using one or more protocols, such as Wi-Fi, BLUETOOTH, BLUETOOTH Low Energy (BLE), ZigBee, Z-Wave, Thread, Matter, etc. In a specific example, the smart home devices **3804** may be discoverable by the vehicle **102** when the vehicle **102** is networked to the home, and the vehicle **102** may query for smart home devices **3804** that are connected to the user's home Wi-Fi. A list of available devices may be shown to the user, and the user may select from those devices to add them to the arriving actions lists **4108**.

Similar to the arriving actions list **4108**, to configure the set of departing actions, the mode configuration screen **4102** includes a departing actions list **4118**. The list includes a set of items that correspond to the different actions to be automatically performed when transitioning to the away mode. Each departing action may be shown in the departing actions list **4118**. For instance, for each action a smart device description **4120** may be shown, consistent with the display of the smart device description **4110**. The departing actions list **4118** may also include a function selector **4122** configured to operate as the function selector **4112** for departing actions, toggles **4124** that allows the user to select how to set the chosen function when the user departs from the home, and an add devices control **4126** for adding additional devices to the departing actions list **4118**. The function selector **4122**, toggles **4124**, and add devices control **4126** may operate as discussed above with respect to the function selector **4112**, toggles **4114**, and add devices control **4116**, respectively.

Once the user has completed configuring the arriving actions list **4108** and departing actions list **4118**, the user may select the close button **4128** to return to the main screen **4002**. In some examples, the mode configuration screen **4102** may confirm with the user to save changes before applying the changes to the function mappings **3812**. If the user confirms to update the actions, the HMI display system **104** may use the telematics control module **130** to send the updated function mapping **3812** to the cloud server **3808**.

FIG. **42** illustrates an example process **4200** for the configuration of the physical toggle button **3802**. In an example, the process **4200** may be performed by may be performed by the components of the vehicle **102** and cloud server **3808**, as discussed in detail herein.

At operation **4202**, the HMI display system **104** identifies available smart home devices **3804**. In an example, the user may select the add devices control **4116** or the add devices control **4126** from the mode configuration screen **4102**. Responsive to the selection, the HMI display system **104** may use the telematics control module **130** to wirelessly discover the smart home device **3804** local to the vehicle **102**. In some examples, the communication may further include discovering the available functions of the discovered smart home devices **3804**.

At operation **4204**, the HMI display system **104** adds available devices to the arriving actions list **4108** and/or the departing actions list **4118**. In an example, a list of available devices identified at operation **4202** may be shown to the user, and the user may select from those devices to add them to the arriving actions lists **4108** and/or the departing actions list **4118**, as selected by the user.

At operation **4206**, the HMI display system **104** receives selection of desired functions for the added devices. In an example, the user may utilize the function selector **4112** to configure the desired functions for the smart home devices **3804** added to the arriving actions list **4108** and may use the

function selector **4122** to configure the desired functions for the smart home devices **3804** added to the departing actions list **4118**.

At operation **4208**, the HMI display system **104** saves the function mappings **3812**. In an example, the user may use the close button **4128** to close the mode configuration screen **4102**. The user may also in some cases confirm that the changes to the function mapping **3812** should be applied to the cloud server **3808**. If confirmed, the HMI display system **104** may use the telematics control module **130** to send the updated function mapping **3812** to the cloud server **3808**. After operation **4208**, the process **4200** ends.

FIG. **43** illustrates an example process **4300** for the execution of a mode change triggered by the physical toggle button **3802**. In an example, the process **4200** may be performed by the components of the vehicle **102** and cloud server **3808**, as discussed in detail herein.

At operation **4302**, the HMI display system **104** detects whether the state of the physical toggle button **3802** has changed. In an example, the HMI display system **104** may be in communication with the physical toggle button **3802** internal to the vehicle **102** and may receive a signal indicative of the current state of the physical toggle button **3802**. If that state changes, control passes to operation **4304**. Otherwise, control remains at operation **4302**.

At operation **4304**, the HMI display system **104** indicates the state transition to the cloud server **3808**. In an example, the HMI display system **104** may direct the telematics control module **130** to send a state change request **3806** to a cloud server **3808** over a communications network **3810**. The state change request **3806** may indicate whether to change to the home mode or to the away mode.

At operation **4306**, the cloud server **3808** uses the function mapping **3812** to configure the smart home devices **3804**. In an example, the cloud server **3808** uses the function mapping **3812** configured using the process **4200** to identify the actions to be performed. The cloud server **3808** may then communicate with the smart home devices **3804** to implement the identified actions. By sending the state change request **3806** to the cloud server **3808** and having the cloud server **3808** forward the commands to the smart home devices **3804**, the vehicle **102** may be able to configure the mode of the smart home devices **3804** without having to be in Wi-Fi or other communications with the home itself.

At operation **4308**, the HMI display system **104** receives confirmation from the cloud server **3808**. In an example, the cloud server **3808** may further provide the state change result **3814** to the telematics control module **130** as a confirmation to the vehicle **102** indicating the result of the configuration of the smart home devices **3804**. For instance, the state change result **3814** may indicate whether or not the state change is completed. In another example, the state change result **3814** may indicate which functions from the function mapping **3812** could or could not be completed.

At operation **4310**, the HMI display system **104** indicates the confirmation in the HMI display system **104**. In some examples, the HMI display system **104** may provide a chime and/or an on screen icon to confirm the change in status. In another example, the physical toggle button **3802** may include a light, and the light may be activated responsive to the physical toggle button **3802** being confirmed by the state change result **3814** as being in sync with the home. After operation **4310**, the process **4300** returns to operation **4302**.

FIG. **44** illustrates an example process **4400** for the configuration of a modular programmable widget **116**. In an example, the process **4400** may be performed by the com-

ponents of the vehicle **102**, as discussed in detail herein, when operating in a configuration mode.

At operation **4402**, the HMI display system **104** detects the presence of a modular programmable widget **116**. In an example, responsive to connection of the modular programmable widget **116** to the configurable display **114**, the widget controller **112** of the HMI display system **104** may receive signals indicative of the connection. In a first example, the connection may include the contact pads **306** of the modular programmable widget **116** being electrically connected to the dock connectors **514** of the dock **202**. The modular programmable widgets **116** and docks **202** may be of any of the designs discussed in detail herein.

Regardless of example, the modular programmable widget **116** may provide information to the configurable display **114** via the electrical connection of the contact pads **306** to the dock connectors **514**. This information may be indicative of the type of the modular programmable widget **116**. This information may indicate, for example, that the modular programmable widget **116** is an HMI device. The information may further include information such as model of the modular programmable widget **116**, color of the modular programmable widget **116**, quantity of programmable dials, quantity of programmable toggles, quantity of programmable switches, quantity of hidden buttons or other controls, etc., which may be used to allow the configurable display **114** to display an image consistent with the modular programmable widget **116** that is attached. In some cases, the modular programmable widgets **116** may be preconfigured to have a specific function. In such an example, the information may indicate the preconfigured function.

At operation **4404**, the HMI display system **104** presents an HMI for the configuration of the modular programmable widget **116**. In an example, the configurable display **114** may provide a screen for the configuration of the modular programmable widget **116**.

At operation **4406**, the HMI display system **104** receives input assigning functions to controls of the modular programmable widget **116**. In an example, the HMI display system **104** may present an HMI to receive selection of a function to assign to the control (or controls) of the modular programmable widget **116**. The functions available for assignment may include, as some examples, hot keys (e.g., quick selection of menu operations within the menu structure of the vehicle), utility controls (e.g., towing functions), climate control functions (e.g., temperature settings, fan settings, etc.), gig work function (e.g., rideshare commands), media control functions (e.g., volume, station presets, media input selection), smart home settings (e.g., garage door operation, home climate control settings), vehicle settings (e.g., interior lighting settings, gauge cluster display modes), etc.

At operation **4408** the HMI display system **104** saves the function assignment to the display settings **134** in the memory **108**. For instance, the modular programmable widget **116** may have a unique identifier, and the display settings **134** may store the unique identifier the modular programmable widget **116**, an identifier of the control and a corresponding identifier of the assigned function corresponding to the unique identifier of the modular programmable widget **116** and the identifier of the control. In another example, each control of the modular programmable widget **116** may have a unique identifier, and the display settings **134** may store the unique identifier the control and a corresponding identifier of the assigned function corresponding to the unique identifier of the control. After operation **4408**, the process **4400** ends.

FIG. 45 illustrates an example process 4500 for the use of a configured modular programmable widget 116. In an example, as with the process 4400, the process 4500 may be performed by the components of the vehicle 102, as discussed in detail herein, when operating in a runtime mode.

At operation 4502, the HMI display system 104 receives input from a control of a modular programmable widget 116. In an example, a user may press, turn, or otherwise interact with configurable button 308, display 310, configurable dial 312, additive power/data port 314, configurable touch screen 316, toggle button 318, camera 320, biometric sensor 322, controls of the modular programmable widgets 116 installed to the HMI display system 104. Responsive to receipt of the input, the modular programmable widget 116 may send a message to the configurable display 114 via the electrical connection to the configurable display 114. This message may indicate, for example, an identifier of the modular programmable widget 116 and/or an identifier of the control of the modular programmable widget 116.

At operation 4504, the HMI display system 104 retrieves a function assignment corresponding to the selected control. In an example, the HMI display system 104 may access the memory 108 to identify, from the display settings 134, an identifier of the selected feature corresponding to the identifier of the control and/or the identifier of the modular programmable widget 116. At operation 4506, the HMI display system 104 executes the function identified at operation 4504. After operation 4506, the process 4500 ends.

The processes, methods, or algorithms disclosed herein can be deliverable to/implemented by a processing device, controller, or computer, which can include any existing programmable electronic control unit or dedicated electronic control unit. Similarly, the processes, methods, or algorithms can be stored as data and instructions executable by a controller or computer in many forms including, but not limited to, information permanently stored on non-writable storage media such as read-only memory (ROM) devices and information alterably stored on writeable storage media such as floppy disks, magnetic tapes, compact discs (CDs), random access memory (RAM) devices, and other magnetic and optical media. The processes, methods, or algorithms can also be implemented in a software executable object. Alternatively, the processes, methods, or algorithms can be embodied in whole or in part using suitable hardware components, such as Application Specific Integrated Circuits (ASICs), Field-Programmable Gate Arrays (FPGAs), state machines, controllers or other hardware components or devices, or a combination of hardware, software and firmware components.

While exemplary embodiments are described above, it is not intended that these embodiments describe all possible forms encompassed by the claims. The words used in the specification are words of description rather than limitation, and it is understood that various changes can be made without departing from the spirit and scope of the disclosure. As previously described, the features of various embodiments can be combined to form further embodiments of the invention that may not be explicitly described or illustrated. While various embodiments could have been described as providing advantages or being preferred over other embodiments or prior art implementations with respect to one or more desired characteristics, those of ordinary skill in the art recognize that one or more features or characteristics can be compromised to achieve desired overall system attributes, which depend on the specific application and implementation. These attributes can include, but are not limited to strength, durability, life cycle, marketability, appearance,

packaging, size, serviceability, weight, manufacturability, case of assembly, etc. As such, to the extent any embodiments are described as less desirable than other embodiments or prior art implementations with respect to one or more characteristics, these embodiments are not outside the scope of the disclosure and can be desirable for particular applications.

With regard to the processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. In other words, the descriptions of processes herein are provided for the purpose of illustrating certain embodiments and should in no way be construed so as to limit the claims.

Accordingly, it is to be understood that the above description is intended to be illustrative and not restrictive. Many embodiments and applications other than the examples provided would be apparent upon reading the above description. The scope should be determined, not with reference to the above description, but should instead be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. It is anticipated and intended that future developments will occur in the technologies discussed herein, and that the disclosed systems and methods will be incorporated into such future embodiments. In sum, it should be understood that the application is capable of modification and variation.

All terms used in the claims are intended to be given their broadest reasonable constructions and their ordinary meanings as understood by those knowledgeable in the technologies described herein unless an explicit indication to the contrary is made herein. In particular, use of the singular articles such as "a," "the," "said," etc. should be read to recite one or more of the indicated elements unless a claim recites an explicit limitation to the contrary. The term "substantially," "generally," or "about" may be used herein and may modify a value or relative characteristic disclosed or claimed. In such instances, "substantially," "generally," or "about" may signify that the value or relative characteristic it modifies is within $\pm 0\%$, 0.1% , 0.5% , 1% , 2% , 3% , 4% , 5% or 10% of the value or relative characteristic (e.g., with respect to degrees of offset from an angle when referring to substantially perpendicular or parallel).

The abstract of the disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

While exemplary embodiments are described above, it is not intended that these embodiments describe all possible

forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

What is claimed is:

1. A modular programmable widget for a vehicle, comprising:

a front housing and a back housing defining an enclosure; one or more physical controls provided on a front face of the enclosure;

a contact pad exposed by the enclosure for electrical connection to a dock, the contact pad providing for transmission of data and power between the modular programmable widget and the dock; and

a toe-in alignment mechanism configured to facilitate guided engagement of the modular programmable widget with the dock, wherein a top face of enclosure defines a locating groove providing a concave surface along the length of the top face of the enclosure, the locating groove being sized to receive a corresponding protrusion of the dock during installation of the modular programmable widget to the dock, the groove and protrusion defining a pivot point for guiding rotation of the modular programmable widget into place such that the contact pad is in electrical contact with a dock connector of the dock.

2. The modular programmable widget of claim 1, wherein the locating groove is defined by the front housing.

3. The modular programmable widget of claim 1, wherein the locating groove is defined by the back housing.

4. The modular programmable widget of claim 1, wherein the back housing defines an attachment body extending rearward from a vertical surface of a rear wall of the back housing.

5. The modular programmable widget of claim 4, wherein the attachment body is sized to conform with a profile of a securing channel of the dock.

6. The modular programmable widget of claim 4, wherein the attachment body forms a frustrum shape extending rearwards from the back housing.

7. The modular programmable widget of claim 4, wherein the attachment body defines a contact window at the top of the attachment body to expose the contact pad through the top of the back housing for the electrical connection to the dock.

8. The modular programmable widget of claim 1, wherein the back housing defines a horizontal channel extending along the length of the back housing, the horizontal channel configured to receive a mounting bar of the dock, the mounting bar defining a contour to guide the modular programmable widget to a proper height for connection of the contact pad to the dock.

9. The modular programmable widget of claim 1, further comprising an internal magnet mounted within the enclosure, the internal magnet being configured to retain the modular programmable widget in position within the dock.

10. The modular programmable widget of claim 9, wherein an inside rear wall of the back housing defines a magnet recess lengthwise across the inside rear wall, sized to receive the internal magnet and hold the internal magnet in position.

11. The modular programmable widget of claim 1, wherein the contact pad includes leaf spring contacts.

12. The modular programmable widget of claim 1, wherein the contact pad includes a magnetic connector for securing the contact pad to the dock.

13. The modular programmable widget of claim 1, wherein the contact pad includes a set of pins extending rearwards towards the back of the modular programmable widget.

14. The modular programmable widget of claim 1, wherein the one or more physical controls include one or more of a configurable button, a configurable dial, a power/data port, a configurable touch screen, a toggle button, a camera, a biometric sensor, a credit card reader, an radio frequency identification (RFID) sensor, an inventory tracker, an air quality sensor, a light emitting diode (LED) light array, a joystick, a four-way button pad, an high-definition multimedia interface (HDMI) port, an audio jack, a trailer control, a multi-directional control, a wireless charger, an access pass button pad, an analog clock, or a companion device.

15. The modular programmable widget of claim 1, further comprising a removable signifier cap mounted to one of the one or more physical controls, the signifier cap indicating a function assigned to the one of the physical controls.

16. The modular programmable widget of claim 15, further comprising:

a rotary encoder defining a cylindrical aperture;

a switch mounted within the cylindrical aperture;

a switch stem defining an elongate body with an inner end aimed towards the switch to transfer pressure provided by a user onto the signifier cap onto the switch; and

a switch guide installed within the aperture defining a circular outer surface sized to allow for insertion into the aperture as well as a guide aperture open to receive the elongate body of the switch stem to limit the switch stem to linear motion towards or away from the switch.

17. The modular programmable widget of claim 1, wherein the one or more physical controls include a plurality of knobs and one or more power/data ports.

18. The modular programmable widget of claim 1, wherein the one or more physical controls include a predefined second row bundle of controls, including volume, temperature, and fan speed controls.

19. The modular programmable widget of claim 1, wherein a main circuit board of the modular programmable widget is be preprogrammed to control predefined functions by communication with a widget controller of the vehicle when the modular programmable widget is electrically connected to the dock.

20. A vehicle system for utilizing modular programmable widgets, comprising:

a dock comprising a widget controller;

a modular programmable widget, comprising an enclosure,

one or more physical controls provided on a front face of the enclosure,

a contact pad exposed by the enclosure for electrical connection to the widget controller of the dock, the contact pad providing for transmission of data and power between the modular programmable widget and the dock, and

a toe-in alignment mechanism configured to facilitate guided engagement of the modular programmable widget with the dock, wherein a top face of enclosure defines a locating groove providing a concave surface along the length of the top face of the enclosure, the locating groove being sized to receive a corresponding protrusion of the dock during installation

45

of the modular programmable widget to the dock, the groove and protrusion defining a pivot point for guiding rotation of the modular programmable widget into place such that the contact pad is in electrical contact with a dock connector of the dock; and

a human-machine interface (HMI) display system, in communication with a configurable display, configured to

in a configuration mode, present a user interface to the configurable display for the configuration of the one or more physical controls of the modular programmable widget, and

in a runtime mode, perform an assigned function responsive to user input to the one or more physical controls.

21. The vehicle system of claim 20, wherein a main circuit board of the modular programmable widget is be preprogrammed to control predefined functions by communication

46

with the widget controller when the modular programmable widget is electrically connected to the dock.

22. The vehicle system of claim 20, wherein the one or more physical controls include a predefined second row bundle of controls, including volume, temperature, and fan speed controls.

23. The vehicle system of claim 20, wherein the modular programmable widget is keyed to a credential, such that the credential is required to be received to the HMI display system to enable the modular programmable widget to operate.

24. The vehicle system of claim 20, wherein the modular programmable widget is keyed to a credential, such that the credential is required to be received to the HMI display system to lock the modular programmable widget to the dock or unlock the modular programmable widget from the dock.

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